



AFE4900 Evaluation Module

This user's guide describes the characteristics, operation, and use of the AFE4900 EVM demonstration kit. This demonstration kit is an evaluation module for the AFE4900 device. The family of devices are fully-integrated AFE, ideally suited for *Optical Heart Rate* monitoring, *Pulse Oximeter*, and *Electrocardiogram* (ECG) applications. The EVM is intended for prototyping and evaluation. This user's guide includes a complete circuit description, schematic diagram, and bill of materials.

The following related documents are available through the Texas Instruments web site at www.ti.com:

Device	Literature Number
AFE4900	SBAS861

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1 AFE4900 EVM Overview

1.1 Important Disclaimer Information

CAUTION

The AFE4900 EVM is intended for feasibility and evaluation testing only in laboratory and development environments. This product is not for diagnostic use. This product is not for use with a defibrillator.

Only use the AFE4900 EVM under the following conditions:

- The AFE4900 EVM demonstration kit is intended only for electrical evaluation of the features of the AFE4900 devices in a laboratory, simulation, or development environment.
- The AFE4900 EVM demonstration kit is not intended for direct interface with a patient, or patient diagnostics.
- The AFE4900 EVM demonstration kit is intended for development purposes only. It is not intended to be used as all or part of an end-equipment application.
- The AFE4900 EVM demonstration kit should be used only by qualified engineers and technicians who are familiar with the risks associated with handling electrical and mechanical components, systems, and subsystems.
- The user is responsible for the safety of themselves, fellow employees and contractors, and co-workers when using or handling the AFE4900 EVM. Furthermore, the user is fully responsible for the contact interface between the human body and electronics; consequently, the user is responsible for preventing electrical hazards such as shock, electrostatic discharge, and electrical overstress of electric circuit components.

2 Overview

2.1 Introduction

The EVM is intended for evaluating the AFE4900 device. The family of devices consist of a low-noise receive channel and the LED transmit section. The AFE4900 supports four LEDs and three photodiodes. The AFE4900 has a highly-configurable timing controller, enabling complete control of the timing characteristics of the device. The device can be clocked by an external clock or by an internal oscillator. The device communicates to an external host processor using the I²C or serial-peripheral interface (SPI) (default). The purpose of the EVM is to expedite evaluation and system development activities related to AFE4900 devices. The demonstration kit is shown in [Figure 1](#).



Figure 1. AFE4900 Demonstration Kit

Throughout the document, the phrase *demonstration kit* is synonymous with AFE4900 EVM.

2.2 AFE4900 EVM Kit Contents

- AFE4900 EVM Demonstration Kit
- USB-to-micro USB cable
- 10-pin to 10-pin sensor cable

NOTE: The sensor cable is provided as a convenience to enable measurements and cannot be used for any signal-to-noise ratio (SNR) measurements.

- AFE4900 sensor board

2.3 Features Supported in This Version

1. Acquire data at up to 1000 samples per second (sps) in evaluation mode
2. USB-based power and PC application connectivity
3. Access to all AFE4900 registers via an easy-to-use GUI
4. Built-in time domain, histogram, and FFT on the PC application
5. USB-based firmware upgrade option

2.4 AFE4900 EVM Evaluation

Use the following steps as a quick start guide for AFE4900 EVM reflective sensing:

1. Install the PC application software:
 - (a) Before installing the software, make sure the AFE4900 EVM is **not** connected to the PC.
 - (b) Run the executable and follow the prompts to install the software.
 - (c) The default installation directory will be:
 - On a Microsoft® Windows® XP machine – *C:\Program Files\Texas Instruments\AFE4900 EVM*
 - On Windows 7, Windows 8, and Windows 10 machines – *C:\Program Files(x86)\Texas Instruments\AFE4900 EVM*
 - (d) The detailed steps for installing the USB drivers are outlined in [Section 3.3](#).
2. Install the USB driver:
 - (a) The communication interface between the AFE4900 EVM board and PC is through the USB, using the CDC profile. A one-time installation of the USB driver is required for the communication between the AFE4900 EVM and PC application.
 - (b) Plug in the USB-to-mini USB cable to J12 of AFE4900 EVM and the other end to the USB port on the PC.
 - (c) Update the driver software from the *Device Manager*. The USB driver file AFE44xx.inf is located in the following directory:
 - On a Windows XP machine – *C:\Program Files\Texas Instruments\AFE4900 EVM\USB Driver*
 - On Windows 7, Windows 8, and Windows 10 machines – *C:\Program Files(x86)\Texas Instruments\AFE4900 EVM\USB Driver*
 - (d) The detailed steps for installing the USB drivers are outlined in [Section 3.3](#).
3. Update the MSP430 firmware:
 - (a) Run AFE4900 EVM.exe which is found in the chosen installation directory
 - On a Windows XP machine – *C:\Program Files\Texas Instruments\AFE4900 EVM*
 - On Windows 7, Windows 8, and Windows 10 machines – *C:\Program Files(x86)\Texas Instruments\AFE4900 EVM*
 - (b) Click on *File* at the top left of the window followed by *Firmware Upgrade*. A window pops up, click the **Continue** button.
 - (c) Click the **Browse** button and choose the desired hex file. This file should have a .txt extension (example: AFE4900EVM_FW_V1.0_SPI_LDO_EN.txt).
 - On a Windows XP machine – *C:\Program Files\Texas Instruments\AFE4900 EVM\Firmware Updater*
 - On Windows 7, Windows 8, and Windows 10 machines – *C:\Program Files(x86)\Texas Instruments\AFE4900 EVM\Firmware Updater*

4. Test setup for PPG:

- (a) The AFE4900 sensor board has the optical sensor module: OSRAM SFH7072. The OSRAM sensor has two green LEDs, one red LED and one infrared LED. All the LEDs are configured in push-pull (common anode mode). By default, both the green LEDs are connected to LED1 (the TX1 pin of AFE4900), the red LED is connected to LED2 (the TX2 pin of AFE4900), and the infrared LED is connected to LED3 (the TX3 pin of AFE4900). One out of two green LEDs (that is, G1) can be connected to LED4 after some board changes (that is, remove R1 and populate R2). The OSRAM sensor connections are shown in [Figure 2](#).

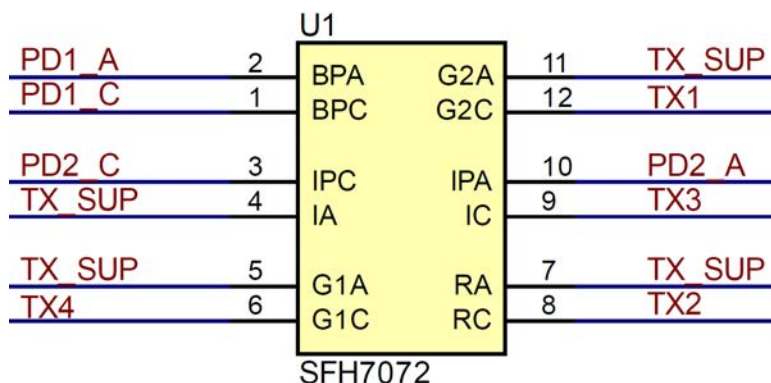


Figure 2. AFE4900 Sensor Board - OSRAM SFH7072 Connections

- (b) Connect the sensor module to the EVM. Make sure that the sensor is connected in the correct orientation – pin 1 should line up with the marking on the cable.
- (c) Place the sensor side of the sensor board on the wrist and tie it snugly. [Figure 3](#) shows the sensor board being held with a velcro strap. Holding the sensor down with a finger is not recommended because a high level of motion noise is likely to occur due to small movements and changes in pressure.

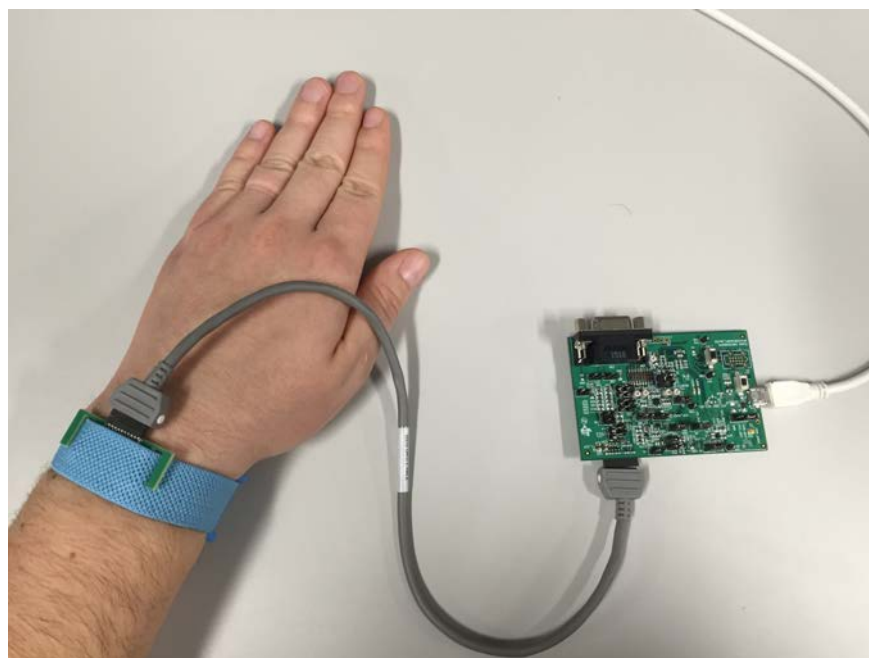


Figure 3. Setup for Obtaining Measurements From the Wrist

5. Test Setup for ECG:

- Make sure jumpers J17, J8, and J13 are populated.
- An ECG simulator can be interfaced through the DB9 connector (that is, J20) with jumpers J19 and J18 being connected. If needed, RLD can be fed to the simulator using J15 after removing J17.
- To connect external electrode, remove the shunt jumpers on J19 and J18. Connect the two electrodes directly to J19 (pin #2) and J18 (pin #2). The RLD electrode, remove J17 and install J15.

NOTE: [Figure 4](#) through [Figure 7](#) help get to [Figure 3](#).

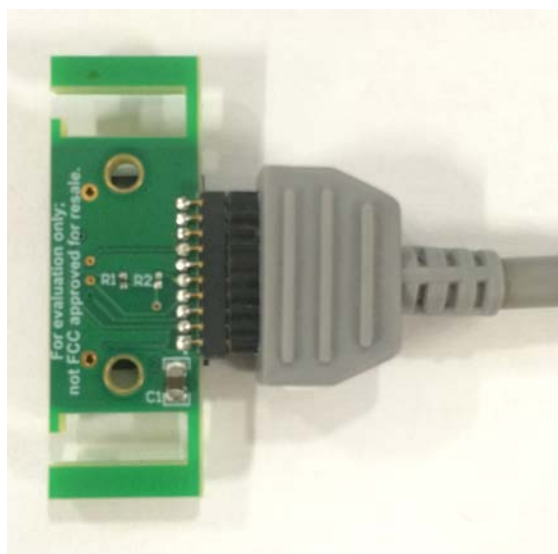


Figure 4. Pin 1 Connections



Figure 5. Connections

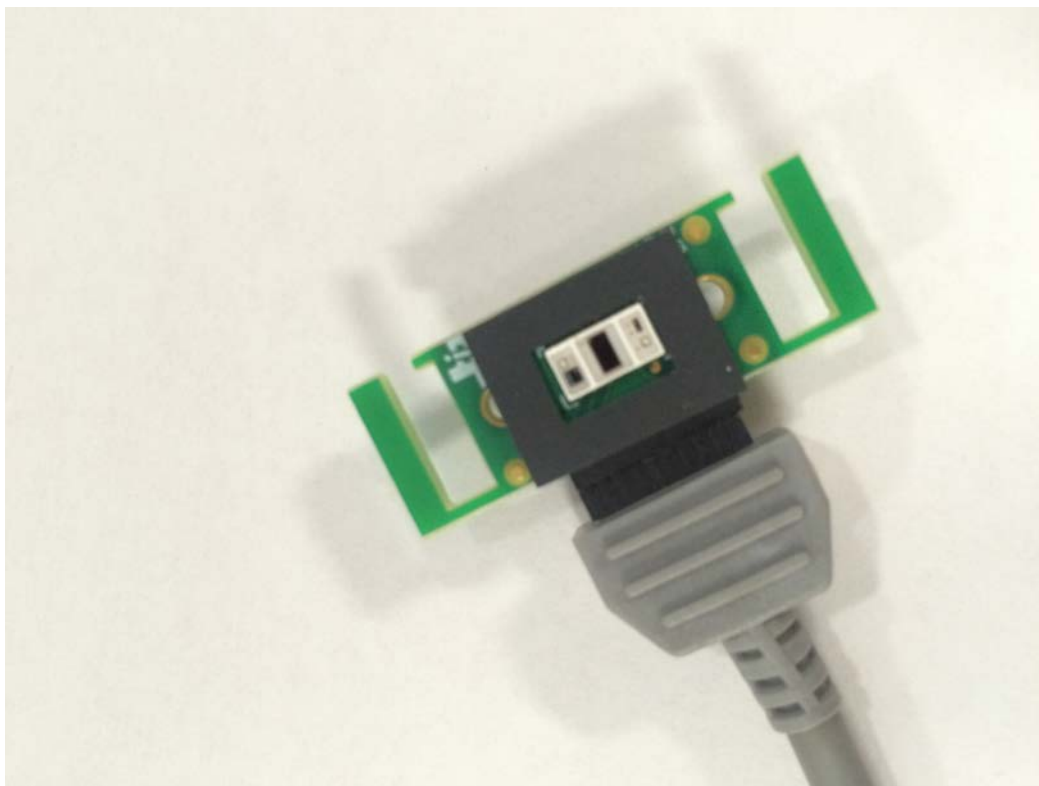


Figure 6. Sensor On With Black Rubber Cover

NOTE: The rubber cover is used to provide ground isolation.



Figure 7. Setting Sensor Board on the Wrist

6. Capturing data:
 - (a) Run the *AFE4900 EVM* found in the installation directory.
 - (b) Click the *ADC Capture & Analysis* tab near the top of the window.
 - (c) Click the **Capture** button to begin capturing data. [Figure 8](#) and [Figure 9](#) show a sample waveform.
 - (d) In case of saturation (that is, output is constant 1.2 V) reduce the LED current from the *TX Stage* subtab.

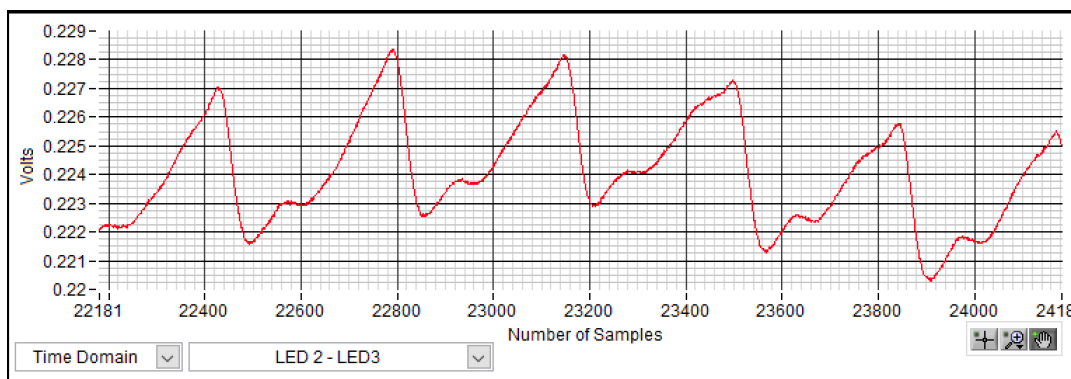


Figure 8. Sample PPG Waveform of Green LED Captured on the Wrist

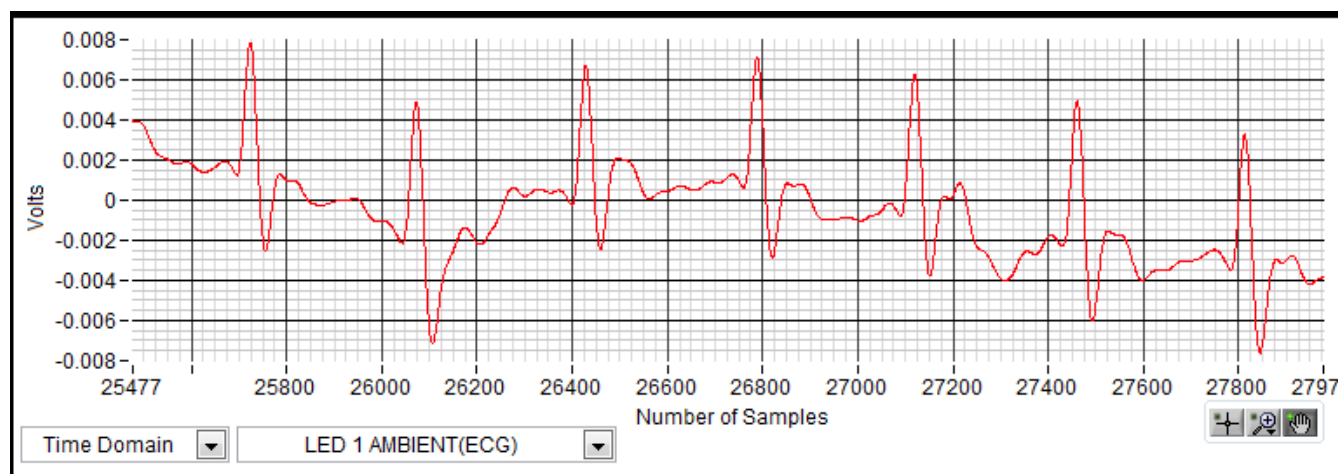


Figure 9. Sample ECG Waveform

NOTE: [Figure 8](#) and [Figure 9](#) are only for illustration. To get waveforms like this, some of the signal chain parameters (LED current, TIA Gain, timing registers) have to be tweaked. Also noise pickup from mains must be avoided.

3 Software Installation

The latest AFE4900 EVM PC application software (GUI) is available at www.ti.com. Download the latest file.

3.1 Minimum Requirements

Before installing the software, verify that your PC meets the minimum requirements outlined in this section.

3.1.1 Required Setup for AFE4900 EVM Demo Software

- IBM PC-compatible computer
- Pentium® III, Celeron® 866 MHz or equivalent processor
- Minimum 256MB of RAM (512MB or greater recommended)
- Hard disk drive with at least 200MB free space
- Microsoft Windows XP SP2, Windows 7, Windows 8, and Windows 10 operating systems (OS) only
- 1280 × 1024 or greater display screen resolution

3.2 Installing the Software (PC Application)

Before installing the software, make sure the AFE4900 EVM is **not** connected to the PC. If using a machine with Windows 7 or Windows 8 OS, TI recommends having administrator rights to avoid problems during installation. Double click the executable to install the software. Unless otherwise specified during the install process, the software installs at the following location:

- On a Windows XP machine – *C:\Program Files\Texas Instruments\AFE4900 EVM*
 - On Windows 7, Windows 8, and Windows 10 machines – *C:\Program Files(x86)\Texas Instruments\AFE4900 EVM*
1. Double click the executable from the *Volume* directory and follow the prompts to continue with the installation process.
 2. Click the **Next>** button to continue.

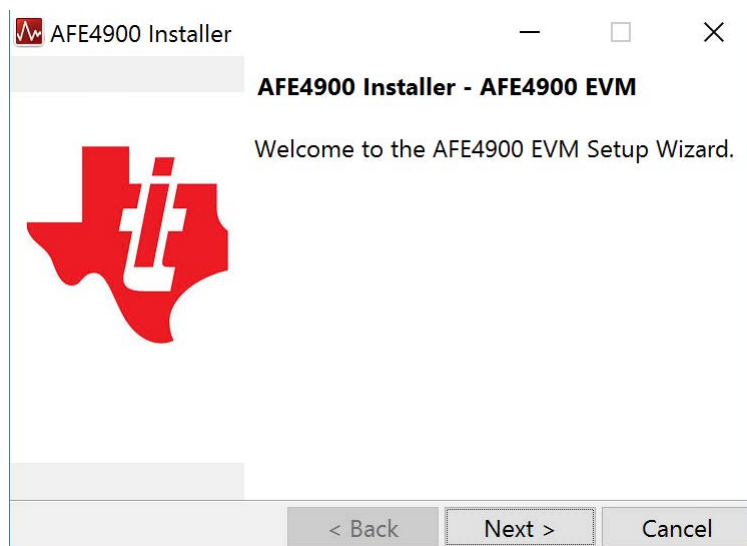
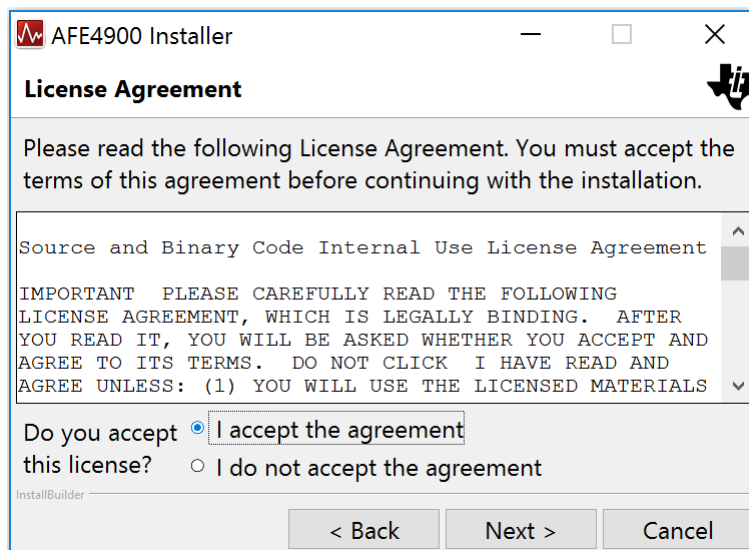
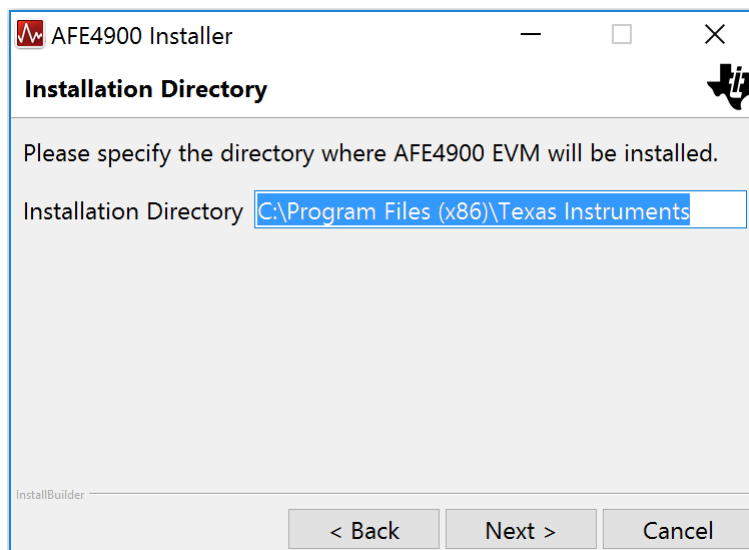


Figure 10. PC Application Installation – Welcome Wizard

3. Accept all the license agreements and click the **Next >** button.

**Figure 11. PC Application Installation – License Agreements**

4. Select the installation directory and click the **Next >** button to begin the installation.

**Figure 12. PC Application Installation – Installation Directory**

5. Click **Next >** button to continue.

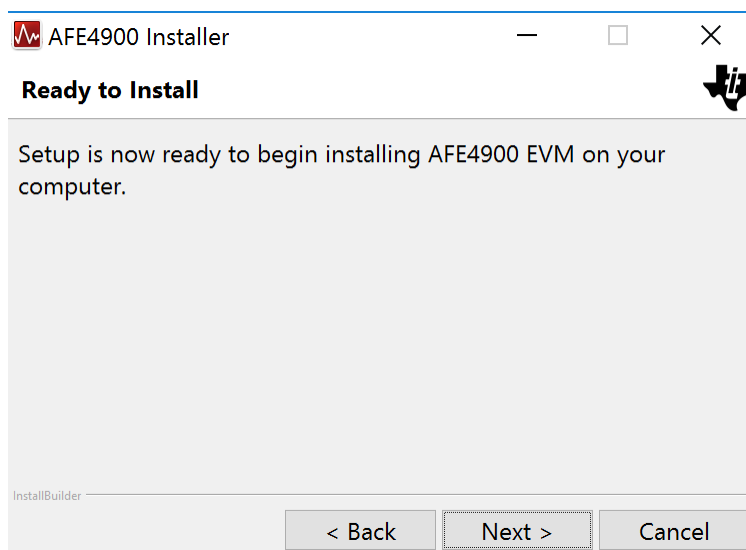


Figure 13. PC Application Installation – Begin Installation

6. The application software is now installed. Choose whether or not to create a Desktop Icon and click the **Finish** button.

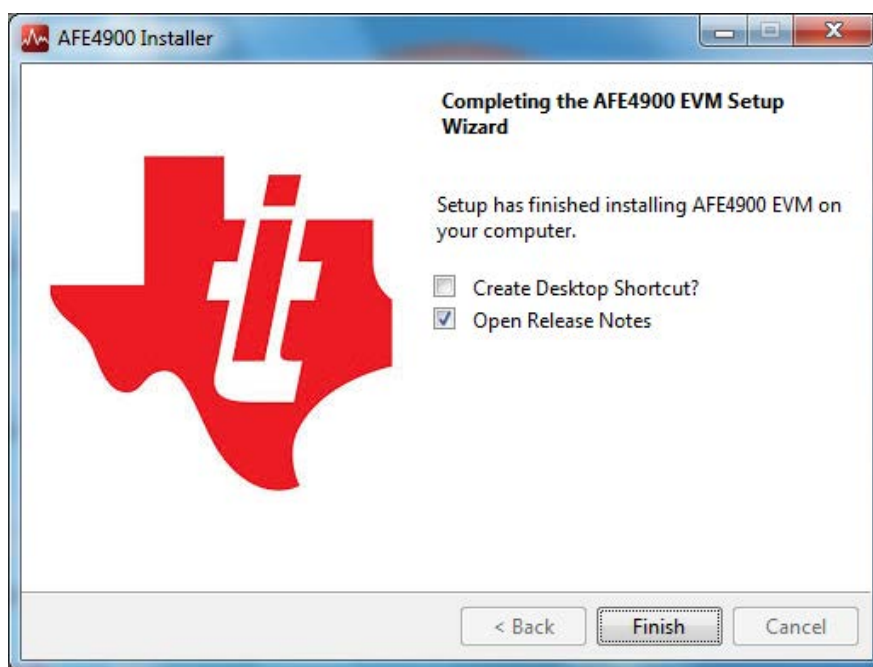


Figure 14. PC Application Installation – Installation Complete

The steps in the following sections ensure proper installation of the PC application:

3.3 Installing the USB Drivers

The communication interface between the AFE4900 EVM board and PC is through the USB, using the CDC profile. A one-time installation of the USB driver is required for the communication between the AFE4900 EVM and PC application.

Using the following steps ensures proper installation of the USB drivers:

1. Plug in the USB-to-mini USB cable to J12 of the AFE4900 EVM and the other end to the USB port on the PC.
2. The Windows XP OS starts up the *New Hardware Wizard* to enable the user to install the USB driver for the new hardware. The Windows 7 or Windows 8 OS attempts to find the driver for the *new hardware found* automatically and if the driver is not found, there is no pop-up message to indicate that the driver installation failed. In the Windows 7, Windows 8, or Windows 10 OS, click on *Device Manager*, right click on the *MSP430-USB* example under *Other devices* and click on *Update Driver Software* as shown in [Figure 15](#). This step is not required for the Windows XP OS.

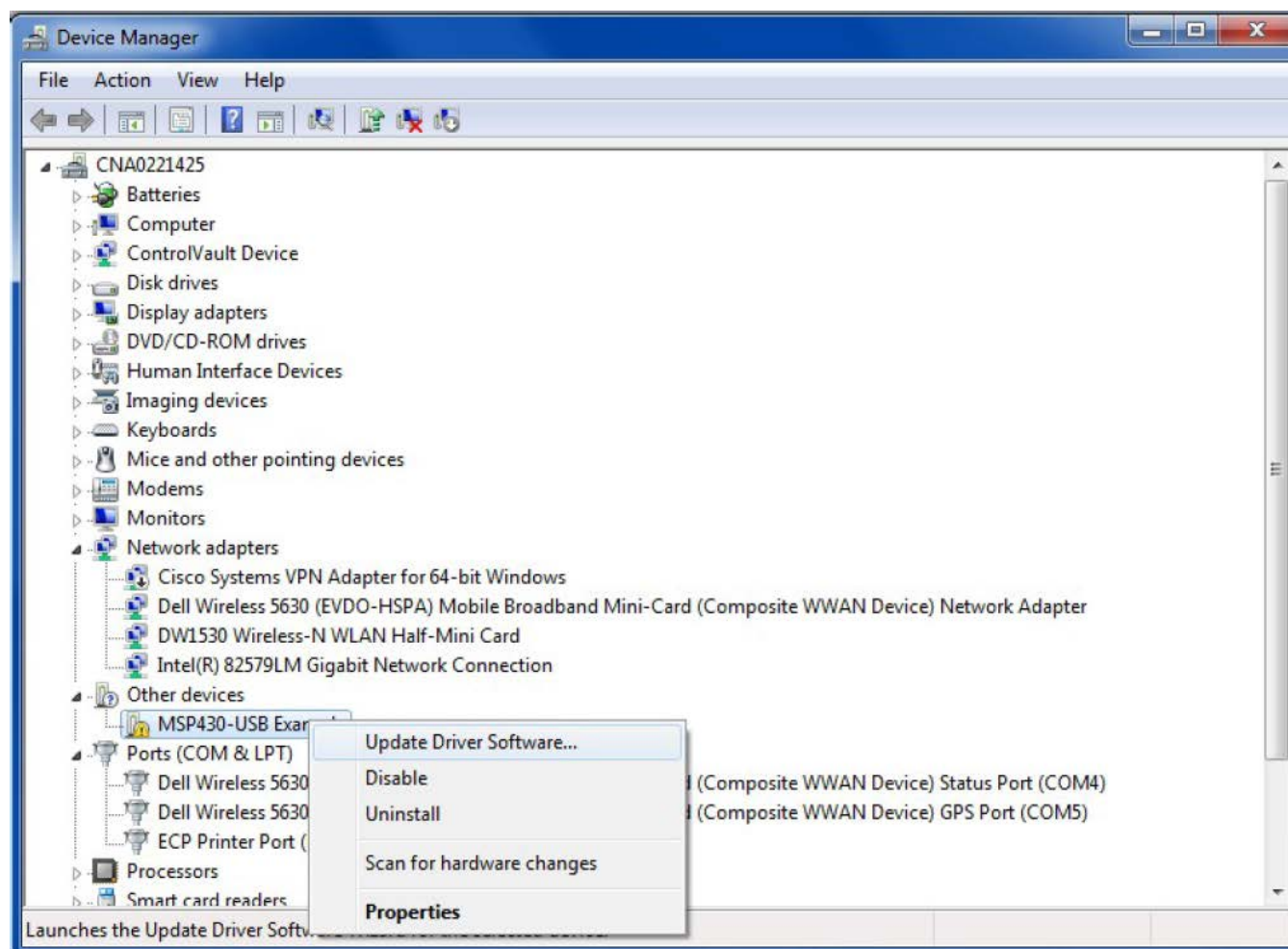


Figure 15. USB Driver Installation - Screen 1 (Windows® 7 or Windows® 8 OS only)

NOTE: Steps for Windows 10 are outlined in [Section 3.3.2](#).

3. Select the *Browse my computer for driver software* option

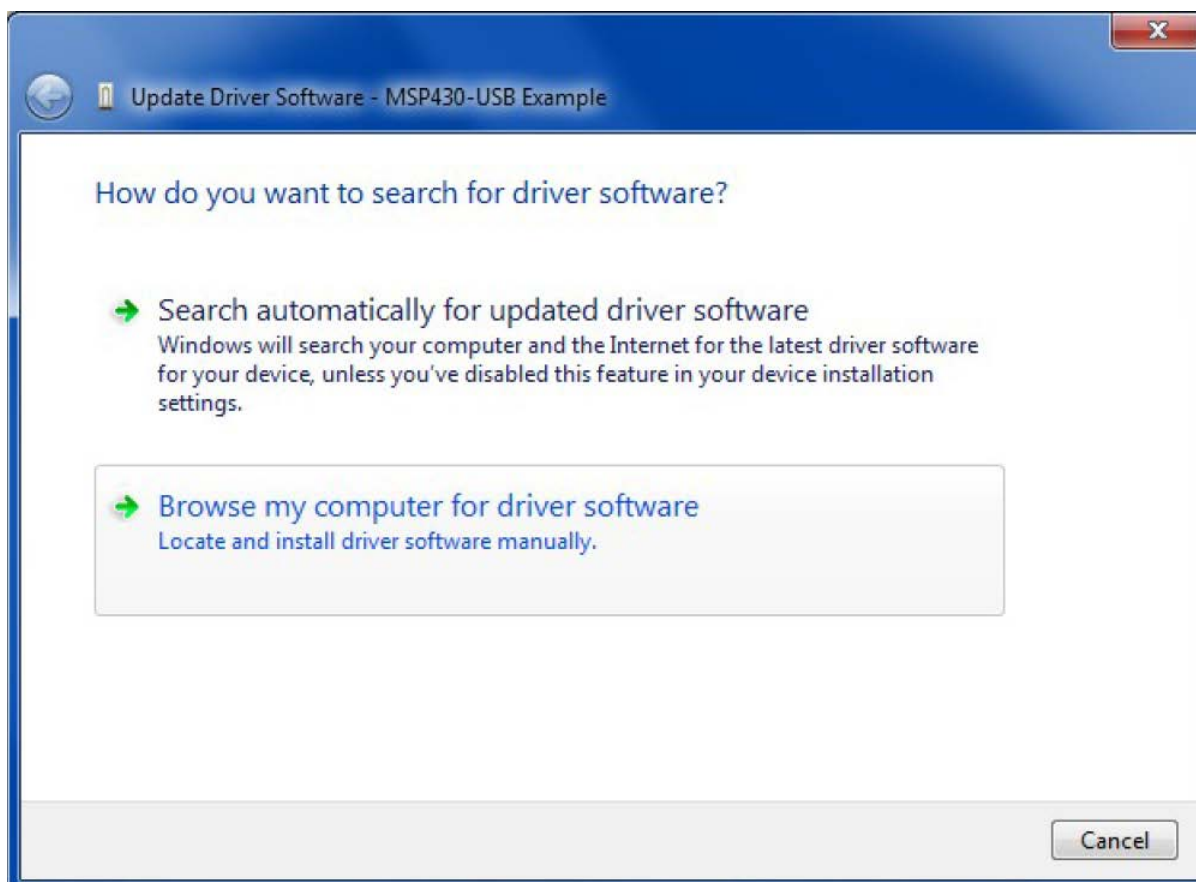


Figure 16. USB Driver Installation - Screen 2

4. As shown in [Figure 17](#), navigate to the directory where the *AFE44xx.inf* file is located by clicking the **Browse** button. The file is located at the following path:
- On a Windows XP machine – *C:\Program Files\Texas Instruments\AFE4900 EVM\USB Driver*
 - On Windows 7, Windows 8, and Windows 10 machines – *C:\Program Files(x86)\Texas Instruments\AFE4900 EVM\USB Driver*

Click the **Next** button to continue. The *Driver* file is copied to the system directory after clicking the **Next** button.

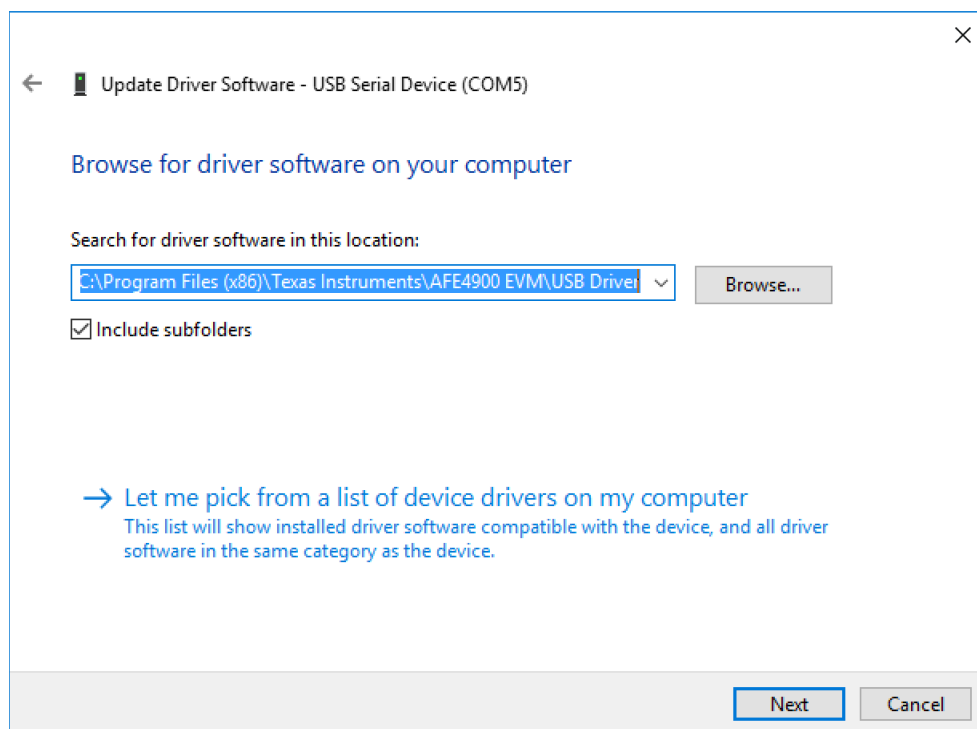


Figure 17. USB Driver Installation - Screen 3

5. There may be a warning that Windows cannot verify the publisher of this driver software, as shown in [Figure 18](#). Choose *Install this driver software anyway* to proceed.

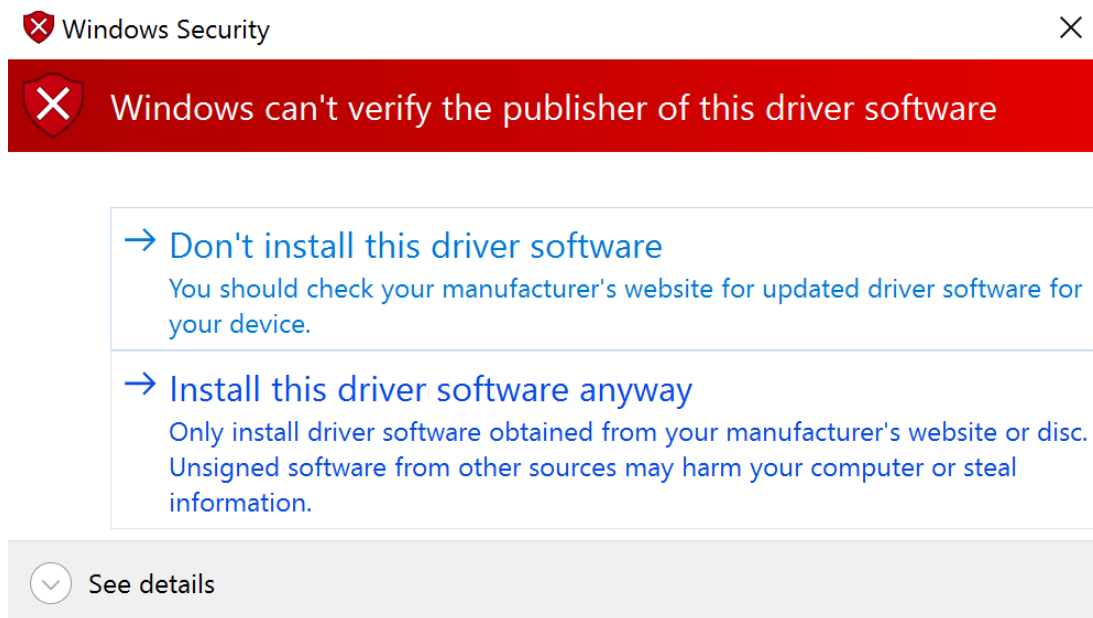


Figure 18. Windows® Publisher Verification Warning

- Click the **Close** button once the driver installation is complete (Figure 19).

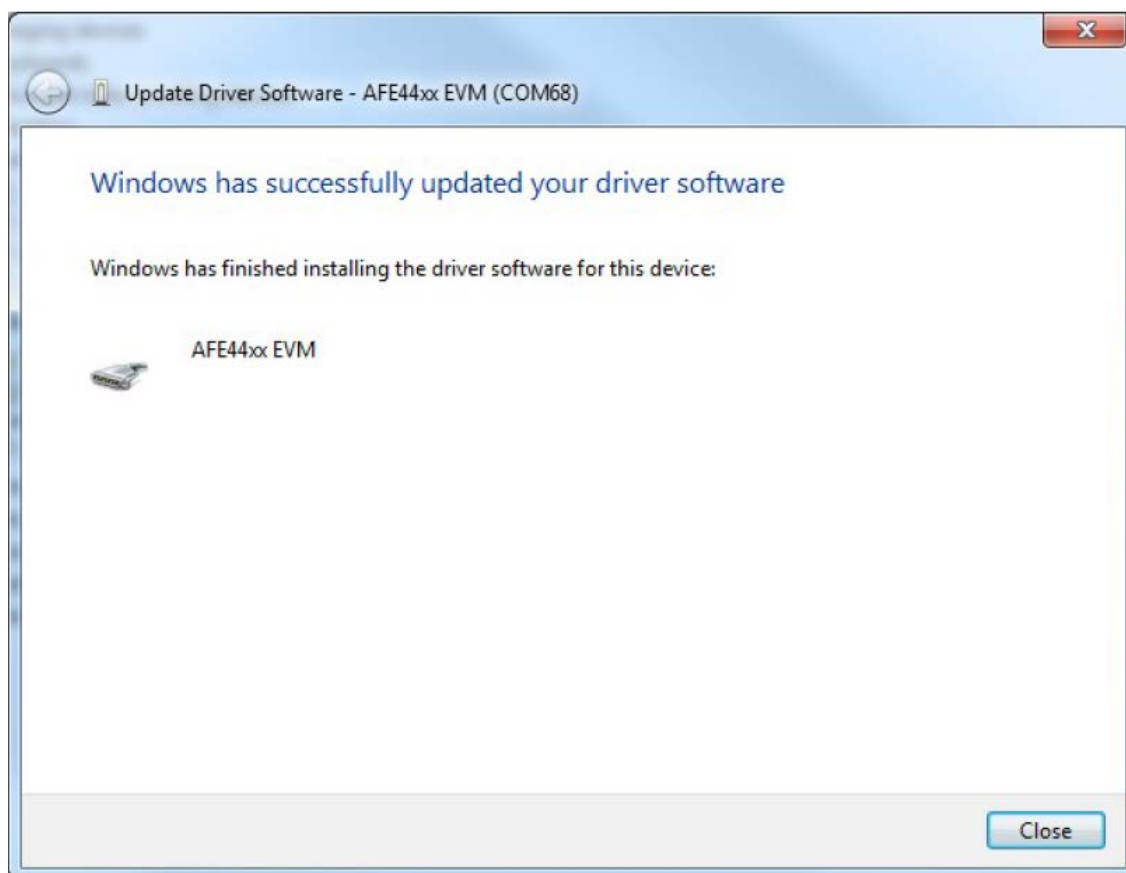


Figure 19. USB Driver Installation - Screen 4

- The AFE4900 EVM is now recognized as Virtual COM Port under the *Device Manager* as shown in Figure 20.

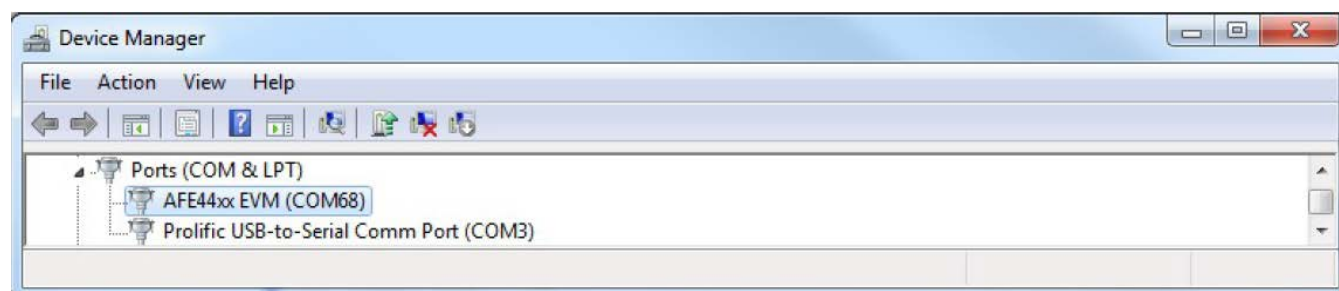


Figure 20. Device Manager Screen

The USB driver installation is now complete and the EVM is ready to use.

3.3.1 Windows® 8 Installing Unsigned Driver

Perform an advanced startup sequence to let Windows 8 install the unsigned driver.

Move the cursor to the top right of the screen, click *Settings*, then *Power*, then *hold down the shift key* and click *Restart* as shown in [Figure 21](#).

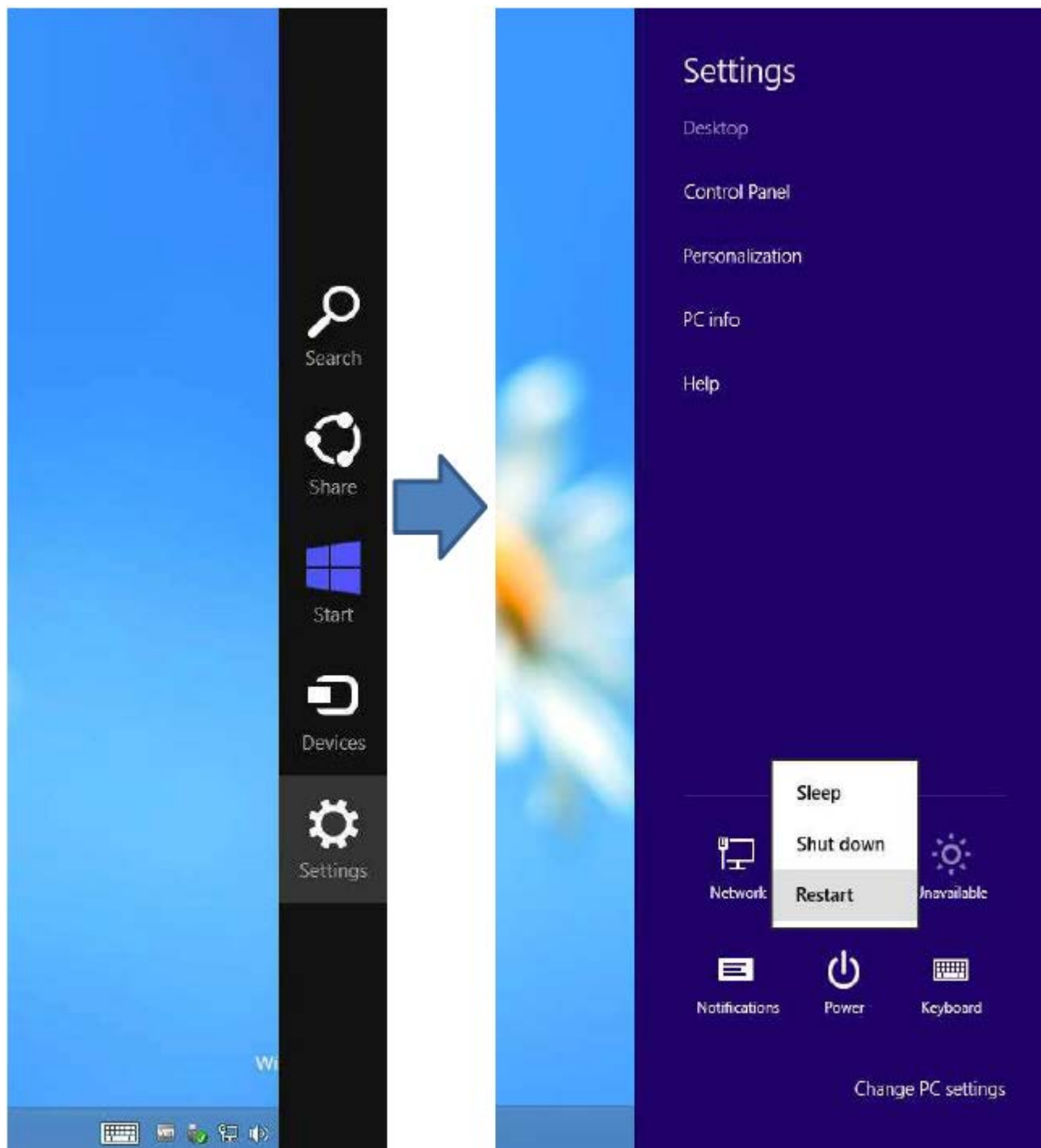


Figure 21. Windows® 8 Installing Unsigned Driver - Screen 1

After a loading screen, three options appear. Choose *Troubleshoot* as shown in [Figure 22](#).

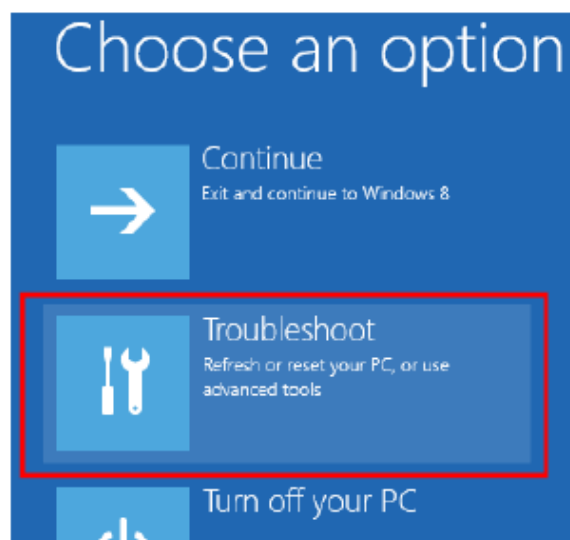


Figure 22. Windows® 8 Installing Unsigned Driver - Screen 2

Choose *Advanced Options* as shown in [Figure 23](#).

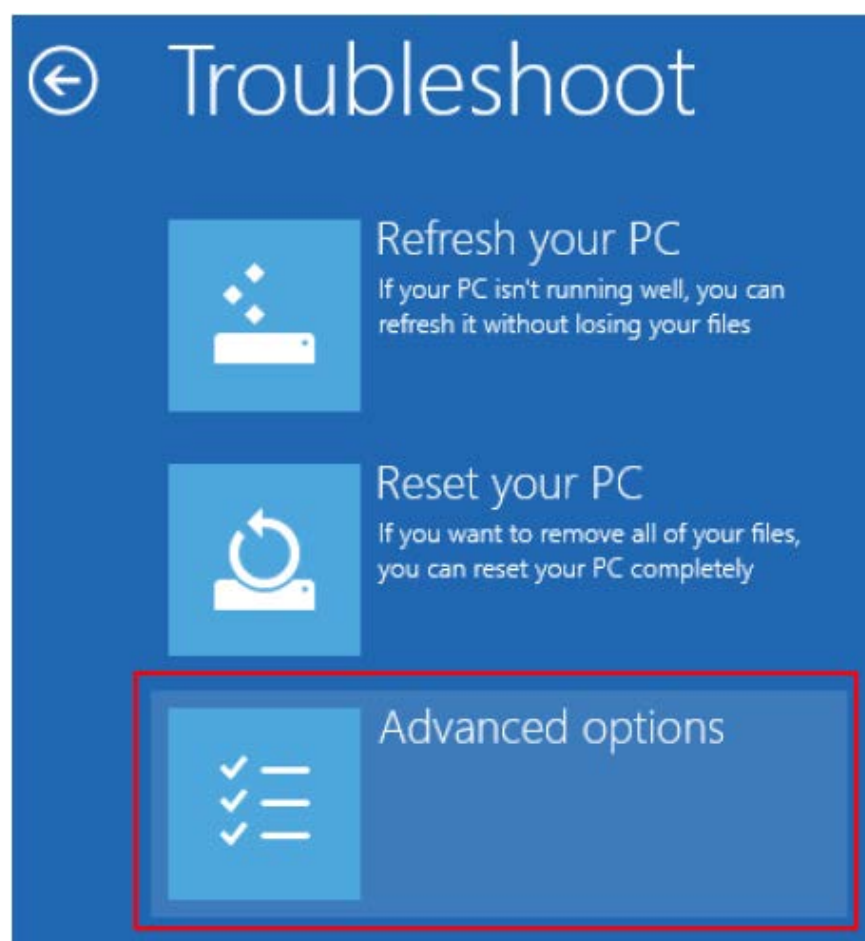


Figure 23. Windows® 8 Installing Unsigned Driver - Screen 3

Choose *Startup Settings* as shown in [Figure 24](#).

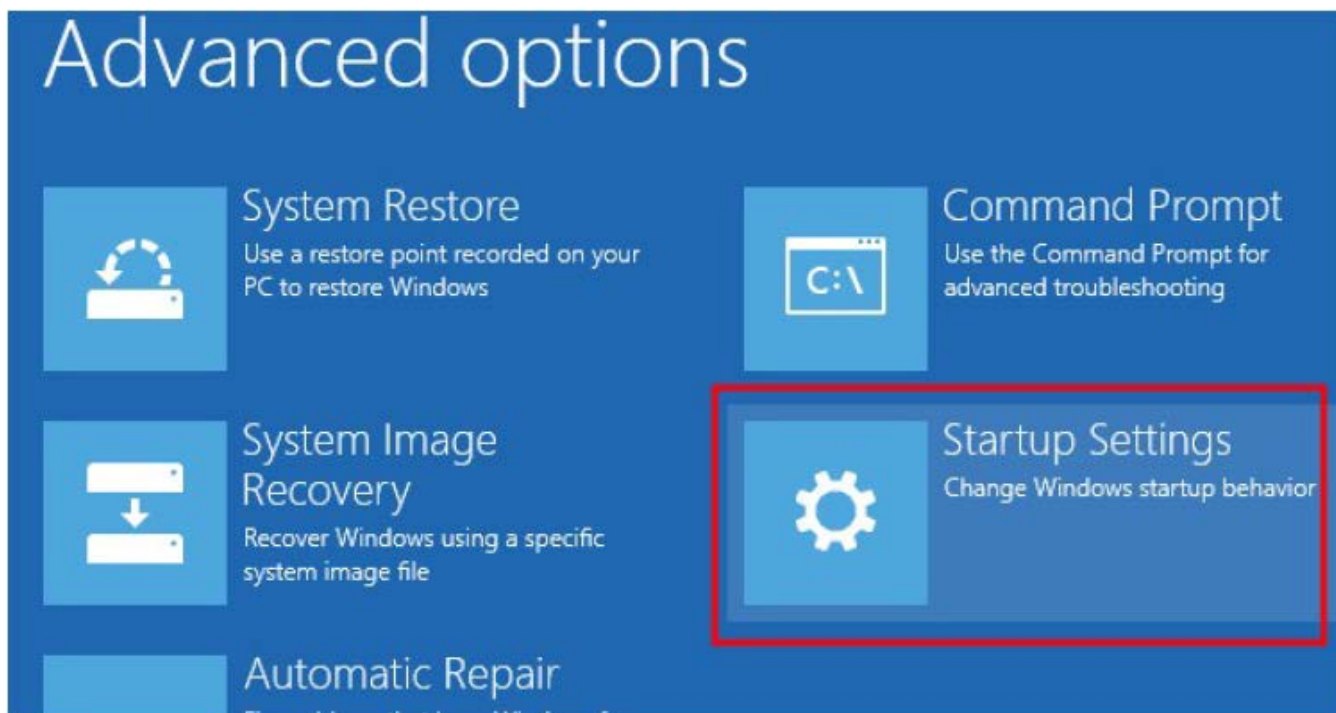


Figure 24. Windows® 8 Installing Unsigned Driver - Screen 4

Next a list of options displays. Click *Restart* at the bottom right as shown in [Figure 25](#).

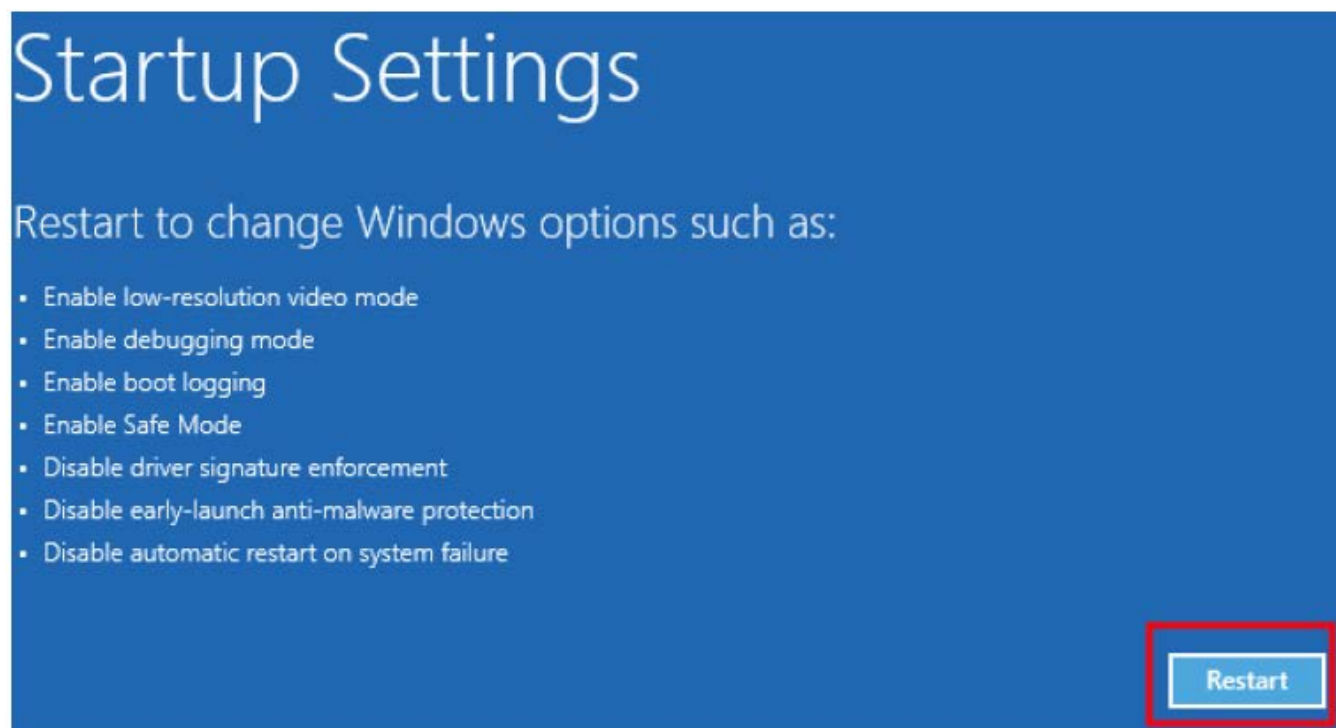


Figure 25. Windows® 8 Installing Unsigned Driver - Screen 5

After the computer restarts, the following screen appears (Figure 26). Press F7 to disable driver signature enforcement.

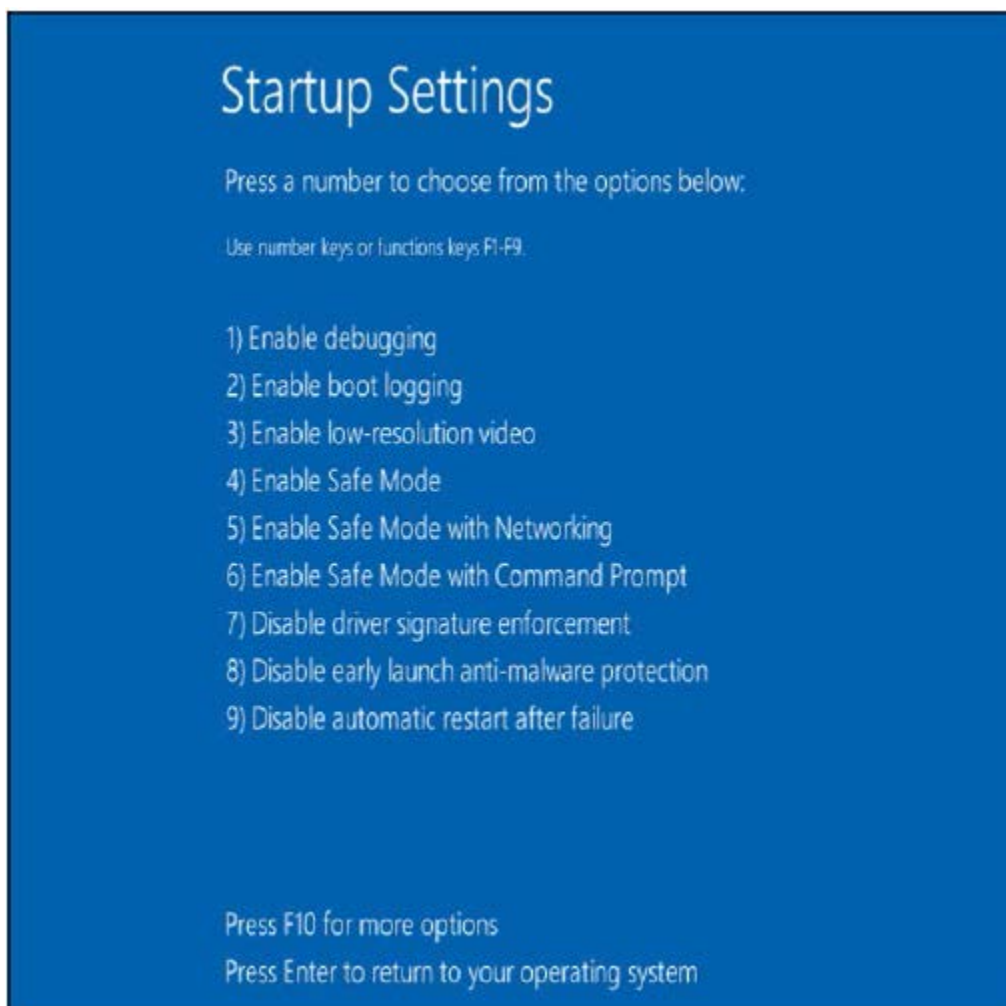


Figure 26. Windows® 8 Installing Unsigned Driver - Screen 6

Now, install the unsigned drivers. A warning may appear as shown in [Figure 27](#); choose *Install this driver software anyway*.

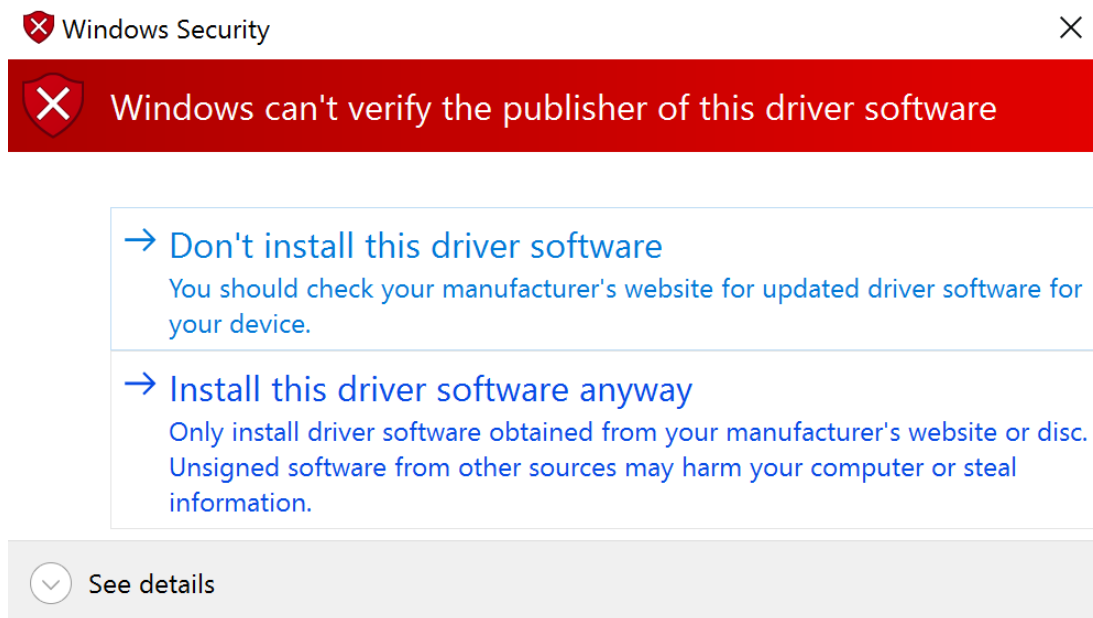


Figure 27. Windows® 8 Installing Unsigned Driver - Screen 7

Restart the computer again to re-enable driver signature enforcement after the installation is complete.

3.3.2 Windows® 10 Installing Unsigned Driver

In addition to the steps mentioned in [Section 3.3.1](#), use the following steps:

1. Plug in the USB-to-mini USB cable to J12 of the AFE4900 EVM and the other end to the USB port of the PC.
2. Open the *Device Manager*, right click on the USB Serial Device (COMx) under the Ports (COM and LPT) and click on *Update Driver Software* as shown in [Figure 28](#).

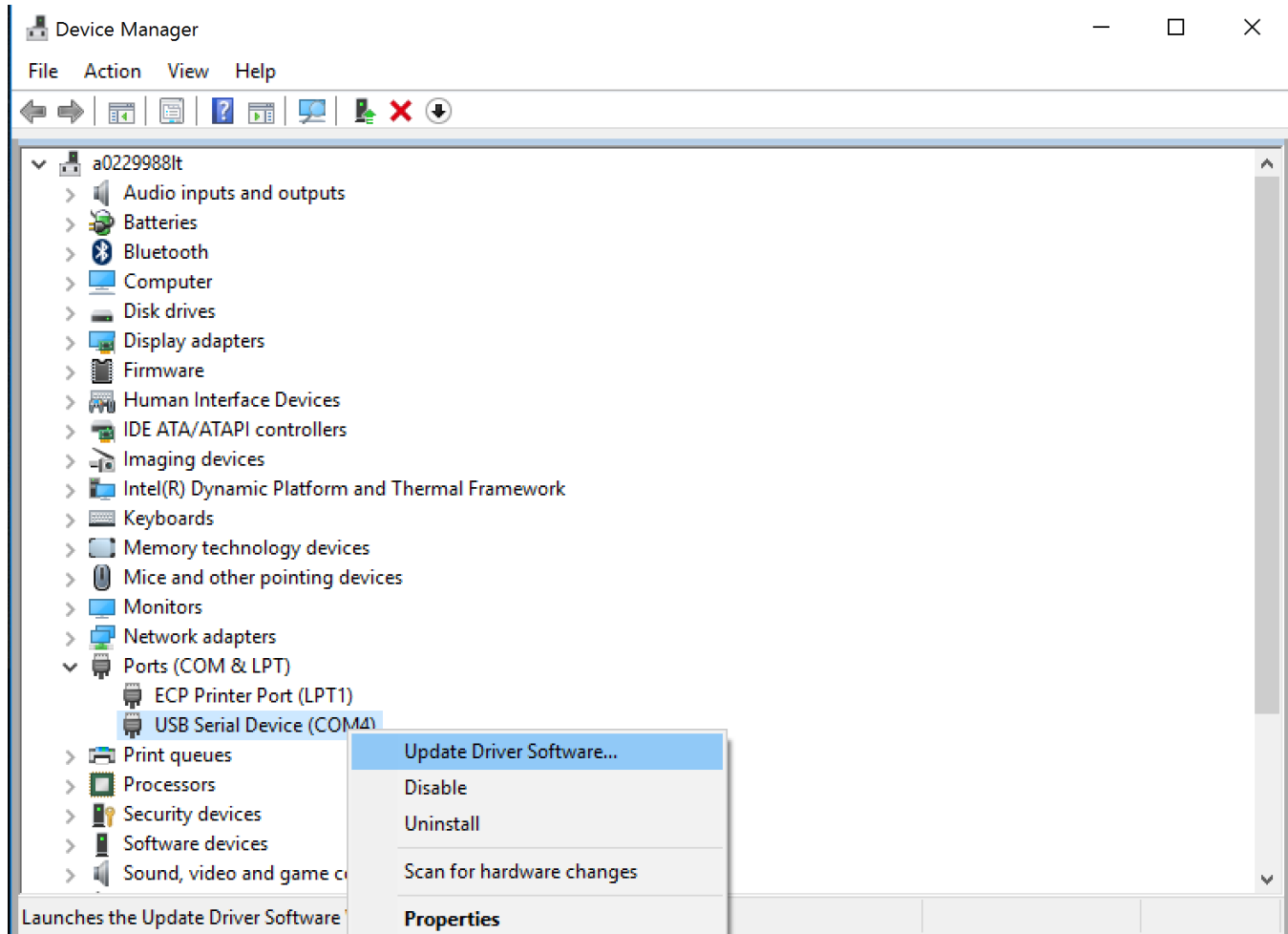


Figure 28. USB Driver Installation – Screen 1 (Windows® 10 OS only)

3. Select the *Browse my computer for driver software* option as shown in [Figure 29](#).



←  Update Driver Software - USB Serial Device (COM4)

How do you want to search for driver software?

→ Search automatically for updated driver software
Windows will search your computer and the Internet for the latest driver software for your device, unless you've disabled this feature in your device installation settings.

→ Browse my computer for driver software
Locate and install driver software manually.

Cancel

Figure 29. USB Driver Installation – Screen 2

4. Select *Let me pick from a list of drivers on my computer* option as shown in [Figure 30](#).

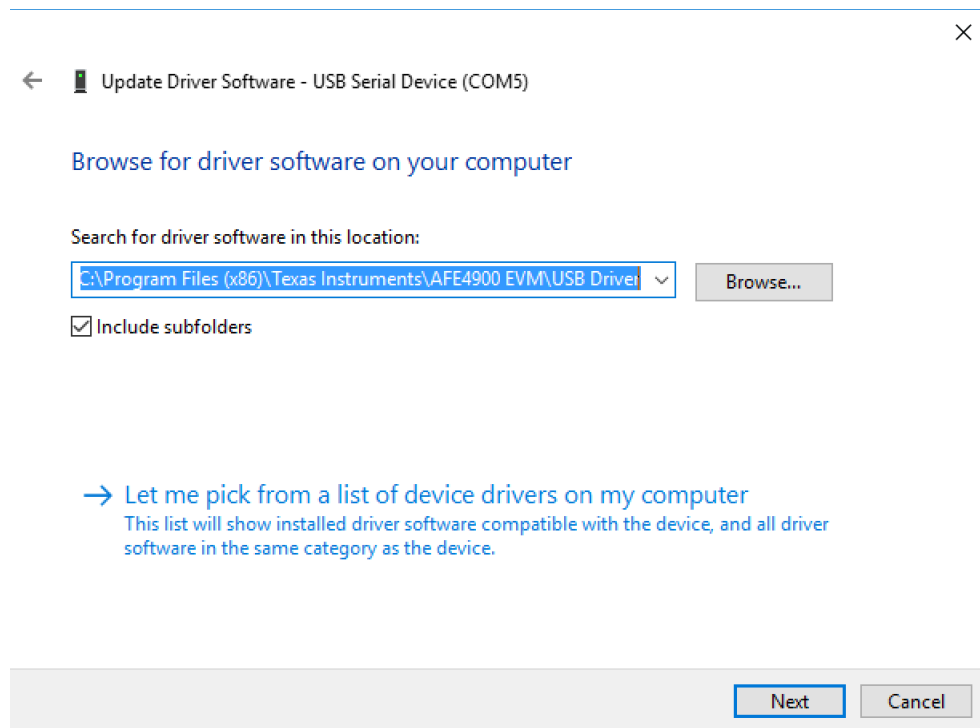


Figure 30. USB Driver Installation – Screen 3

5. Select the *Have Disk...* option.



←  Update Driver Software - USB Serial Device (COM4)

Select the device driver you want to install for this hardware.



Select the manufacturer and model of your hardware device and then click Next. If you have a disk that contains the driver you want to install, click Have Disk.

☒ Show compatible hardware

Model

 USB Serial Device

 USB Serial Device



This driver is digitally signed.

[Tell me why driver signing is important](#)

Have Disk...

Next

Cancel

Figure 31. USB Driver Installation – Screen 4

6. As shown in [Figure 32](#), navigate to the directory where the AFE44xx.inf file is located by clicking the **Browse** button. The file is located at the following path *C:\Program Files(x86)\Texas Instruments\AFE4900EVM GUI\USB Driver*
Click the **OK** button to continue.

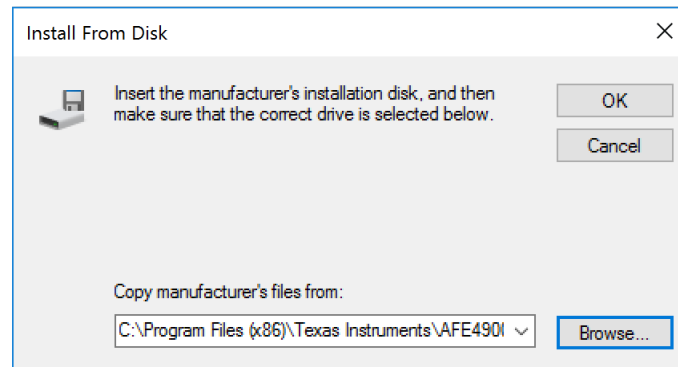


Figure 32. USB Driver Installation – Screen 5

7. Select AFE44xx and click **Next** to continue as shown in [Figure 33](#).



←  Update Driver Software - USB Serial Device (COM3)

Select the device driver you want to install for this hardware.



Select the manufacturer and model of your hardware device and then click Next. If you have a disk that contains the driver you want to install, click Have Disk.

☒ Show compatible hardware

Model

AFE44xx EVM



This driver is not digitally signed!

[Tell me why driver signing is important](#)

Have Disk...

Next

Cancel

Figure 33. USB Driver Installation – Screen 6

8. There may be a warning that Windows cannot verify the publisher of this driver software, as shown in [Figure 34](#). Choose to install the driver software anyway to proceed.

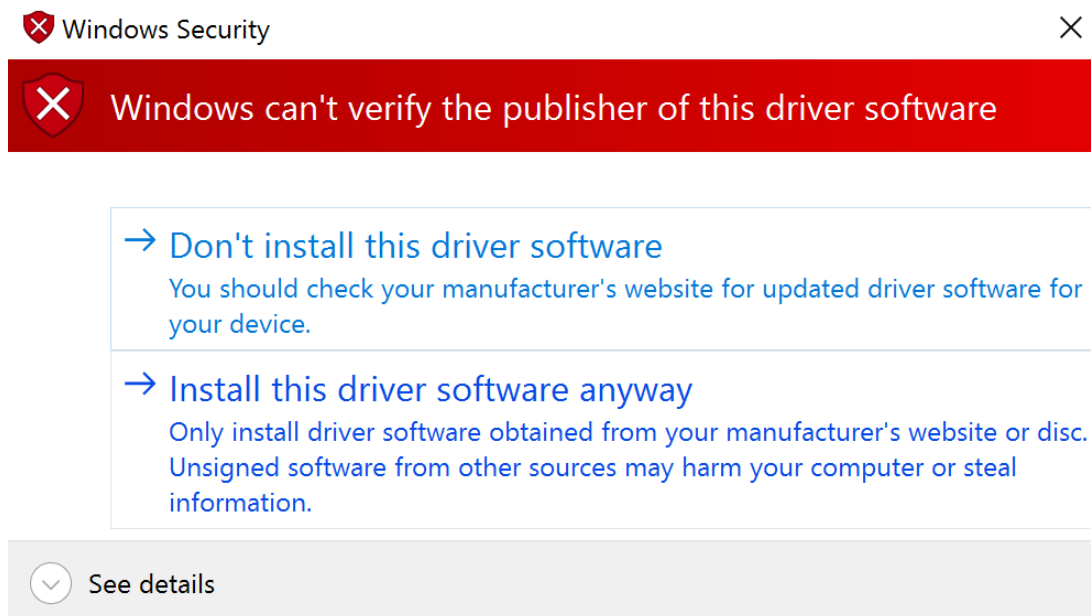


Figure 34. USB Driver Installation – Screen 7

9. Click the **Close** button once the driver installation is complete.

10. The AFE4900 EVM is now recognized as Virtual COM Port under the *Device Manager* as shown in [Figure 35](#).

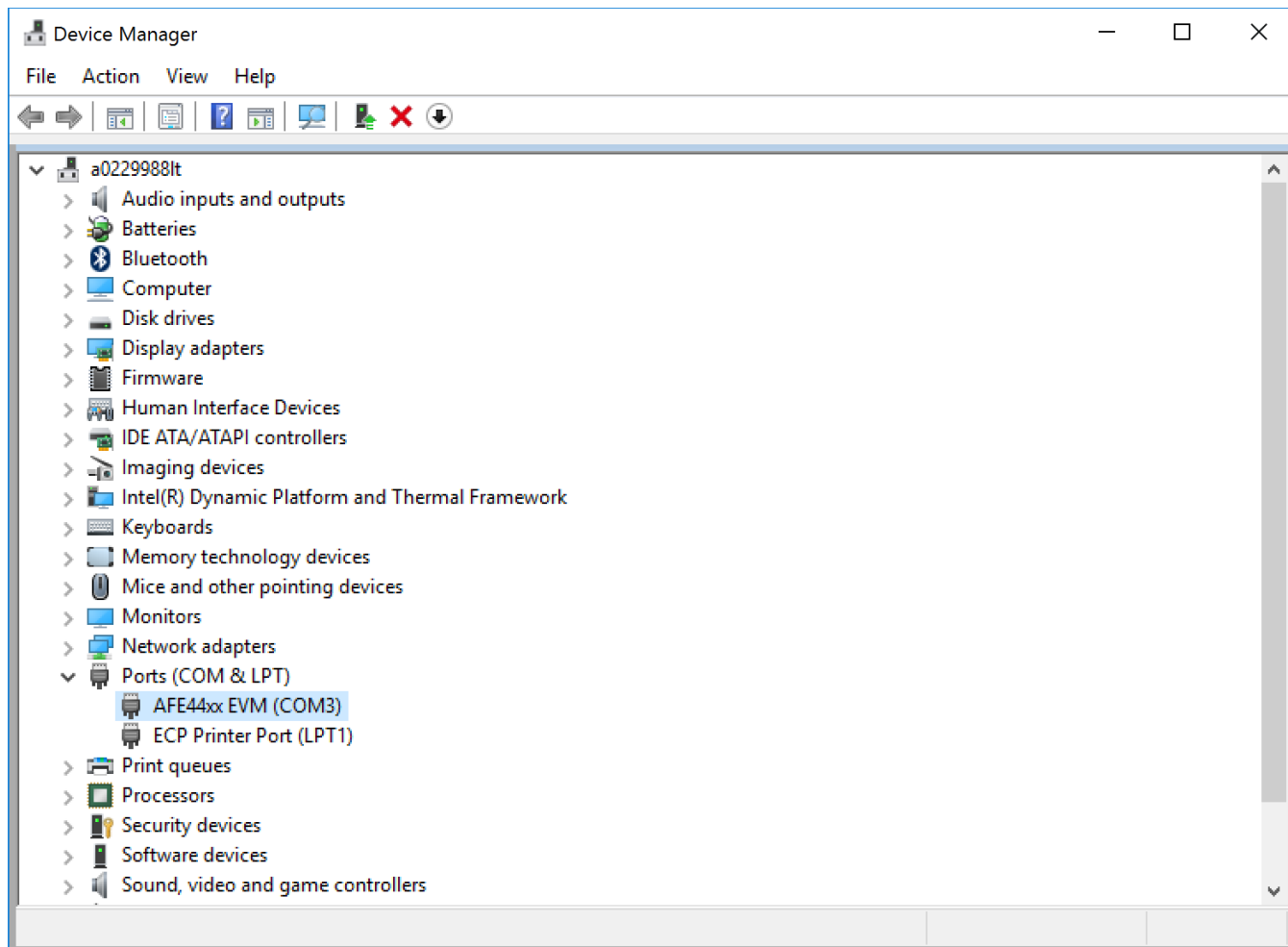


Figure 35. USB Driver Installation – Screen 8

4 Running the Software

Run the AFE4900 EVM software by selecting the *AFE4900 EVM.exe* which is found in the chosen installation directory (by default, this is *C:\Program Files(x86)\Texas Instruments\AFE4900 EVM*). Unless the hardware has been disconnected, observe messages that confirm the connection has been established and the program waits in idle mode for user input.

If the connection to the AFE4900 EVM board is not established, the program prompts to continue to run the GUI in *Simulation* mode, or to *Stop & Close* the GUI and check if the AFE4900 EVM is connected to the PC.

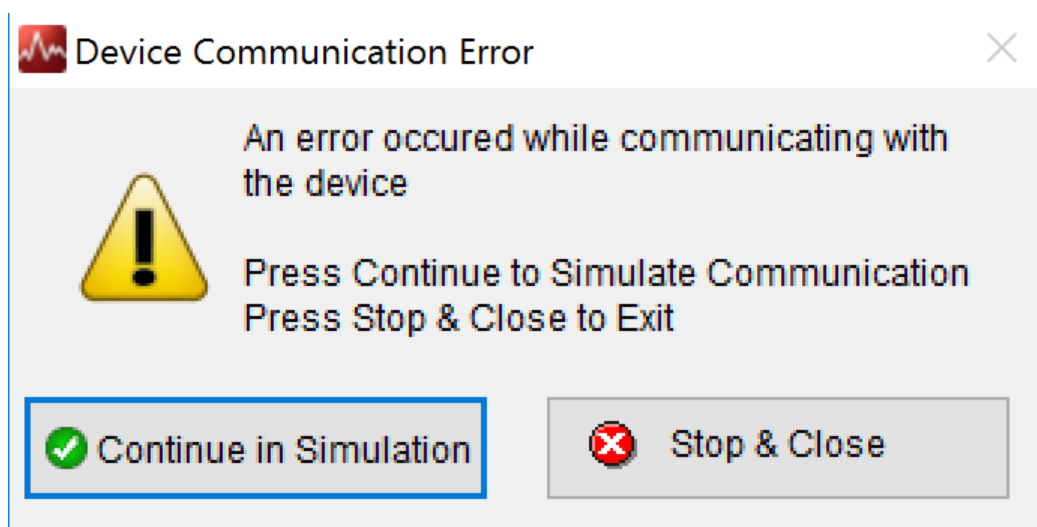


Figure 36. AFE4900 EVM not Connected Error Message

4.1 Overview of the Features

This section provides a quick overview of the various features and functions of the AFE4900 EVM software GUI. The GUI allows the user to easily configure the various functions of the AFE, such as receiver gain, bandwidth settings, LED current settings, and timing or clocking control settings.

Operations in the GUI should only be performed after the status bar (located at the bottom of the GUI) displays **Ready For New Command**.

The main tabs consist of:

- *Product Safety Warnings* – Product Safety Warnings, Restrictions and Disclaimers (see [Figure 37](#)).
- *Device Configuration* – Configures all the AFE4900 user registers in a series of related subtabs.
 - *Global Settings*
 - *Tx Stage*
 - *Rx Stage*
 - *Timing Controls*
 - *ECG Signal*
 - *Low Level Configuration*
- *ADC Capture & Analysis* – For viewing and analyzing the raw data.
- *Save* – For writing data samples and analysis results to a file.

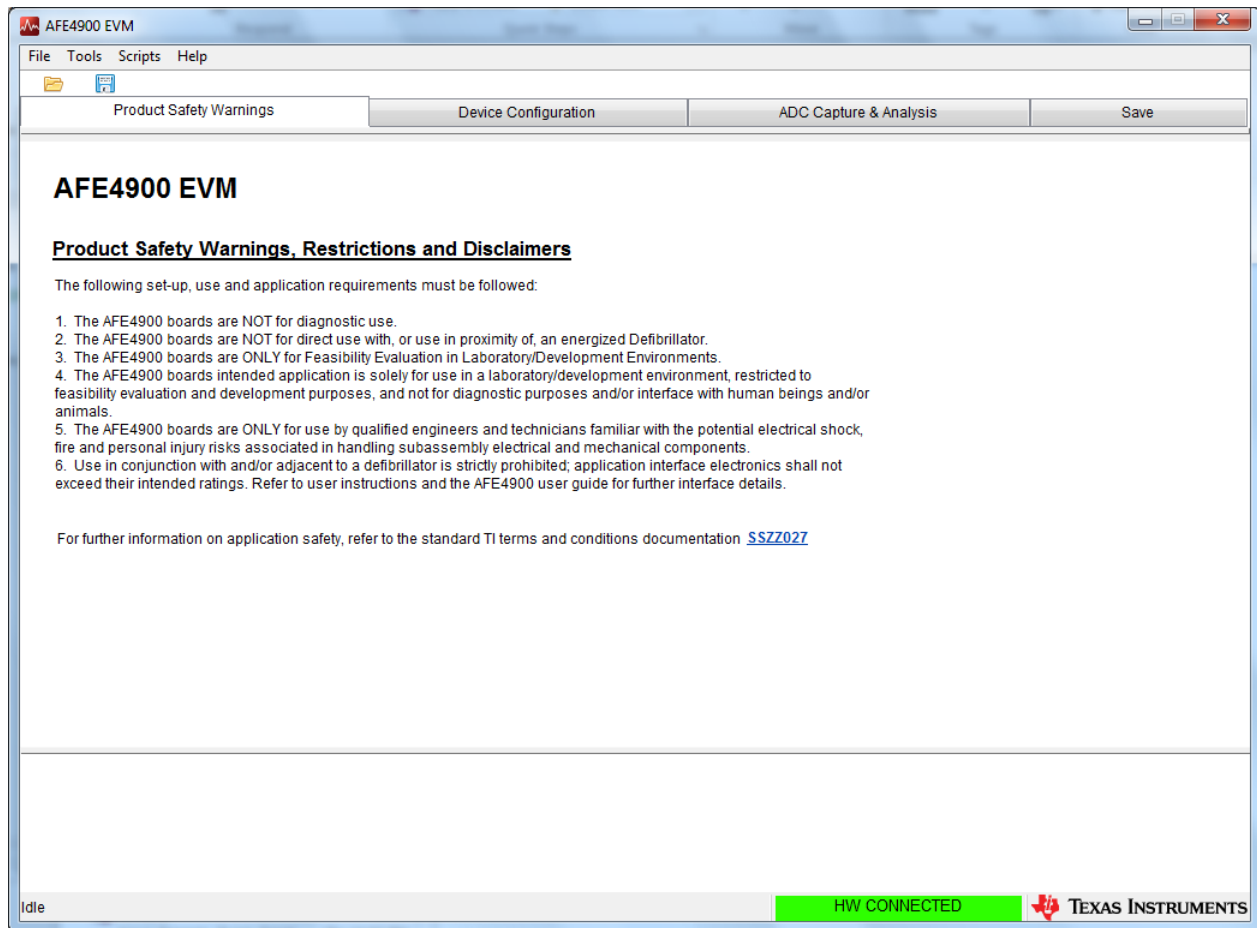


Figure 37. Product Safety Warnings, Restrictions, and Disclaimers

4.1.1 Device Configuration Tab

The *Device Configuration* tab allows configuration of the various registers of the AFE4900 device. This tab contains five subtabs: *Global Settings*, *Tx Stage*, *Rx Stage*, *Timing Controls*, *ECG Signal*, and *Low Level Configuration*.

4.1.1.1 Global Settings Subtab

The Global Settings subtab for the AFE4900 device shown in [Figure 38](#) has the following features:

1. View the Device ID and Firmware Version
2. **Device Reset** button that resets the device. (Please note that after a device reset is issued, the AFE4900 device registers must be programmed correctly for the PC application GUI to function properly. See **Reset to EVM Defaults** on how to issue a device reset and also program the AFE4900 registers to the EVM default register settings.)
3. **Reset to EVM Defaults** button that resets the device and sets up the board to the EVM default register settings.
4. Enables the user to set or reset:
 - (a) Power down settings
 - Power down AFE
 - Power down RX
 - (b) Dynamic Power down settings
 - Controls for dynamic power down

- Enable input short
 - Short the PD1, PD2 to internal common mode voltage when PD1, PD2 is disconnected from TIA
- (c) Clock configuration settings
- Input clock frequency: Multiple options available in External CLK mode.
 - Clock mode
 - Enable CLK output
 - Clock divider ratio for external CLK, pulse repetition frequency (PRF), and CLK output
- (d) FIFO mode settings
- FIFO_EN - Enable FIFO. Ensure FIFO_RDY is selected for INT_MUX1.
 - FIFO_PARTITION setting - selection of phases of data (that is, FIFO_NPHASE)
 - REG_FIFO_PERIOD setting - set the number of periods over which the FIFO gets filled. It should be set such that $((\text{REG_FIFO_PERIOD} + 1) \times \text{FIFO_NPHASE}) \leq 128$
 - FIFO_EARLY setting - advance the FIFO_RDY by FIFO_EARLY number of periods
 - MASK_FIFO_RDY - Masks the FIFO_RDY interrupt generation when operating in watermark FIFO mode.
 - WM_MODE: Puts the FIFO in watermark mode. In this mode, the FIFO_PERIOD should be $(n \times \text{FIFO_NPHASE}) - 1$. Maximum value it can take is 127. For example:
 - (i) If FIFO_PARTITION = 0 (that is, FIFO_NPHASE = 4) FIFO_PERIOD = $(n \times 4) - 1$, where $n = 1, 2, 3..32$.
 - (ii) If FIFO_PARTITION = 4 (that is, FIFO_NPHASE = 2) FIFO_PERIOD = $(n \times 2) - 1$, where $n = 1, 2, 3..64$.
- (e) Decimation mode
- DEC_EN - enable decimation mode
 - FIFO_EN_DEC: Enables the FIFO in decimation mode.
 - DEC_FACTOR - select the number of PRF cycles over which the averaging is done
- (f) Interrupts selection
- EN_PROG_OUT1 - Enable interrupt on PROG_OUT1 pin
 - Multiplexed interrupts using the controls INT_MUX1, INT_MUX2, and INT_MUX3
 - EN_SDOUT_INT - Enable interrupt on SDOUT pin in I2C mode.
- (g) Threshold detect settings
- THR_DET_EN - Enable threshold detect mode. Ensure THR_DET_RDY is selected for INT_MUX1
 - THR_DET_LOW_CODE and THR_DET_HIGH_CODE - The lower and upper limits set as 24-bit codes
 - THR_DET_PHASE_SEL - select the phase of the output data compared against the low and high thresholds
 - CHANGE_SEL_LOGIC - selects whether THR_DET_RDY interrupt is generated by the output code going in-range or out-of-range of the range.
- (h) Ultra Low Power (ULP) settings
- ENABLE_ULP - Enables ULP mode
 - Controls for different blocks (BG, VCM, ALDO, DLDO) during deep sleep.
 - SHORT_ALDO_TO_DLDO_IN_DEEP_SLEEP
- (i) Miscellaneous settings
- SDOUT_TRISTATE - SDOUT buffer is forced to tristate always.
 - PD_DISCONNECT - Disconnects the PD from the TIA.
 - LDO_BYPASS - Enables the LDO bypass mode by forcing CONTROL1 pin high.

- Enables the user to control the clock divider ratio settings. When the user enters an input clock, the GUI will automatically choose a divide by value so that the output clock is within 32–4000 kHz

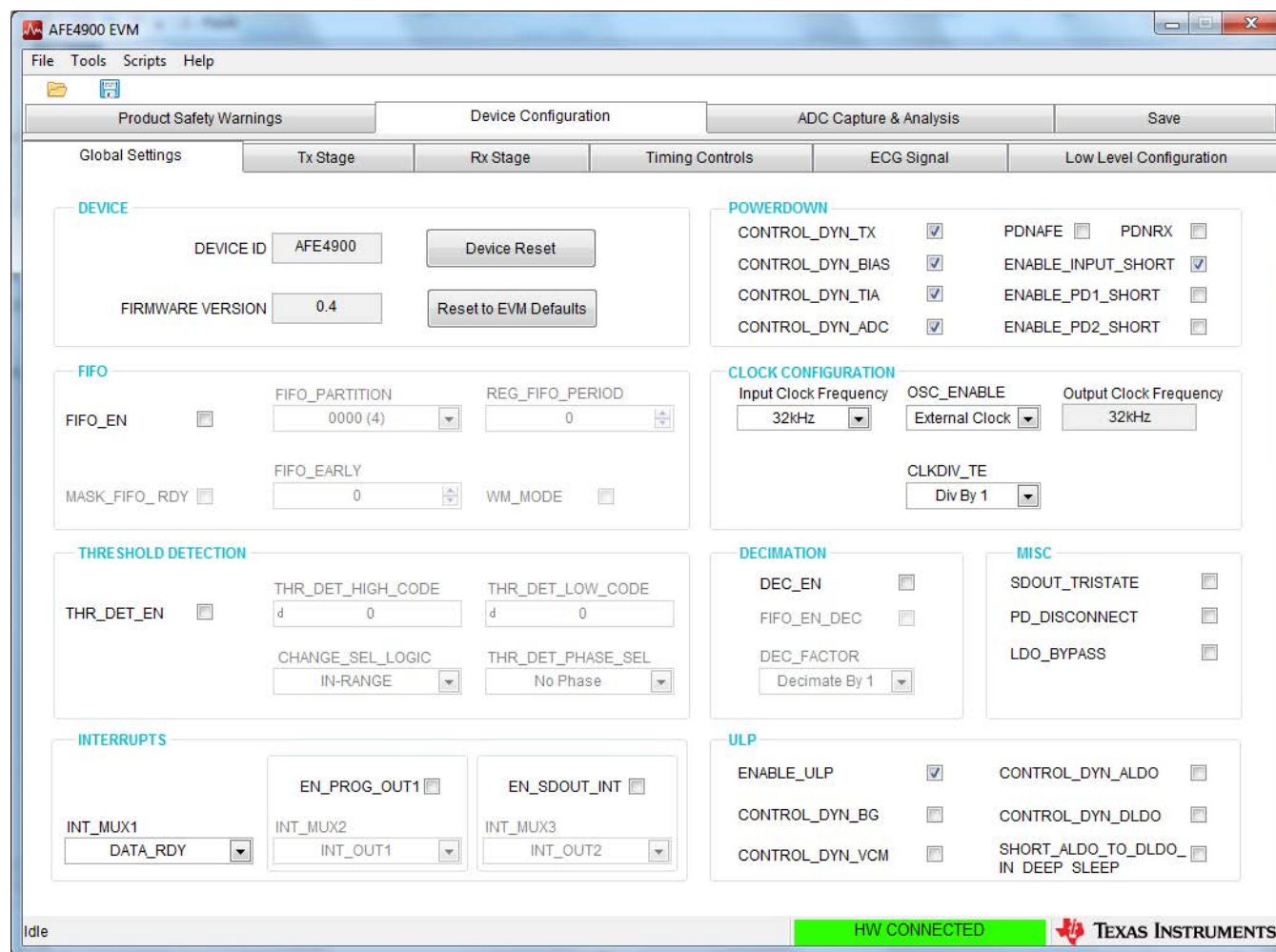


Figure 38. AFE4900: Device Configuration: Global Settings

4.1.1.2 Tx Stage Subtab

Figure 39 shows the Tx Stage subtab under the Device Configuration tab consisting of the settings to:

1. Set LED1, LED2, LED3, and LED4 currents in mA or current setting in codes (8-bit).
2. Enable or disable LED1, LED2, LED3, and LED4 in both the LED drivers.
3. Set the Maximum LED current range.

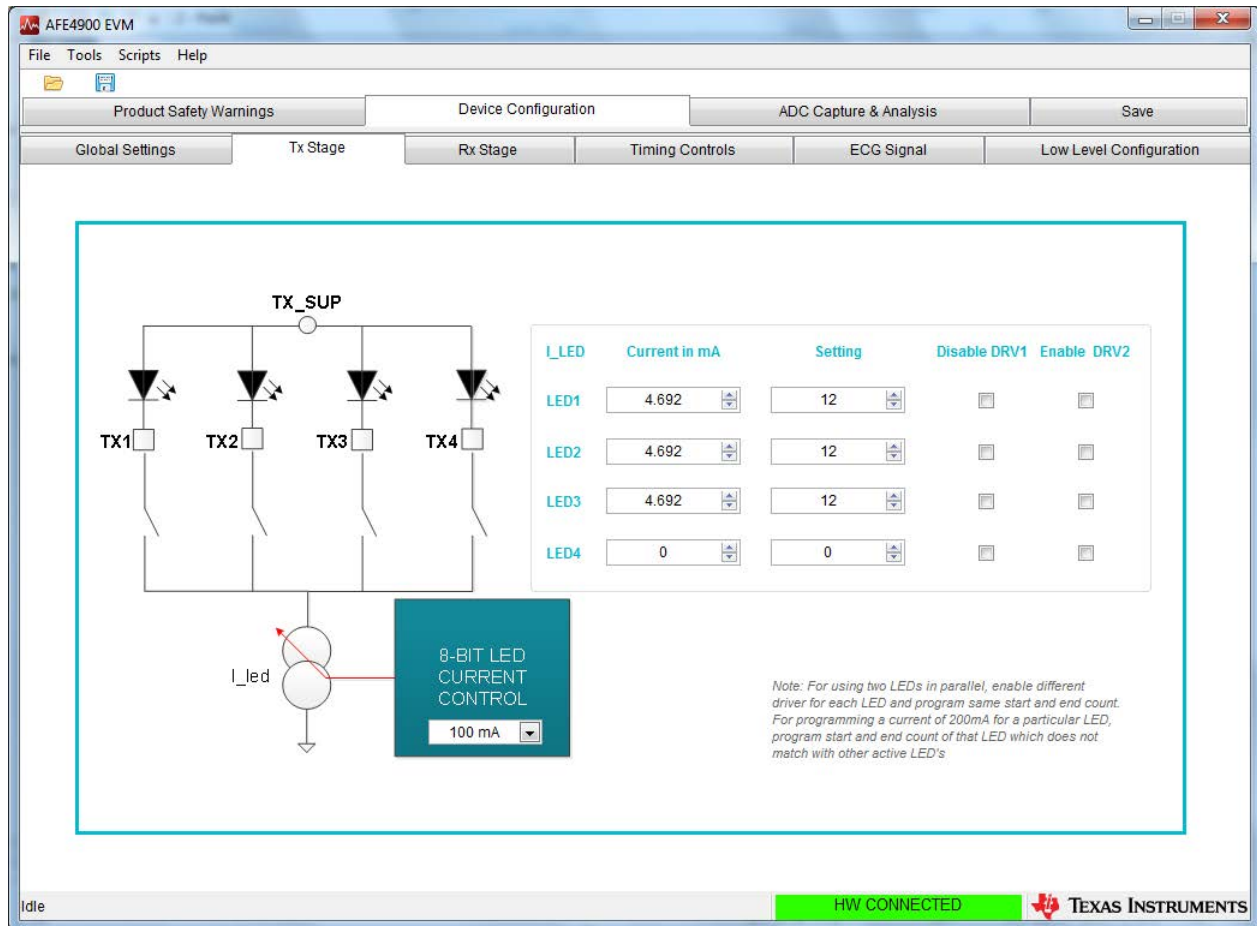


Figure 39. AFE4900: Device Configuration: Tx Stage

The following are a few configurations for TX:

1. Default Configuration
 - (a) All LEDs are enabled in LED_DRIVER1 and all LEDs are disabled in LED_DRIVER2.
 - (b) Current for all the LEDs are controlled by their respective controls (that is, ILEDx).
 - (c) Maximum current through the LED is governed by full scale control.
2. Double the full scale current for an LED (for example, LED1)
 - (a) Enable all LED in LED_DRIVER1 (Default).
 - (b) Enable only LED1 in LED_DRIVER2.
 - (c) Current through LED1 will be double of that set by its respective control.
 - (d) Maximum current for LED will be twice of the full-scale current.
3. Simultaneously switch off the two LEDs (for example, LED1 and LED2).
 - (a) Enable only LED1 in LED_DRIVER1 and only LED2 in LED_DRIVER2.
 - (b) Program same timing controls for LED1 and LED2.
 - (c) Set the same LED current and TIA_GAIN for LED1 and LED2.

4.1.1.3 Rx Stage Subtab

Figure 40 shows the *Rx Stage* subtab under the *Device Configuration* tab consisting of the settings to:

1. Enable separate gain mode.
2. Set the feedback resistance and capacitance for the trans-impedance amplifier with separate gain mode disabled.
3. Set the feedback resistance and capacitance for the trans-impedance amplifier with separate gain mode enabled.
4. Set the ambient DAC current and polarity for LED2, LED3 (LED2 Ambient for dual LED), LED1 Ambient, and LED1.
5. Set full scale of Offset Cancellation DAC.
6. Set the bandwidth of the switched RC filter.
7. Set `EARLY_OFFSET_DAC` to make Offset DAC transition at the start of the LED ON signal.
8. Change the *Bandwidth* setting of the *Noise* reduction filter.

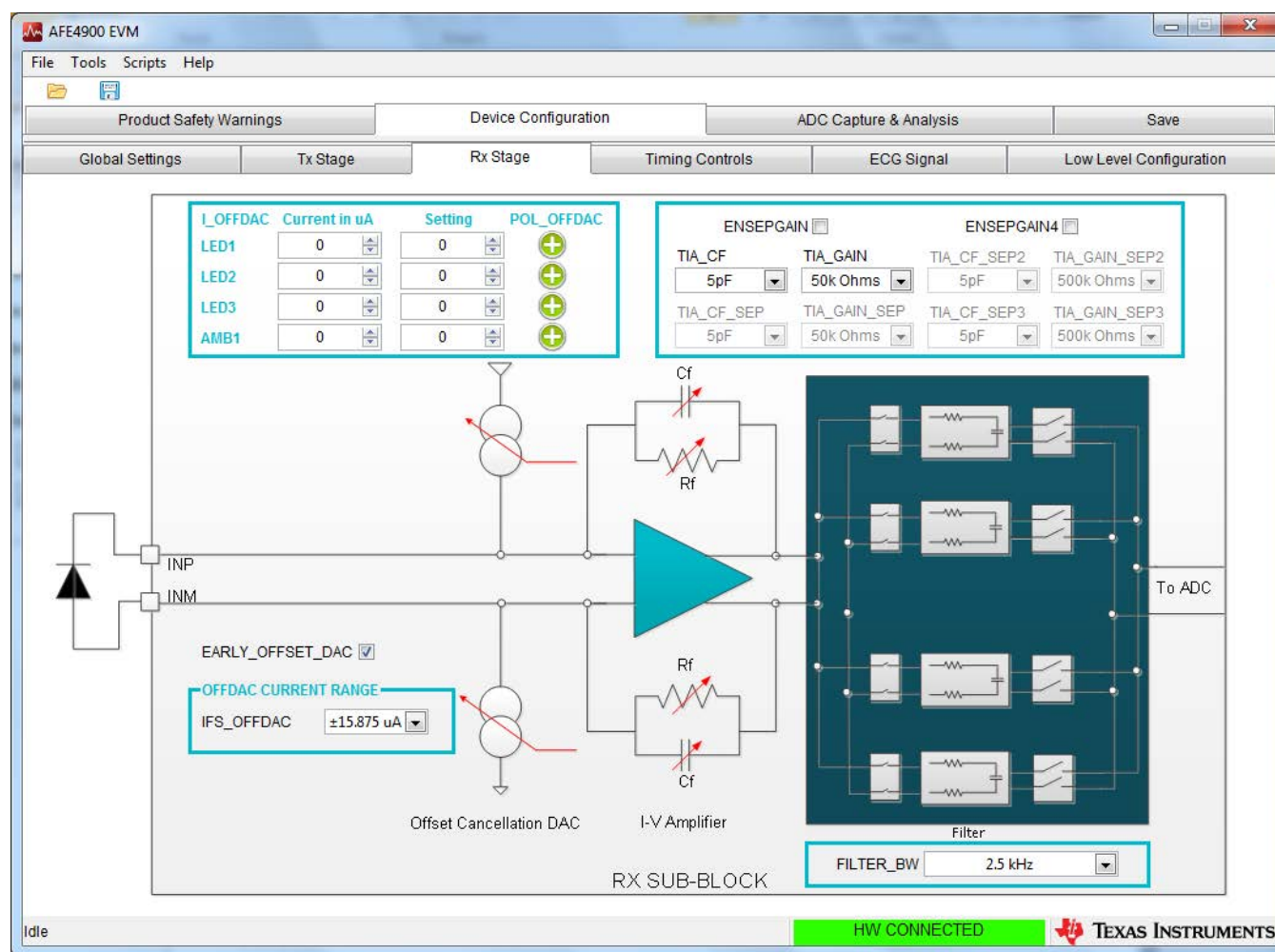


Figure 40. AFE4900: Device Configuration: Rx Stage

4.1.1.4 Timing Controls Subtab

The *Timing Controls* subtab under the *Device Configuration* tab, shown in [Figure 41](#), consists of the following settings:

1. Enter the *Pulse Repetition Frequency* (PRF) to find the *Pulse Repetition Period Count*.
2. Manually control timing settings for on time, sample time, conversion time, and deep sleep cycle time by changing the numbers on the left.
3. Save the timing settings based on PRF and duty cycle to a configuration file.
4. Load the timing settings based on PRF and duty cycle from a configuration file.
5. Load a preset configuration file from a list of options in the drop-down menu.
6. *Timer Enable* selector
7. *Timer Counter* **RESET** button
8. Set the *Number of Averages*.
9. Timing controls for the LED4 signal.
10. Timing controls for the DATA_RDY signal.
11. Timing controls for the PROG_INT1 signal.
12. Timing controls for the PROG_INT2 signal.
13. Timing controls for the PD1 in dual or triple PD mode.
14. Timing controls for the PD2 in dual or triple PD mode.
15. Timing controls for the PD3 in triple PD mode.
16. Timing controls for TIA, 4-MHz oscillator, and ADC for active phase.

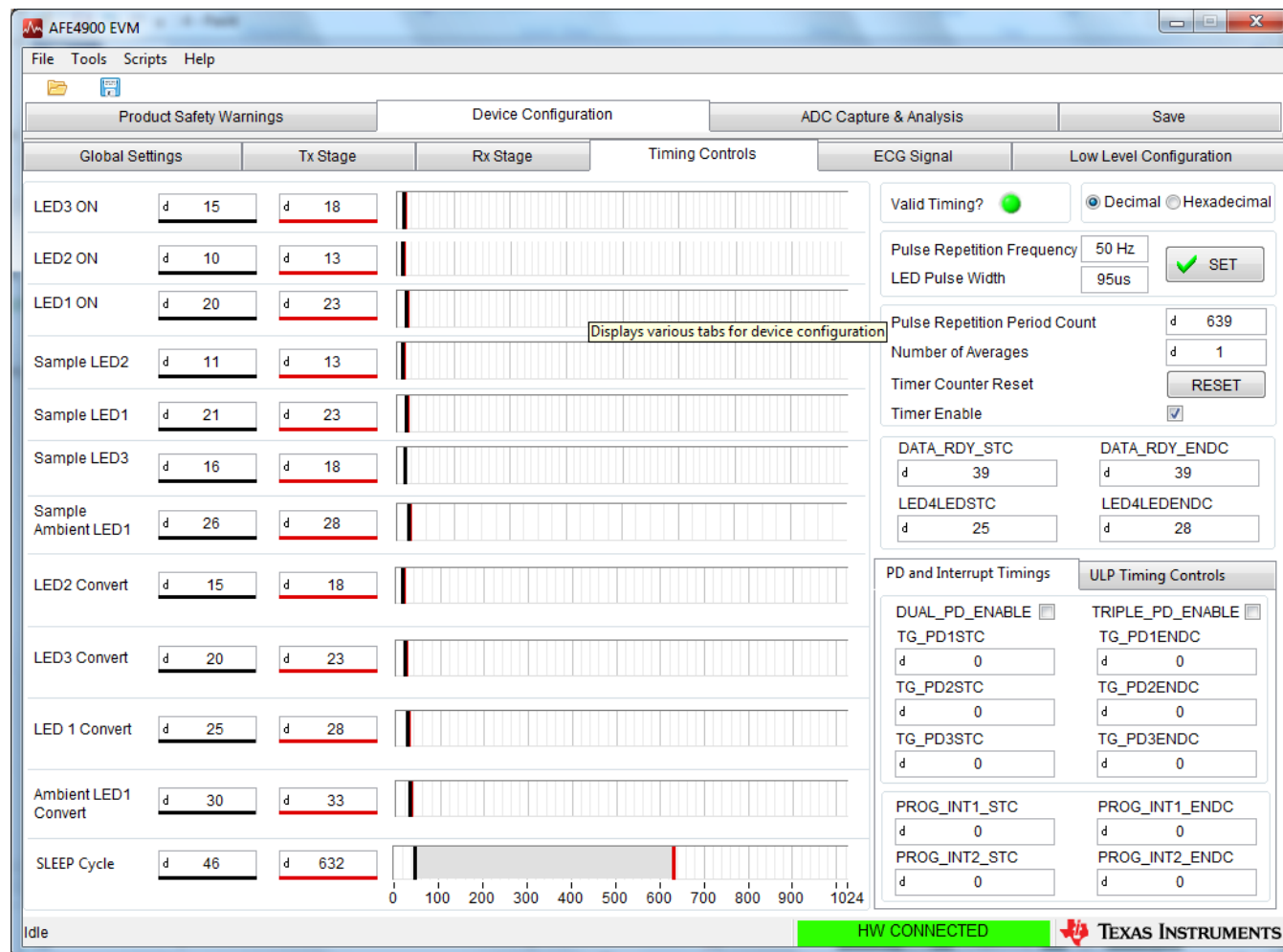


Figure 41. AFE4900: Device Configuration: Timing Controls

4.1.1.5 ECG Signal

Figure 42 shows the *ECG Signal* subtab under the *Device Configuration* tab consisting of the settings to:

1. Enable PTT mode.
2. Set the parameters of ECG signal chain such as the gain of INA, Enable ECG chop and Enable right-leg drive (RLD).
3. Internally short the input of ECG to RLD.
4. Lead-off Detect settings
 - (a) Choose between AC and DC Lead-off detect.
 - (b) Enable the current sources and the voltage comparators
 - (c) Set the direction (source or sink) of current source for INP_ECG and INM_ECG pins.
 - (d) Set the magnitude of Lead-off detect current.
 - (e) Set the threshold levels (VCOMP_HIGH and VCOMP_LOW) for the comparators.
 - (f) Change the frequency of lead-off detect current in AC lead-off detect mode.

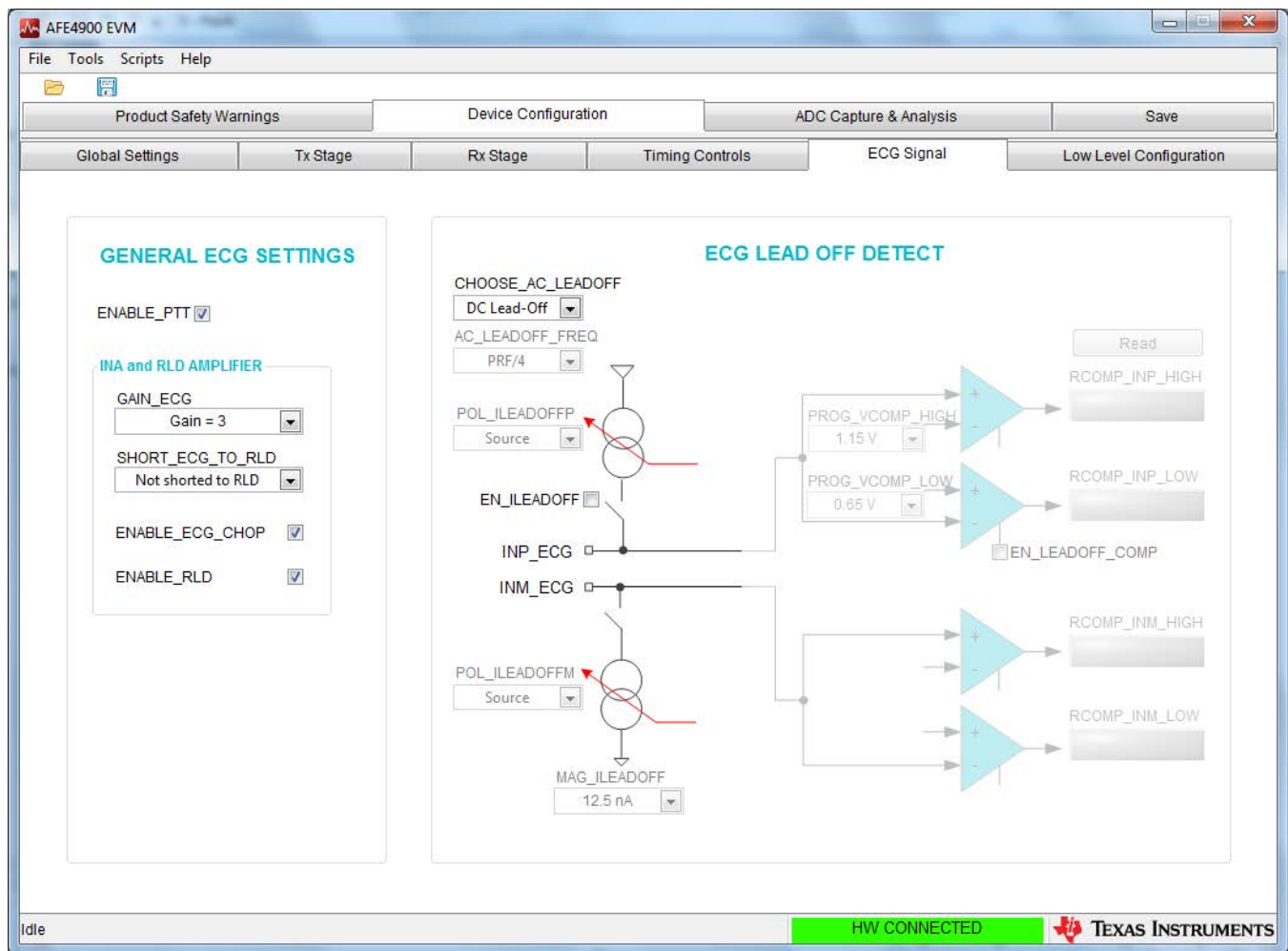


Figure 42. ECG Signal

4.1.1.6 Low Level Configuration Subtab

The *Low Level Configuration* subtab under the *Device Configuration* tab is used to directly configure the various registers of the AFE4900 devices. Refer to the AFE4900 data sheet ([SBAS861](#)) for the register details of the chip.

Figure 43 shows the low-level configuration registers of the AFE4900 devices. The *Register Map* portion of the subtab shows the EVM default values of the registers after the GUI is loaded. From the *Register Map* section, when any register is selected, the bit-level details about the register are explained in the *Register Description* section. The ability to read and write the register and modify the individual bits of the register are changed by clicking on the desired bit. The values of all the registers are read by clicking the **Read All** button.

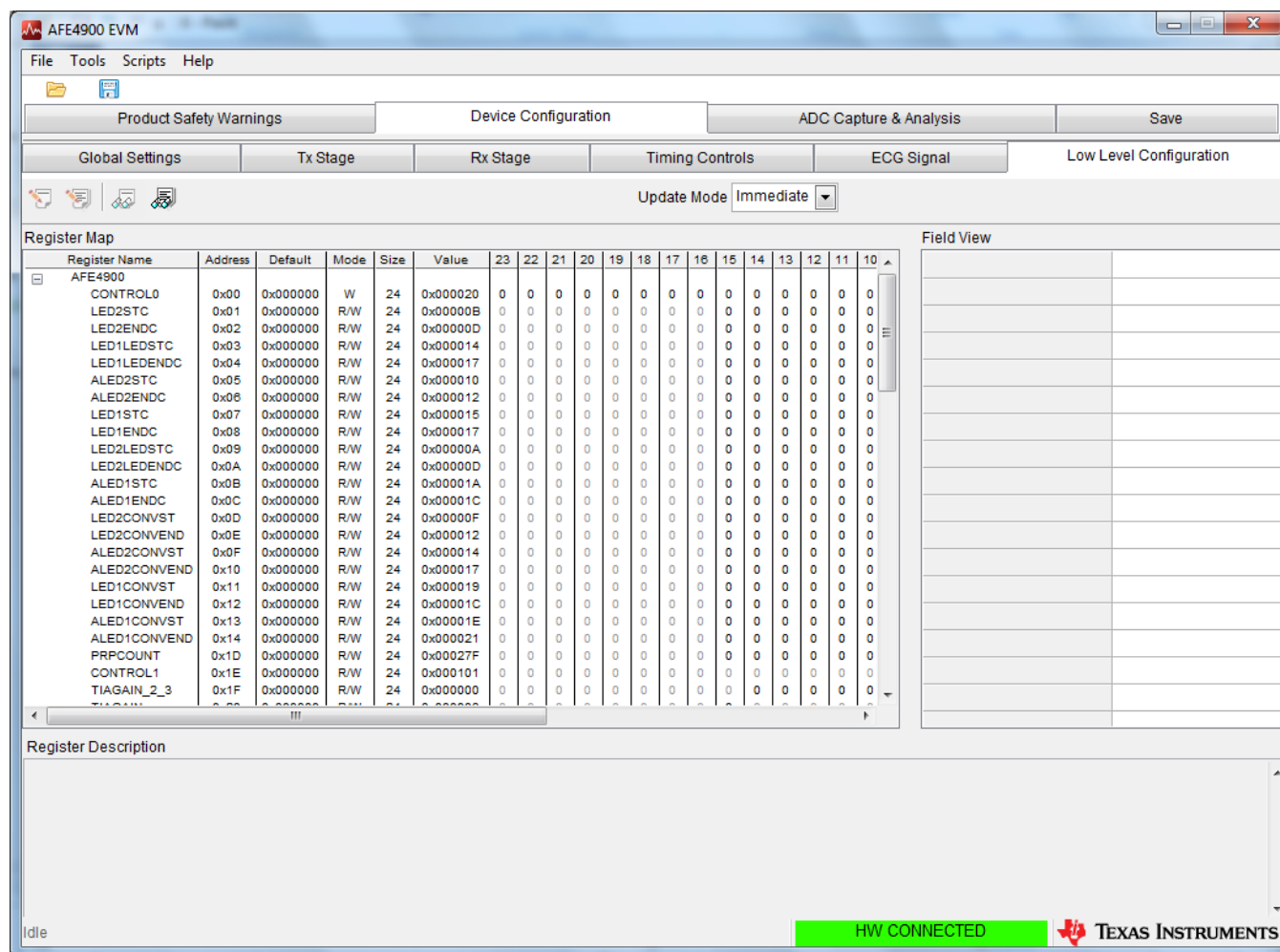


Figure 43. Device Configuration: Low Level Configuration

When a selection is made on any of the tabs on the GUI, multiple fields of various registers are modified. Click on the lower-left corner of the GUI to view the registers that are modified when a selection is made.

4.1.2 ADC Capture and Analysis

4.1.2.1 Options for the “Capture” tab

The *ADC Capture and Analysis* tab consists of various analysis routines and displays. This tab is used to:

- Set the capture mode to finite or continuous
- Set the number of samples (block size) in *Finite Capture* mode

- Set the display to volts or codes
- Set the *Analysis Type* to *All Domain* or *Time Domain* only
- Auto save after capture selector
- Acquire the data by clicking the **Capture** button
- When the user selects the auto save after capture selector under the *ADC Capture & Analysis* tab, the GUI uses the settings selected under *Analysis to Save*, *Channels to Save*, *Data to Save*, and *Save File Settings*. A *Results saved successfully!* notification is given after every capture.

4.1.2.2 Options for the “Filter” tab

- Remove the DC component.
- Enable *Notch* filter and set the *Notch Frequency* to 50 Hz or 60 Hz.
- Enable the *Low Pass Filter* (LPF) and set its order and cutoff frequency.

NOTE: The cutoff frequency for LPF and Notch filter should be smaller than half of sampling frequency (that is, PRF).

The captured data can be analyzed in the time domain and frequency domain; the data can also be displayed in a histogram format. The *ADC Capture & Analysis* tab is shown in [Figure 44](#).

By selecting the *Time Domain* plot, the data are displayed in time domain format. The units can be converted from codes to volts using the drop-down window in the top-left corner of the GUI. For the time domain plot, the mean voltage, root mean square (RMS) voltage, and peak-to-peak voltage are displayed in the *Test Results* section, which is a pop-up window that opens when the **Scope Analysis** button is clicked. The *Scope Analysis: Test Results* section pop-up window is shown in [Figure 45](#).

By selecting the FFT plot, the data are displayed in the frequency domain by performing an FFT on the channel selected. Details of the FFT (including SNR, THD, and so on) are shown in the *Test Results* section located in the left side of the GUI.

Selecting the Histogram plot displays the data in a histogram format for the channel selected. The data are arranged in the total number of histogram bins set within the tab following acquisition. The histogram analysis (shown in the *Test Results* section of the GUI) is used to view the mean voltage, root mean square (RMS) voltage, and peak-to-peak voltage.

Four plot modes can be selected: *Single Plot* mode, *Double Plot* mode, *Three Plot* mode and *Four Plot* mode. In *Single Plot* mode, only one plot (Time, FFT, or Histogram) can be viewed and analyzed for post processing. In *Double Plot* mode, any two plots (Time, FFT, or Histogram) can be viewed and analyzed. In *Three Plot* mode, any three plots and in *Four Plot* mode, any four plots (Time, FFT, or Histogram) can be viewed and analyzed.

The following algorithms have been used to find the # of samples for FFT calculation:

- (a) Number of samples for FFT calc. which is power of 2 $\leq \min((\text{Data rate (sps)} \times N \text{ where } N \text{ is the value in the Show data for the last } N \text{ secs column}), \text{No. of samples})$
- (b) If $((\# \text{ of samples for FFT calc.} == \text{No. of samples}) \ \&\& \ (\text{Filter Type} == \text{"None"}))$ then # of samples for FFT calc. = No. of samples
- (c) If $((\# \text{ of samples for FFT calc.} == \text{No. of samples}) \ \&\& \ (\text{Filter Type} == \text{"Notch"}))$ then # of samples for FFT calc. = No. of samples / 2. This is to allow for filter settling.
- (d) If $(\# \text{ of samples for FFT calc.} < 32 \text{ samples})$ then an error msg “Insufficient # of samples for FFT calculation” will be displayed.

Examples:

1. No. of samples = 8192
Data rate (sps) = 500
Show data for the last 5 secs
Then # of samples for FFT calc. which is power of 2 = $2048 \leq \min((500 \times 5), 8192)$
2. No. of samples = 8192
Data rate (sps) = 500
Show data for the last 8 secs

Then # of samples for FFT calc. which is power of 2 = $2048 \leq \min((500 \times 8), 8192)$

3. No. of samples = 8192

Data rate (sps) = 500

Show data for the last 20 secs

of samples for FFT which is power of 2 = $8192 \leq \min((500 \times 20), 8192)$

Since (# of samples for FFT calc. == No. of samples) and if (Filter Type = None) then # of samples for FFT which is power of 2 = 8192

Since (# of samples for FFT calc. == No. of samples) and if (Filter Type = Notch) then # of samples for FFT which is power of 2 = $8192 / 2$

4. No. of samples = 30

Data rate (sps) = 500

Show data for the last 1 secs

Then display error message "Insufficient # of samples for FFT calculation" since # of samples for FFT which is power of 2 = $16 \leq \min((500 \times 1), 30)$

5. No. of samples = 32

Data rate (sps) = 500

Show data for the last 2 secs

of samples for FFT which is power of 2 = $32 \leq \min((500 \times 2), 32)$

Since (# of samples for FFT calc. == No. of samples) and if (Filter Type = None) then # of samples for FFT calc. which is power of 2 = 32

Since (# of samples for FFT calc. == No. of samples) and if (Filter Type = Notch) then an error msg *Insufficient # of samples for FFT calculation* will be displayed since # of samples for FFT calc. which is power of 2 = $(32 / 2) < 32$ samples.

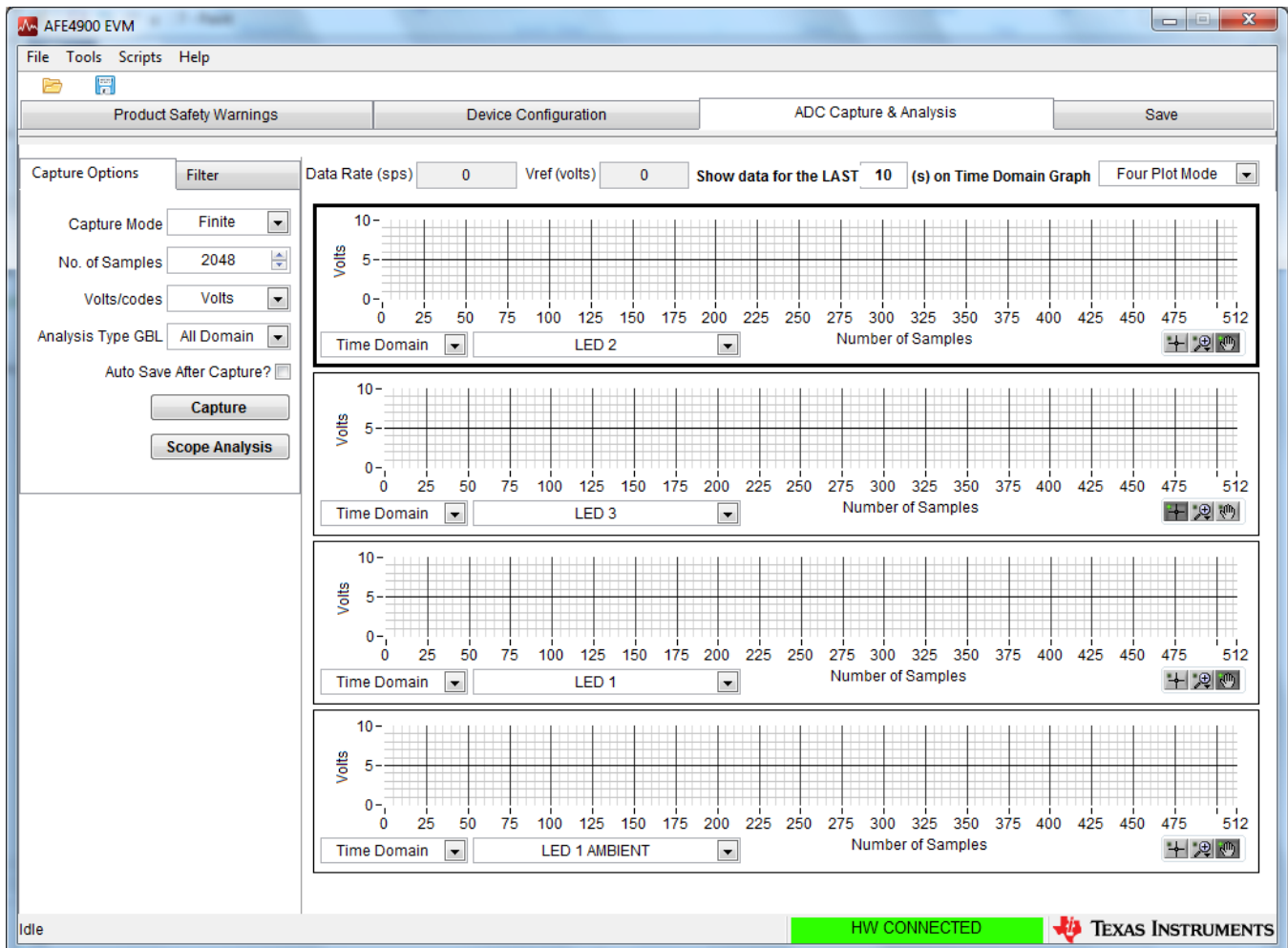


Figure 44. ADC Capture and Analysis Tab

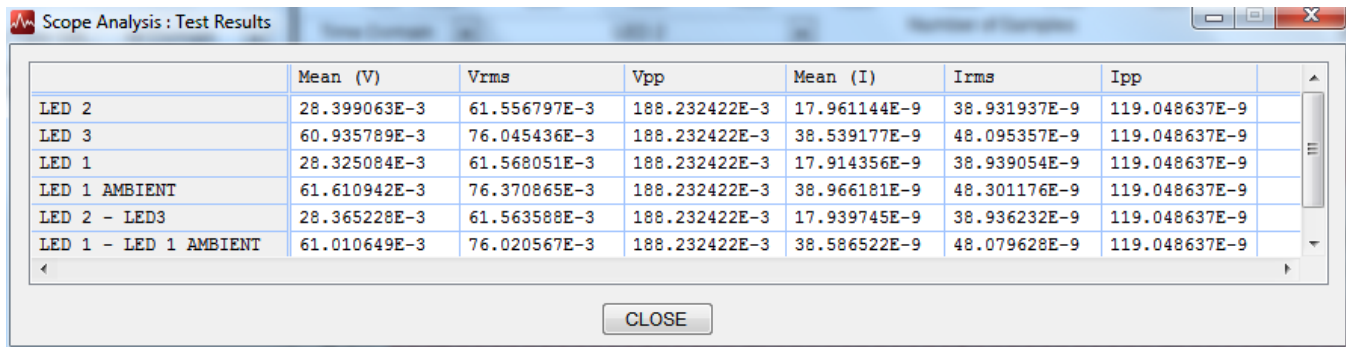


Figure 45. Scope Analysis: Test Results

4.1.2.3 ADC Output Register

The AFE4900 device stores the output of ADC in three locations:

1. Registers 2Ah to 2Fh: Stores the un-decimated output of ADC in 2's complement form.
2. Registers 3Fh and 40h: Stores the decimated output of ADC in 2's complement form when decimation mode is enabled.
3. FIFO: Output of ADC is stored in the FIFO (with address FFh). The data in FIFO can be decimated or un-decimated depending on DEC_EN and FIFO_EN_DEC control bits.

The GUI can capture from any of the output locations based on the settings as described in [Table 1](#).

Table 1. Data Capture Locations for GUI

FIFO_EN	DEC_EN	FIFO_EN_DEC	Location Read by GUI	Data Rate	Frequency of ADC_RDY/FIFO_RDY	Remark
1	1	1	FIFO (FFh)	PRF / DEC_FACTOR	PRF / (FIFO_PERIOD × DEC_FACTOR)	Decimated data
1	1	0	FIFO (FFh)	PRF	PRF / FIFO_PERIOD	Un-decimated data
1	0	X	FIFO (FFh)	PRF	PRF / FIFO_PERIOD	Un-decimated data
0	0	X	2Ah to 2Fh	PRF	PRF	Un-decimated data
0	1	X	3Fh and 40h	PRF / DEC_FACTOR	PRF / DEC_FACTOR	Decimated data

4.1.3 Save Tab

The Save tab shown in [Figure 46](#) provides provisions to save the analysis or data to a file. By default, the data are saved to the following location:

- On a Windows XP machine – *C:\Program Files\Texas Instruments\AFE4900 EVM\Log*
- On Windows 7, Windows 8, and Windows 10 machines – *C:\Program Files(x86)\Texas Instruments\AFE4900 EVM\Log*

Use the *Directory to Save Files* option to select the folder where data are to be saved. In the pop-up window, navigate to the folder where the data file is to be saved and select *Use Current Folder*. Then select *Save to File* to save the file.

When the user selects the auto save after capture selector under the *ADC Capture & Analysis* tab, the GUI uses the settings selected under *Analysis to Save*, *Channels to Save*, *Data to Save*, and *Save File Settings*. A *Results saved successfully!* notification is given after every capture.

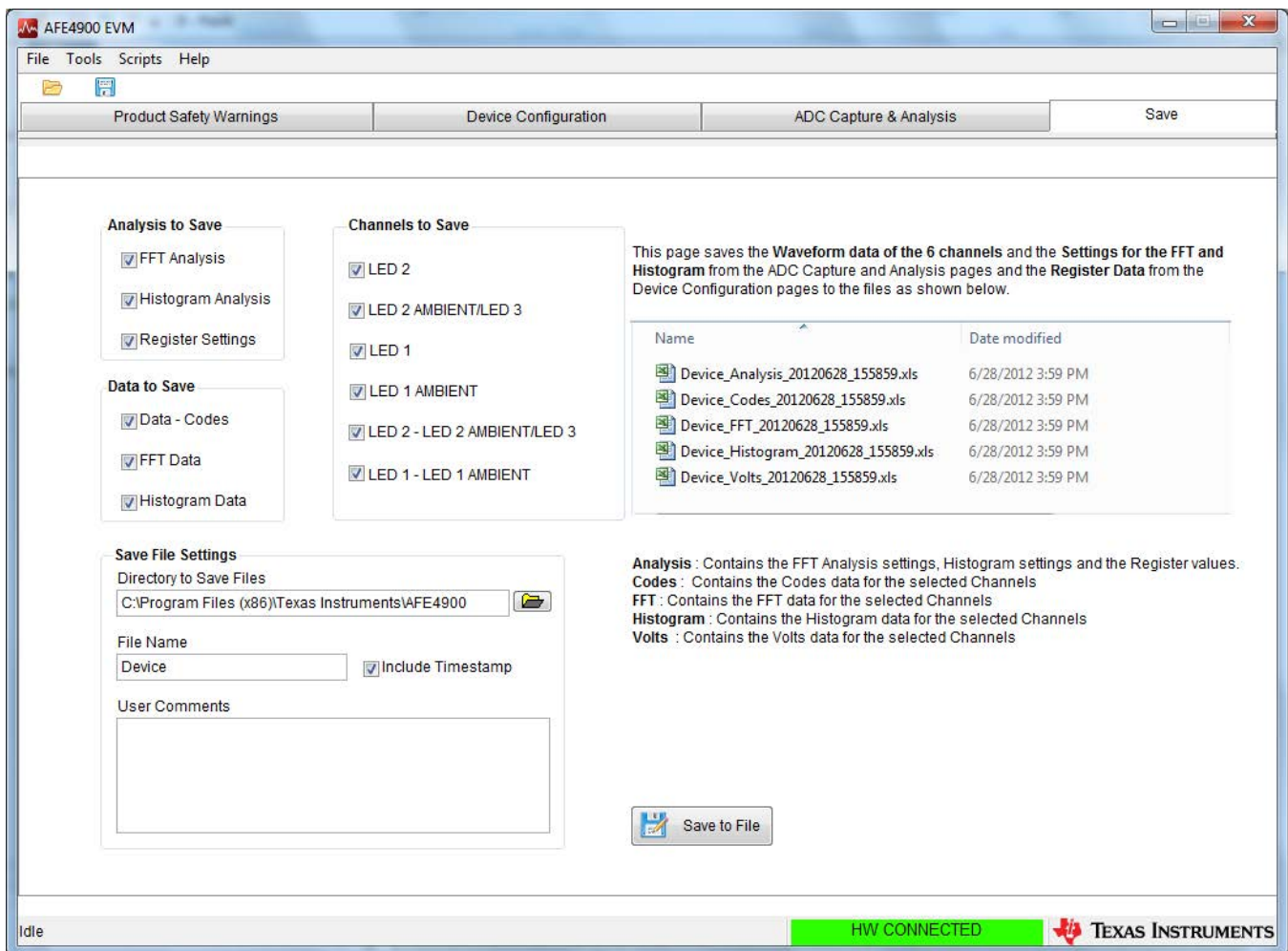


Figure 46. Save Tab

Table 2 contains the Save tab control descriptions.

Table 2. Save Tab Control Descriptions

Button or Control	Description
Scope Analysis	Saves the scope analysis result. The result is saved in the file <i>Device_<record number>_Analysis.xls</i> .
FFT Analysis	Saves the FFT analysis result. The result is saved in the file <i>Device_<record number>_Analysis.xls</i> .
Histogram Analysis	Saves the histogram analysis result. The result is saved in the file <i>Device_<record number>_Analysis.xls</i> .
Register Settings	All the current register values are read from the EVM and stored. The result is saved in the file <i>Device_<record number>_Analysis.xls</i> .
Data - Codes	Acquired data sample values are stored to the file <i>Device_<record number>_Codes.xls</i> .
FFT Data	Acquired FFT values of the data sample are stored to the file <i>Device_<record number>_FFT.xls</i> .
Histogram Data	Acquired histogram values of the data sample are stored to the file <i>Device_<record number>_Histogram.xls</i> .

The *Record Number* saves files with the provided number in the file name. User notes can also be added to the file by typing the notes in the *User Comments* control.

5 AFE4900 EVM Hardware

CAUTION

Many of the components on the AFE4900 EVM are susceptible to damage by electrostatic discharge (ESD). Customers are advised to observe proper ESD handling precautions when unpacking and handling the EVM, including the use of a grounded wrist strap, bootstraps, or mats at an approved ESD workstation. Wear safety glasses.

The key features of the *AFE4900 Analog Front End* demonstration board are:

- Based on MSP430F5529
- Acquire data at up to 1000 Hz in evaluation mode
- SPI Data interface

The AFE4900 EVM board can be used as a demo board for pulse oximeter and heart rate applications. The bill of materials (BOM) is provided in [Section 10](#). The printed-circuit boards (PCB) are shown in [Section 11.1](#) and [Section 11.2](#), and the schematics are shown in [Section 11.3](#).

MSP430F5529 is the microcontroller used on the board. For more details of the MSP430F5529, visit <http://www.ti.com/product/msp430f5529>.

The following sections explain the main hardware components available on the EVM. Figure 47 shows the functional block diagram for the EVM.

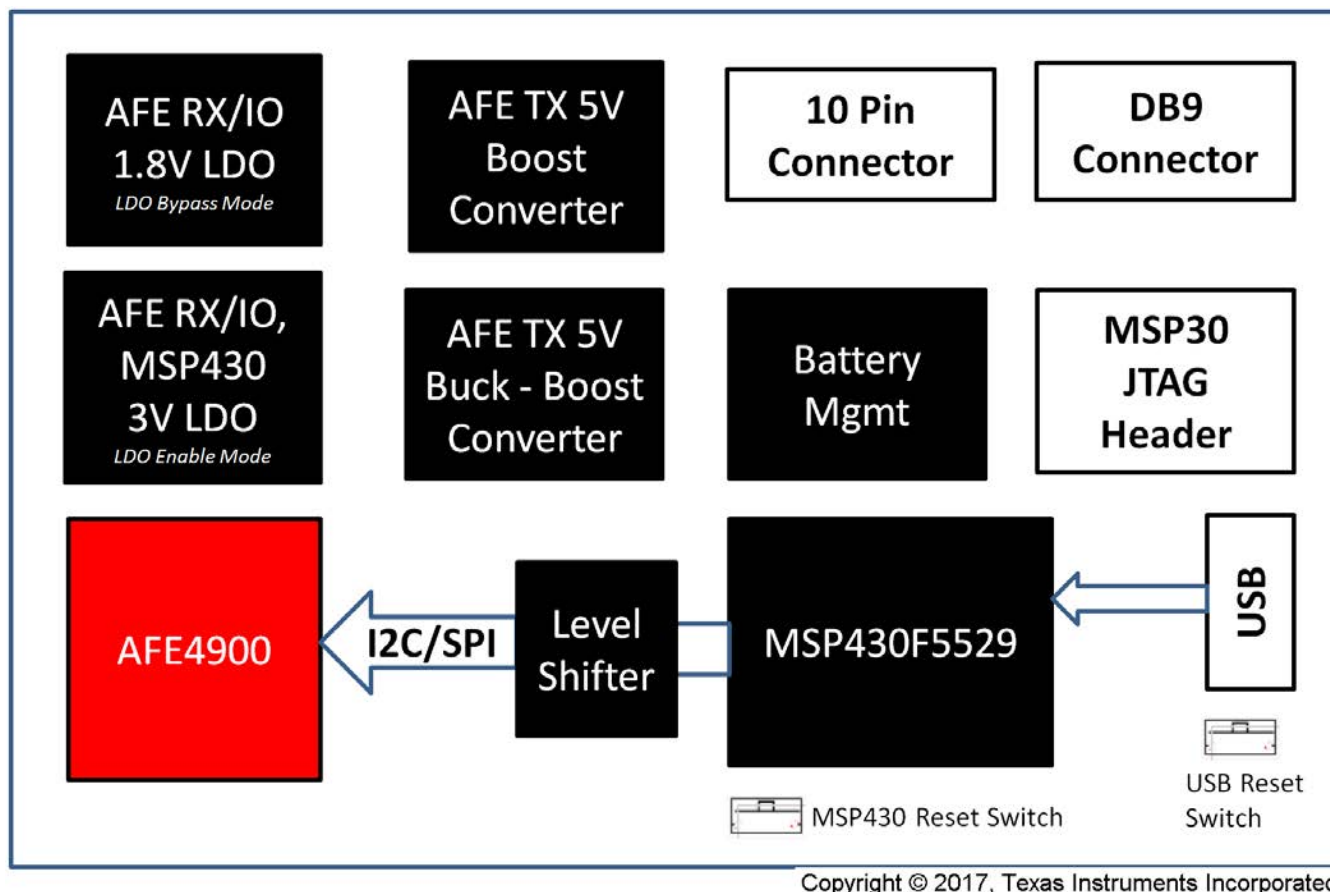


Figure 47. AFE4900 EVM Block Diagram

5.1 Power Supply

The AFE4900 can operate from 2.0-V to 3.6-V RX analog supply (RX_SUP), 1.8-V to 3.6-V IO digital supply (IO_SUP), and 3.0-V to 5.25-V TX Control supply (TX_SUP).

The power for the board is derived from the USB input (J12) through a forward-biased diode (D1) to avoid reverse current flow. The USB data bus is ESD protected using TI's ESD protection diode array TPD4E004DRYR (U20). The USB VBUS is fed to the integrated Li-Ion linear charger and system power-path management module, BQ24032ARHLR (U17), which generates greater than 4.2-V output (VCC_BAT). This output is fed to TI's highly efficient, low quiescent-current buck-boost converter, TPS63050YFFR (U7) for generating 5 V for TX_SUP. TX_SUP (of 5 V) can also be generated using TI's synchronous boost converter, TPS61099YFFR (U8). In default configuration, 5-V output of U7 is used as TX_SUP. The VCC_BAT is also fed to the ultra-low-noise linear voltage regulator TPS7A4901DGN (U18) for generating 3 V (LDO Enable mode) for the MSP_DVCC, MSP_AVCC, RX_SUP, and IO_SUP. For LDO Bypass mode VCC_BAT is fed to TPS72718YFFR (U9) for generating 1.8 V for RX_SUP and IO_SUP.

LDO Enable Mode: Connect J5 and J6, remove J4 and J7.

LDO Bypass Mode: Connect J4 and J7, remove J5 and J6.

Series jumper resistors are provided to ensure the power supplies to the board are correct.

5.2 Clock

The EVM has the option to use the internal clocking using the on-chip 128-kHz oscillator or the clock for the AFE4900 from the MSP430 (device and EVM default).

5.3 Accessing AFE4900 Digital Signals

The AFE4900 SPI and other digital signals with MSP430 can be accessed through the test points given in [Table 3](#).

Table 3. AFE4900 Digital Signals

S. No.	Signal	Description
1	SCLK or I2C_CLK	TP4
2	SDIN or I2C_DAT	TP9
3	SDOUT	TP7
4	SEN	TP11
5	ADC_RDY	TP12
6	RESET	TP8
7	CLK	TP6
8	PROG_OUT1	TP13
9	I2C_SPI_SEL	TP5
10	CONTROL1	TP3

5.4 USB Interface

The EVM has a micro-USB interface for PC application connectivity requiring a standard micro-USB to USB cable for connection. The AFE4900 EVM is designed to work in the slave mode.

5.5 Onboard Key Interface

The EVM has 2 switches. The function of each switch is defined in [Table 4](#).

Table 4. AFE4900 EVM Switches

Switch Number	Description
SW1	This switch is used for hard reset of the board. The board resets and starts again with the firmware loaded.
SW2	This switch is used to enable boot strap loader (BSL) MSP430 firmware. ⁽¹⁾

⁽¹⁾ To enable BSL, disconnect device and then reconnect while holding down SW2. The device will appear as an HID device in the *Device Manager*.

5.6 Visual Indication

The green LED (LED1) indicates the USB power connection. The green LED (LED3) indicates that the microcontroller is busy servicing the requests from the PC application.

6 USB-Based Firmware Upgrade

The firmware on the AFE4900 EVM can be changed from the PC application by selecting the *Firmware Upgrade* menu option on the PC application. At the end of the firmware upgrade, the system issues a reset command and reloads with new firmware. The firmware upgrade process steps are represented in the following screen shots:

- From the PC application, click on *File* → *Firmware Upgrade*
- A pop-up window opens as shown in [Figure 48](#). Follow the instructions to continue to *Firmware Upgrade* or to cancel the operation.

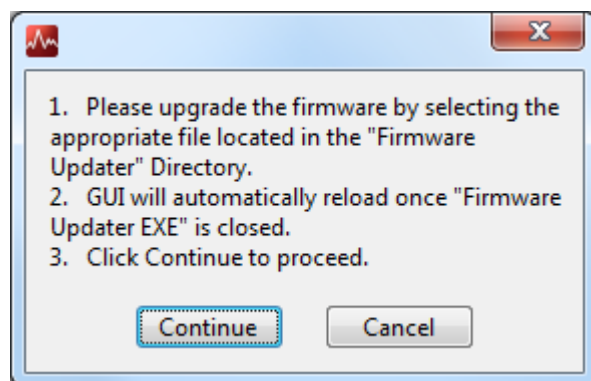


Figure 48. PC Application Firmware Upgrade – 1

- The firmware upgrade application detects the connected EVM ([Figure 49](#)).

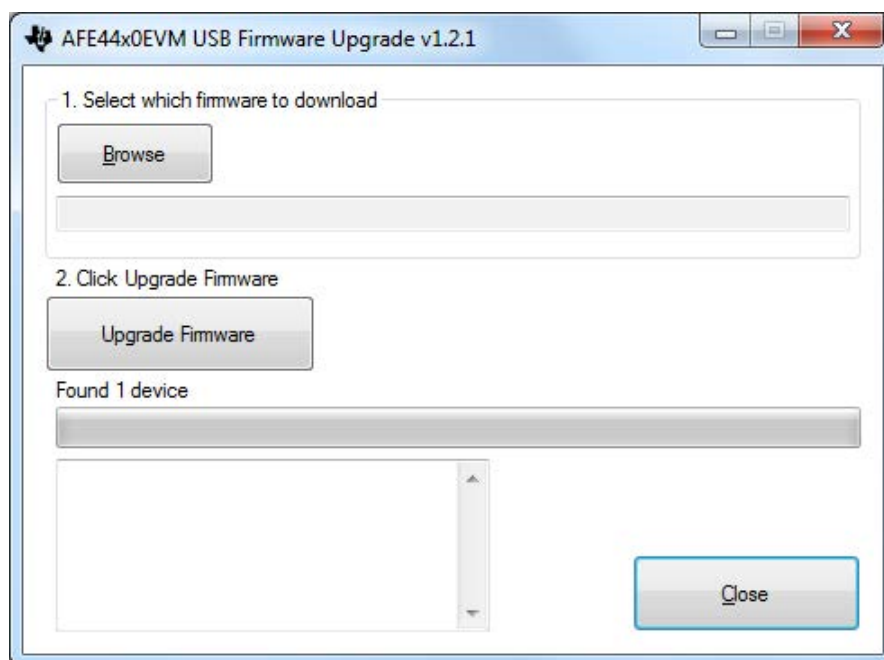


Figure 49. PC Application Firmware Upgrade – 2

- Browse and select the appropriate firmware binary file (example: AFE4900EVM_FW_V1.0_SPI_LDO_EN.txt file) and click *Upgrade Firmware* as shown in [Figure 50](#). The default firmware is available from:
 - On a Windows XP machine:
 - C:\Program Files\Texas Instruments\AFE4900 EVM\Firmware Updater
 - On Windows 7, Windows 8, and Windows 10 machines:
 - C:\Program Files(x86)\Texas Instruments\AFE4900 EVM\Firmware Updater

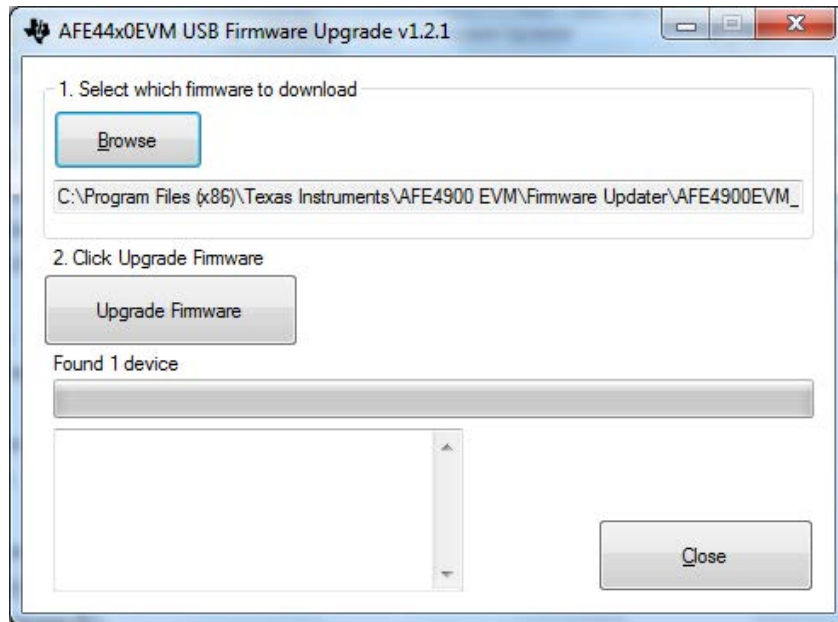


Figure 50. PC Application Firmware Upgrade – 3

- Once the device is programmed successfully, as shown in [Figure 51](#), the device resets and reloads with the new firmware. Close the *Firmware Upgrade* application by clicking on the **Close** button and the PC GUI application automatically restarts the GUI after 4–5 seconds.

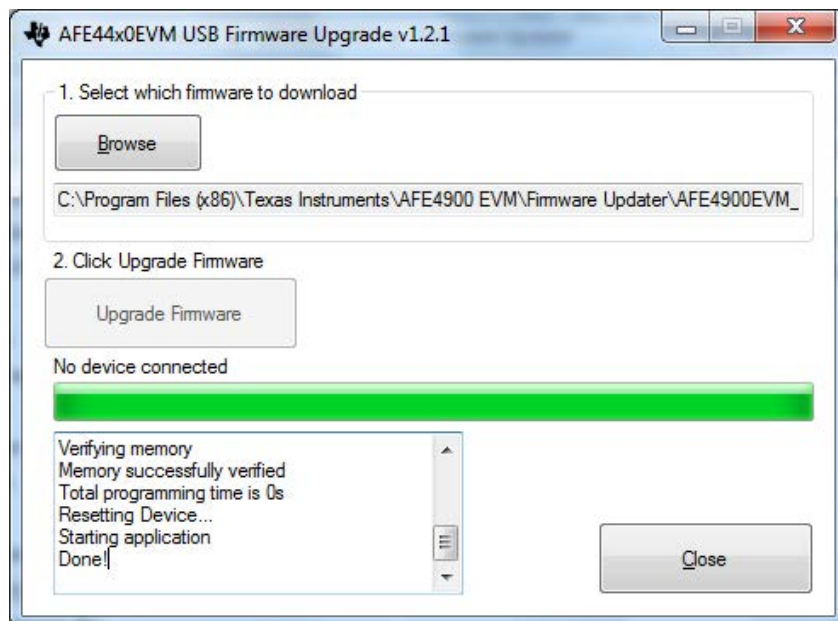


Figure 51. PC Application Firmware Upgrade – 4

7 AFE4900 EVM Firmware Upgrade Without GUI

Use the following steps to upgrade the AFE4900 EVM firmware without the GUI:

1. Open the firmware loader application by clicking the BSL_USB_GUI.exe located at the following location:
 - On Windows 7, Windows 8, and Windows 10 operating systems (OS): "C:\Program Files(x86)\Texas Instruments\AFE4900 EVM\Firmware Updater"
 - On a Windows XP OS: "C:\Program Files\Texas Instruments\AFE4900 EVM\Firmware Updater"
2. Click the **Browse** button and load the AFE4900 firmware. [Figure 52](#) shows the firmware loader application with the appropriate firmware selected. The firmware is located in the "Firmware Updater" directory.

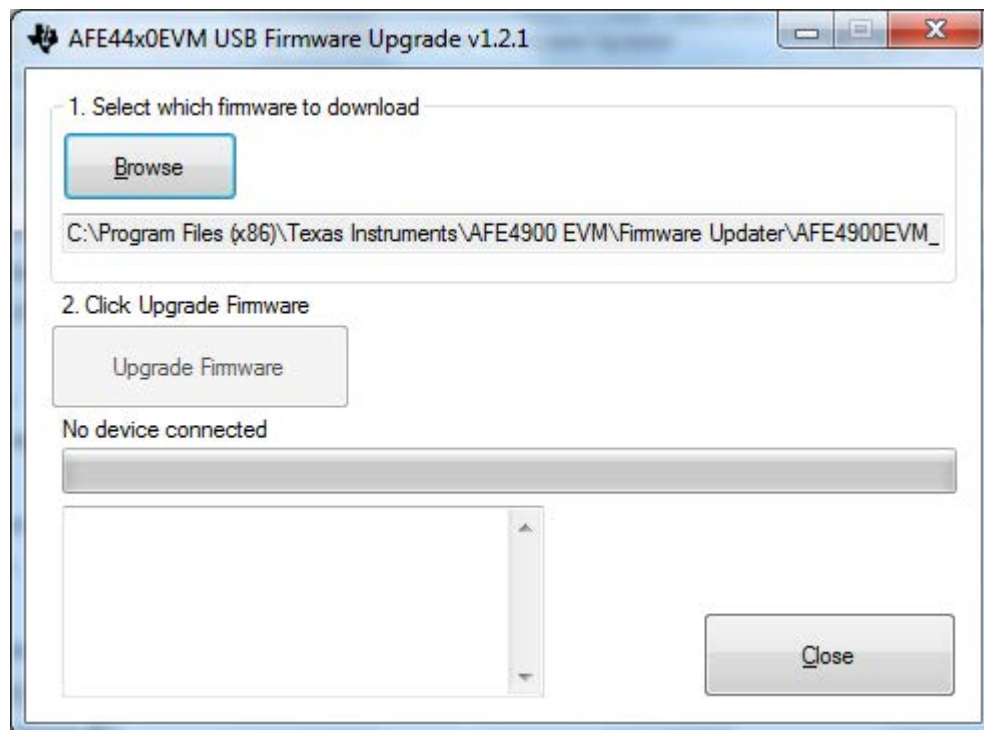


Figure 52. Firmware Loader Application: Select Firmware

3. Press the SW2 switch on the EVM while plugging in the micro-USB interface cable to the J12 micro-USB connector on the EVM.

4. Release the SW2 switch when the application displays *Found 1 device*. If the application does not detect the device and displays *No Device Connected*, then repeat step 3. Click on the *Upgrade Firmware* button (see [Figure 53](#)).

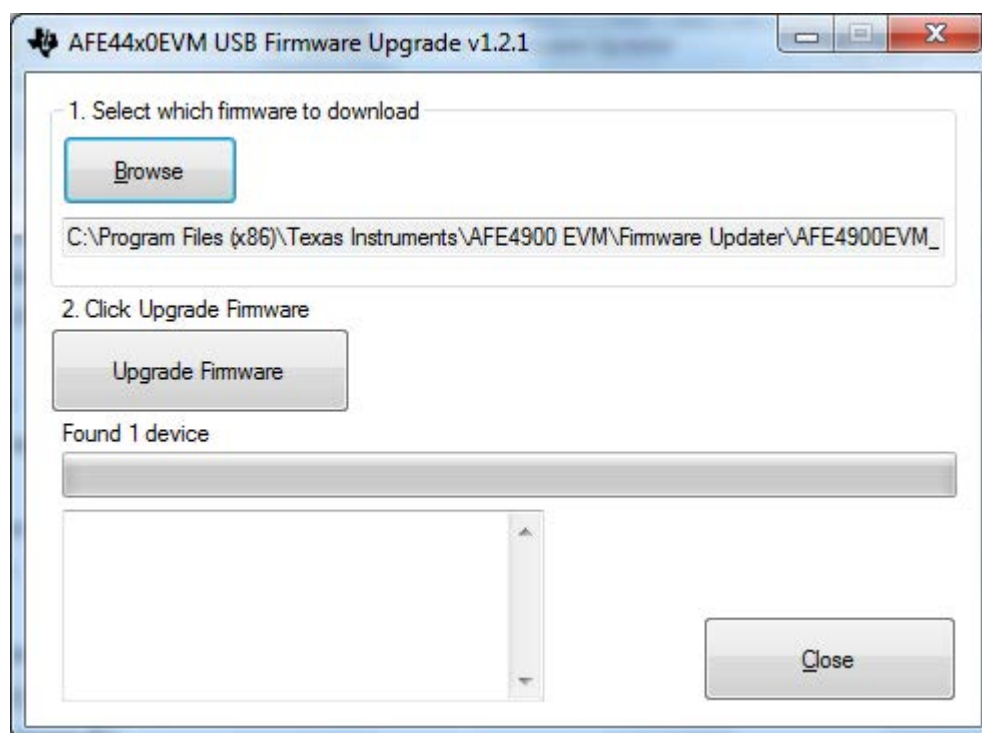


Figure 53. Firmware Loader Application: Found Device

5. The text box will display the status of the firmware programming. If programming is successful, the *Done!* message is displayed in the text box. [Figure 54](#) shows the status of the successful programming.

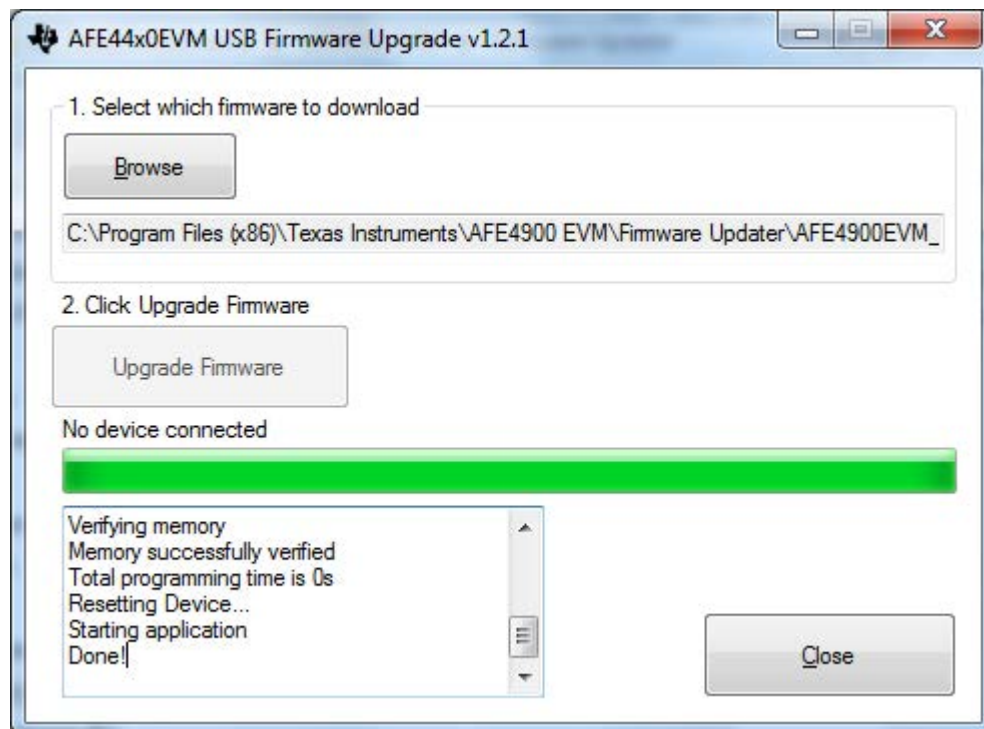


Figure 54. Firmware Loader Application: Programming Status

8 Connector Interface

The following connectors are used for external interface to the AFE4900 Evaluation Module:

- Micro-USB connector
- 10-pin connector

8.1 Micro-USB Connector

The USB micro connector J12 pin-outs are shown in [Figure 55](#). The description of the pin-outs is provided in [Table 5](#).

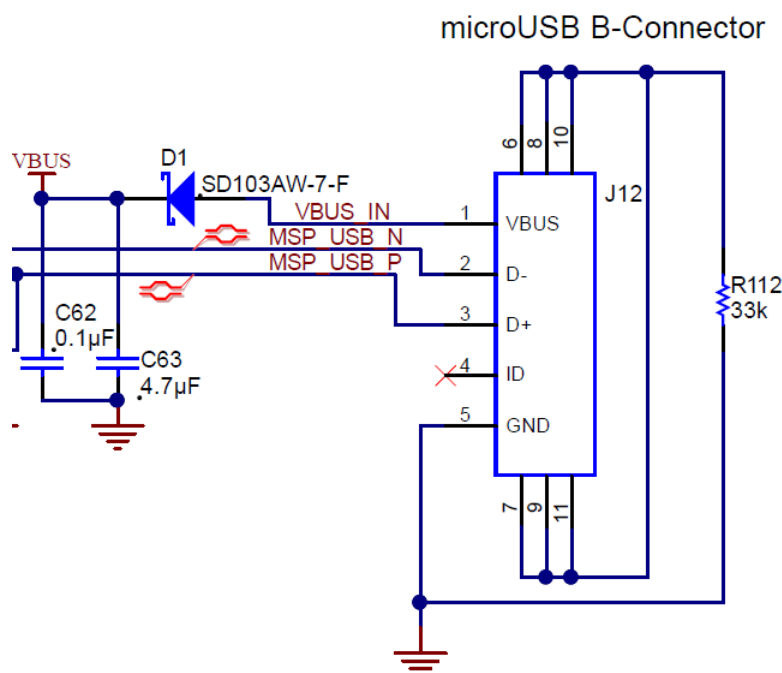


Figure 55. USB Micro Connector Pin Outs

Table 5. USB Micro Connector Pin Out Descriptions

Pin Number	Pin Name	Pin Description
1	VBUS	USB power 5 V
2	D–	USB DM
3	D+	USB DP
4	ID	NC
5	GND	GND

8.2 10-Pin Connector

The 10-pin connector pin-outs are shown in Figure 56. The description is provided in Table 6.

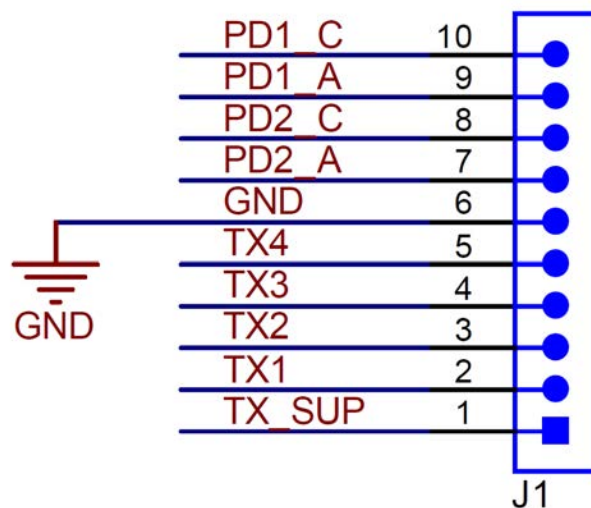


Figure 56. 10-Pin Connector

Table 6. 10-Pin Connector Pin Descriptions

Pin Number	Pin Name	Pin Description
1	TX_SUP	Power Supply for LEDs
2	TX1	Cathode of LED1
3	TX2	Cathode of LED2
4	TX3	Cathode of LED3
5	TX4	Cathode of LED4 (requires hardware modifications)
6	GND	Ground
7	PD2_A	Photodiode2 anode
8	PD2_C	Photodiode2 cathode
9	PD1_A	Photodiode1 anode
10	PD1_C	Photodiode1 cathode

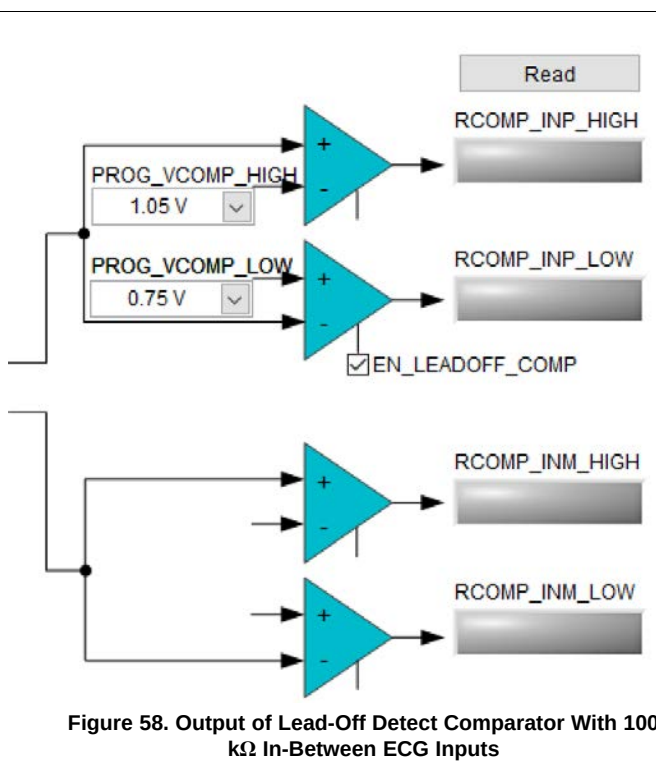
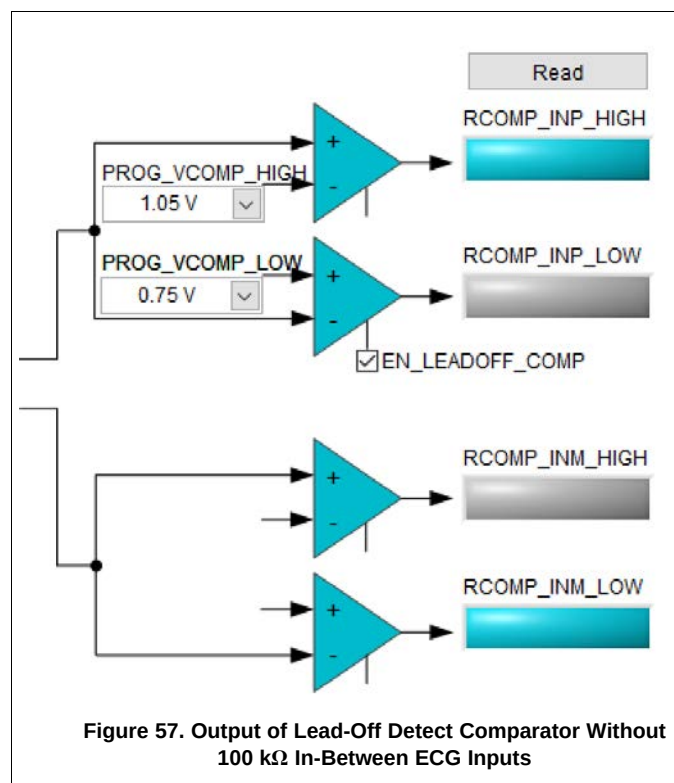
9 AFE4900 EVM FAQs

9.1 ECG Lead-Off Detect

1. Install the shunt jumpers on J17, J8, and J13.
2. Remove J18 and J19.
3. In the *ECG Signal* subtab, set the following parameters.
 - EN_ILEADOFF = Checked
 - POL_ILEADOFFP = Source
 - POL_ILEADOFFM = Sink
 - MAG_ILEADOFF = 50 nA
 - EN_LEADOFF_COMP = Checked
 - PROG_VCOMP_HIGH = 1.05 V
 - PROG_VCOMP_LOW = 0.75 V

9.1.1 DC Lead-Off Detect

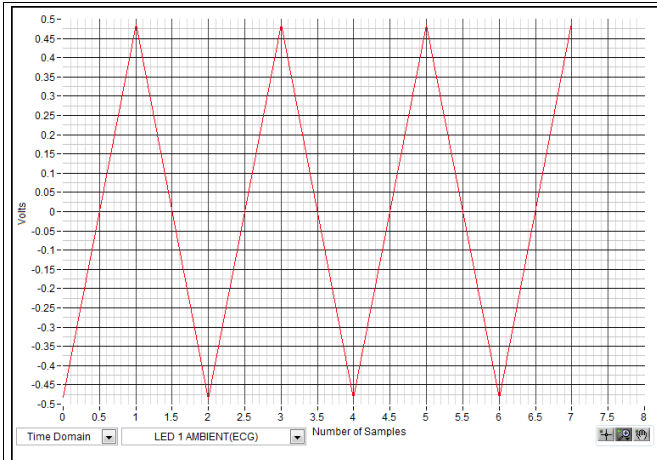
- In the ECG Signal subtab select, CHOOSE_AC_LEADOFF = DC Lead-Off.
- Press the **Read** button, the output should be as shown in [Figure 57](#).
- Now connect a 100-k Ω resistor between J19 (pin #2) and J18 (pin #2).
- If you press the **Read** button this time the output will be as shown in [Figure 58](#)



9.1.2 AC Lead-Off Detect

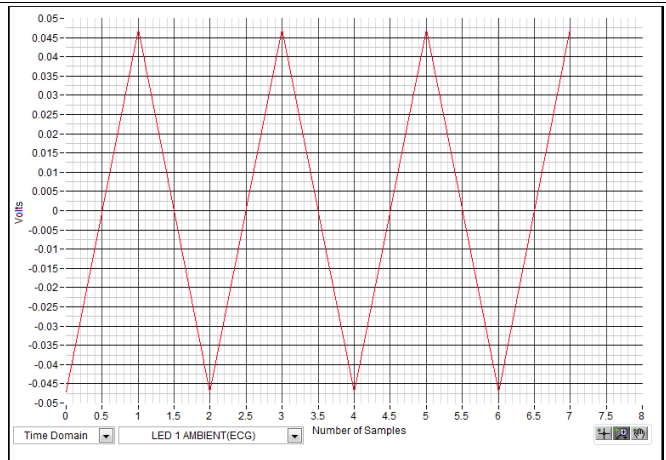
Use the following directions for AC lead-off detect.

- For *AC Lead-Off Detect* select, `CHOOSE_AC_LEADOFF` = AC Lead-Off.
- Select the `AC_LEADOFF_FREQ` as “PRF / 4” or “PRF / 2”.
- Go to *ADC Capture & Analysis* tab and capture the data. The data for ECG phase will have a square (square for `AC_LEADOFF_FREQ` = PRF / 4, whereas it will be triangular for PRF / 2) wave whose swing will depend on the impedance between `INP_ECG` and `INM_ECG` pins: that is, if the body is not in contact, the swing will be higher as compared to the case when body is in contact as shown in [Figure 59](#) and [Figure 60](#).



The swing is 0.91 V, peak-to-peak.

Figure 59. ECG Output Without 100 kΩ In-Between ECG Inputs (With `AC_LEADOFF_FREQ` = PRF / 2)



The swing is 0.91 V, peak-to-peak.

Figure 60. ECG Output With 100 kΩ In-Between ECG Inputs (With `AC_LEADOFF_FREQ` = PRF / 2)

9.2 ADC_RDY Signal

After executing the GUI, observe the `ADC_RDY` waveform at series jumper resistor R18. This should be at the same frequency as the PRF. [Figure 61](#) shows the `ADC_RDY` waveform at 100-Hz PRF.



Figure 61. ADC_RDY Waveform at 100-Hz PRF

9.3 Automation of Register Read and Write Operations

Refer to the *Scripting* document located in the *Documentation* directory for detailed instructions on how to use automation functions for register read and write operations.

The documentation directory is located at the following location:

- On a Windows XP machine – C:\Program Files\Texas Instruments\AFE4900 EVM\Documentation
- On Windows 7, Windows 8, and Windows 10 machines – C:\Program Files(x86)\Texas Instruments\AFE4900 EVM\Documentation

9.4 Optimum Viewing Experience on Windows® 7 or Windows® 8 OS

Change the size of text to *Smaller – 100%* for an optimum viewing experience on Windows 7 or Windows 8 operating system as shown in [Figure 62](#).

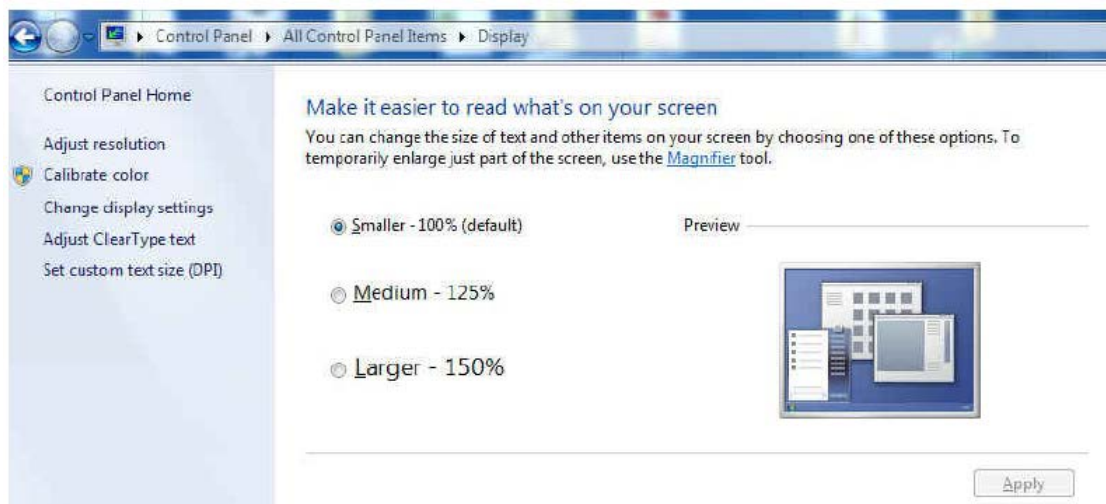


Figure 62. Setting Font Size on Windows® 7 or Windows® 8 Operating System

9.5 Windows® 8 Support for Device GUIs

At GUI start up, sometimes the GUI might show a broken arrow as seen in [Figure 63](#). One of the reasons for this issue may be due to a missing update of Microsoft .NET Framework™ 3.5 (includes .NET 2.0 and .NET 3.0).

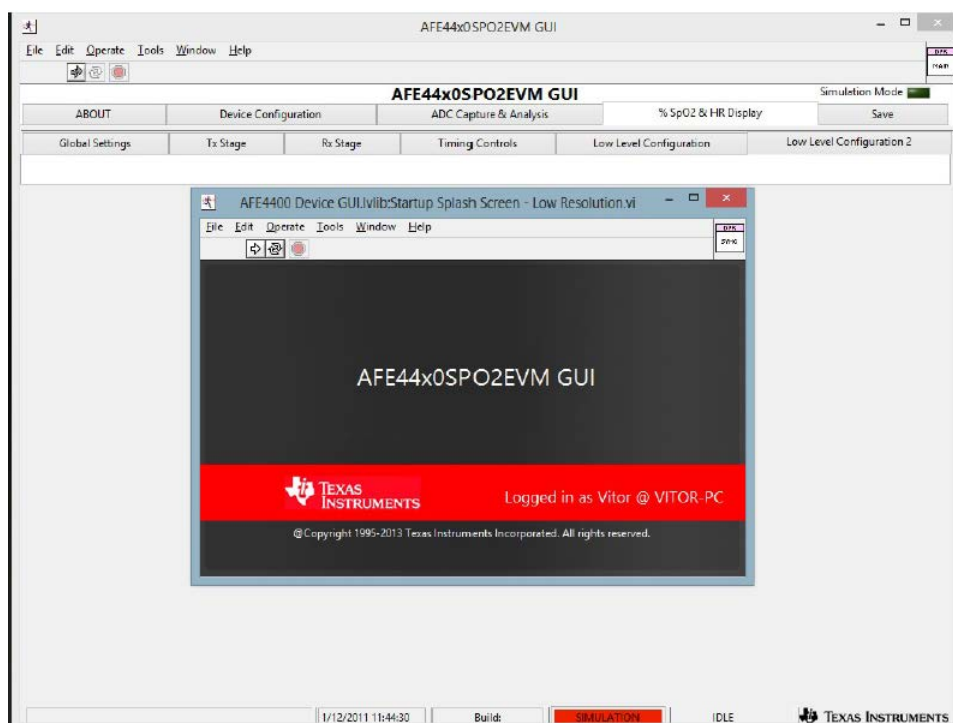


Figure 63. Broken Arrow Error

The .NET Framework 3.5 is needed for the GUI to:

- Check if Python® is installed
- Check and set environment variables needed for the scripting feature in the GUI

Points to Remember:

- There is no download for the .NET Framework 3.5 for Windows 8 or Windows 8.1. The user must enable the .NET Framework 3.5 in *Control Panel* by following the instructions provided in this article.
- Use the .NET Framework 3.5 for apps built for versions 2.0 and 3.0 as well as 3.5.
- Installing a Windows language pack before installing the .NET Framework 3.5 will cause the .NET Framework 3.5 installation to fail. Install the .NET Framework 3.5 before installing any Windows language packs. (Source: [http://msdn.microsoft.com/library/hh506443\(v=VS.110\).aspx](http://msdn.microsoft.com/library/hh506443(v=VS.110).aspx)).

Figure 64 and Section 9.5.2 detail the two methods to resolve this.

9.5.1 Method 1 (Enabling the .NET Framework 3.5 in Control Panel)

In *Control Panel*, choose *Programs and Features*, choose *Turn Windows features on or off*, and then select the *.NET Framework 3.5 (includes .NET 2.0 and 3.0)* check box. This option requires an internet connection. The user does not need to select the child items.

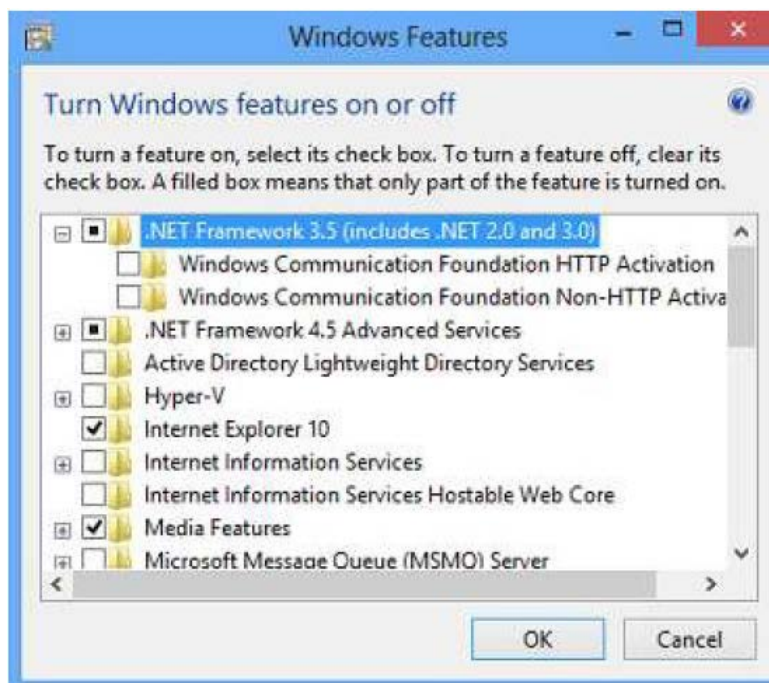


Figure 64. Method 1 (Screen 1)

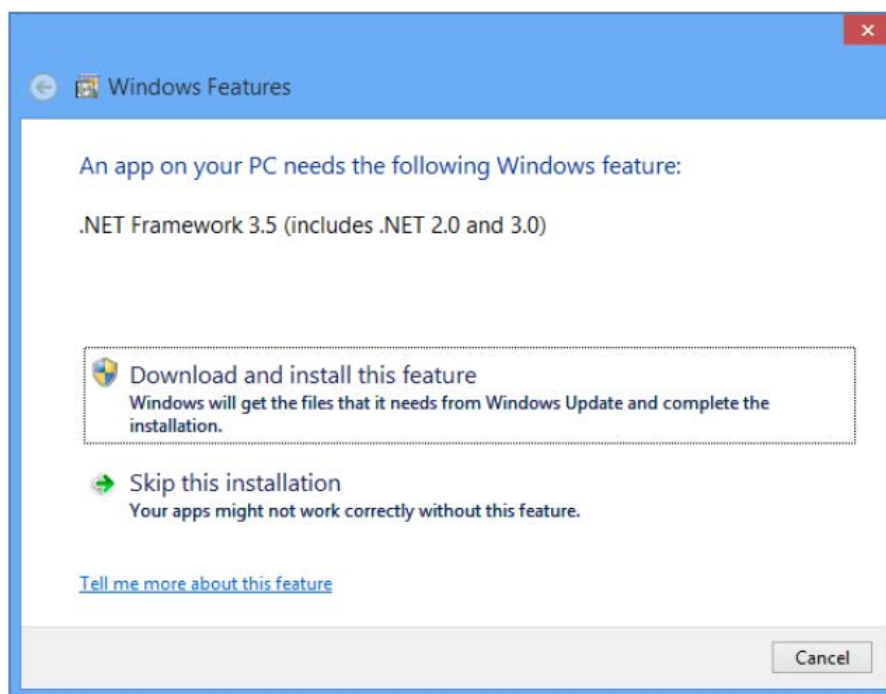


Figure 65. Method 1 (Screen 2)

Select *Download and install this feature*.

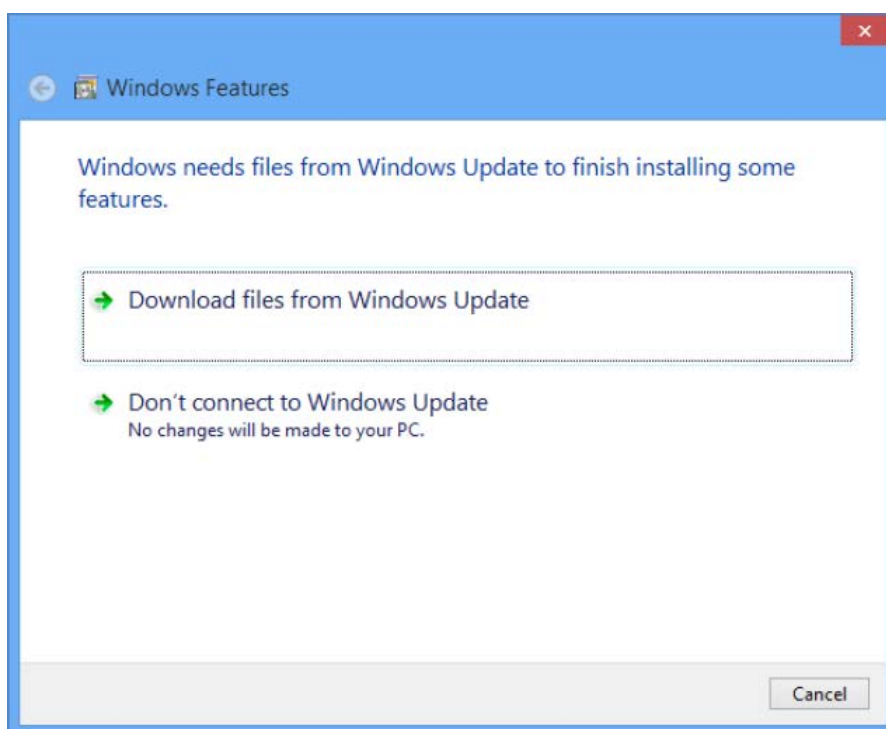


Figure 66. Method 1 (Screen 3)

Select *Download files from Windows Update*.

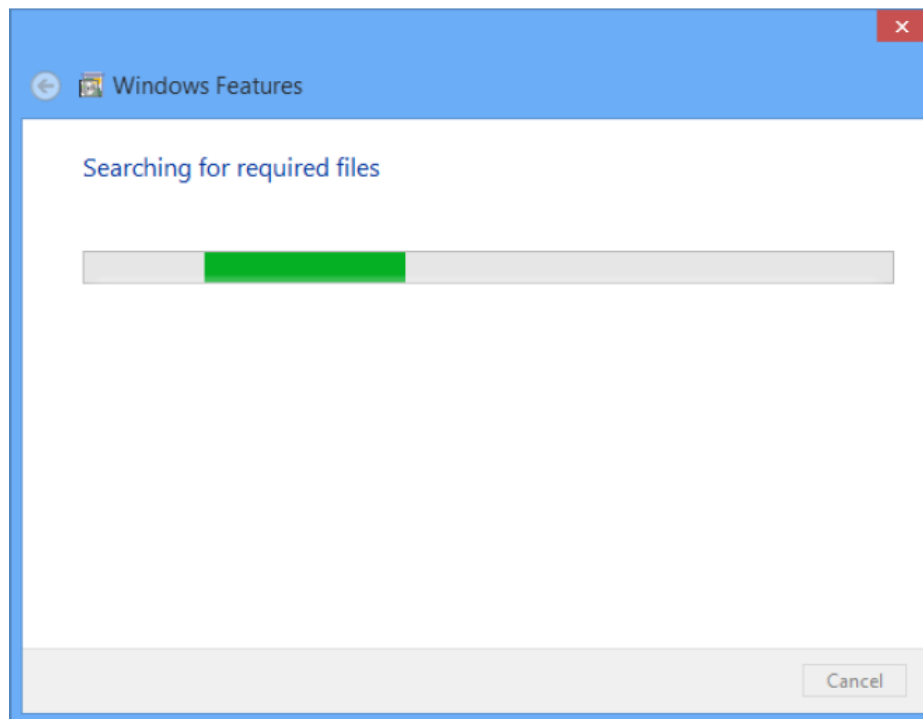
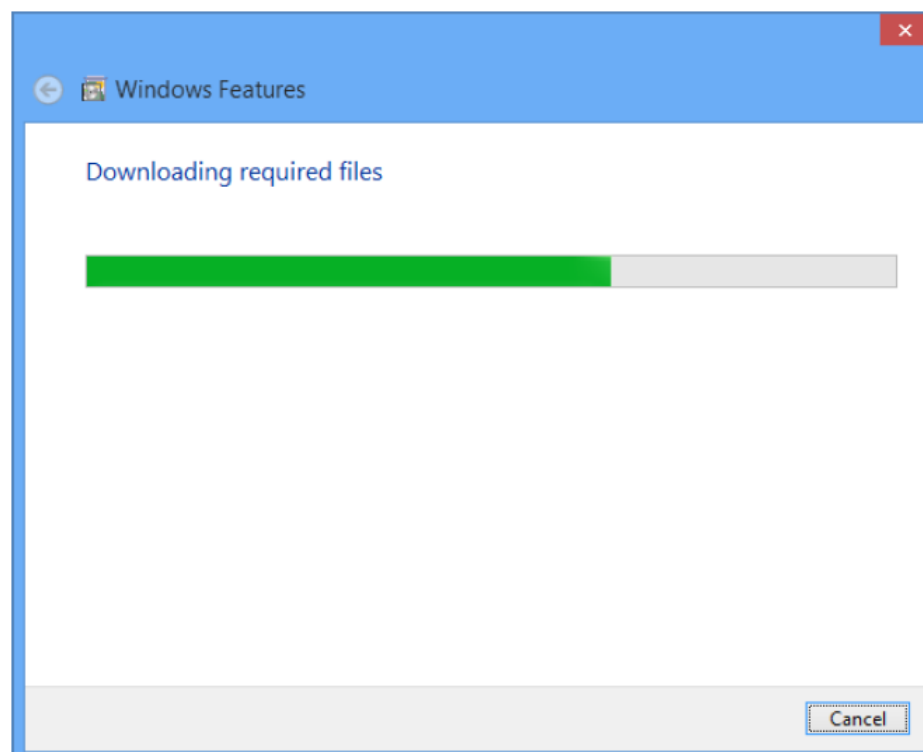
**Figure 67. Method 1 (Screen 4)****Figure 68. Method 1 (Screen 5)**



Figure 69. Method 1 (Screen 6)

9.5.2 Method 2 (Enabling .NET Framework 3.5 on Windows® 8 in Offline Mode)

This describes using the Windows 8 CD to enable and install the .NET Framework 3.5 in the PC. This method does not require an internet connection.

- Step 1. Insert Windows 8 DVD or mount ISO image. The source of this feature can be found in folder E:\sources\sxs. (In this case E: is the user's drive letter on which the user has loaded Windows 8 Media.)

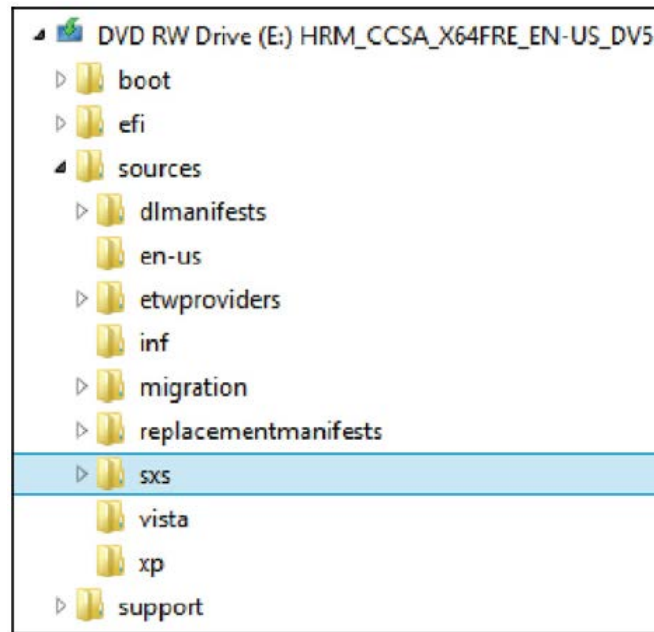


Figure 70. Method 2 (Screen 1)

- Step 2. Open the *Command prompt* as administrator.

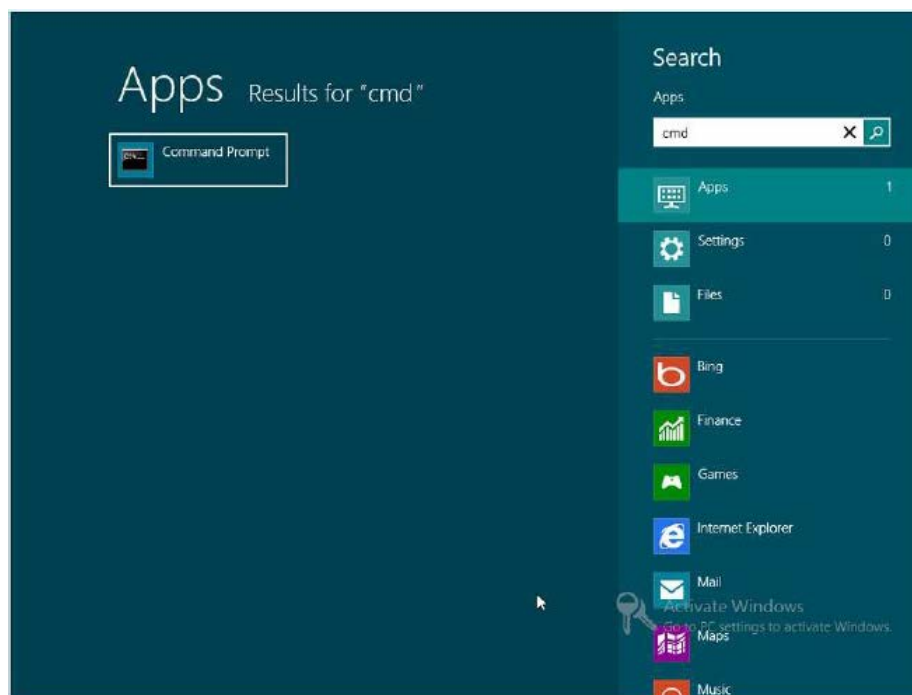


Figure 71. Method 2 (Screen 2)

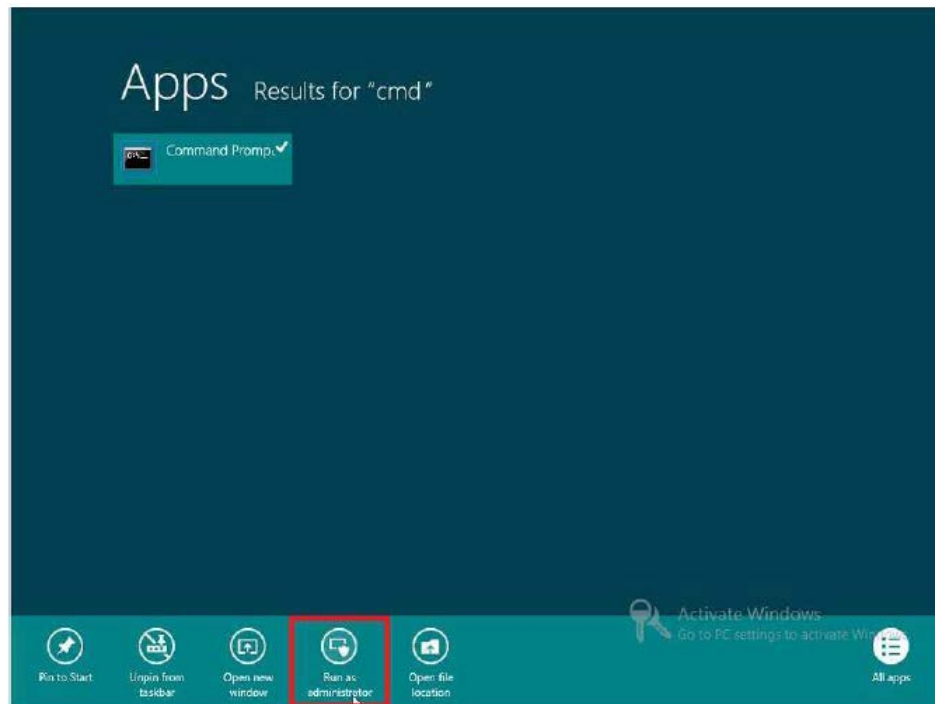


Figure 72. Method 2 (Screen 3)

- Step 3. Run the following command `Dism.exe /online /enable-feature /featurename:NetFX3 /All /Source:E:\sources\sxs /LimitAccess`, and hit Enter. Make sure to choose the appropriate drive letter (in this case it is E:\).

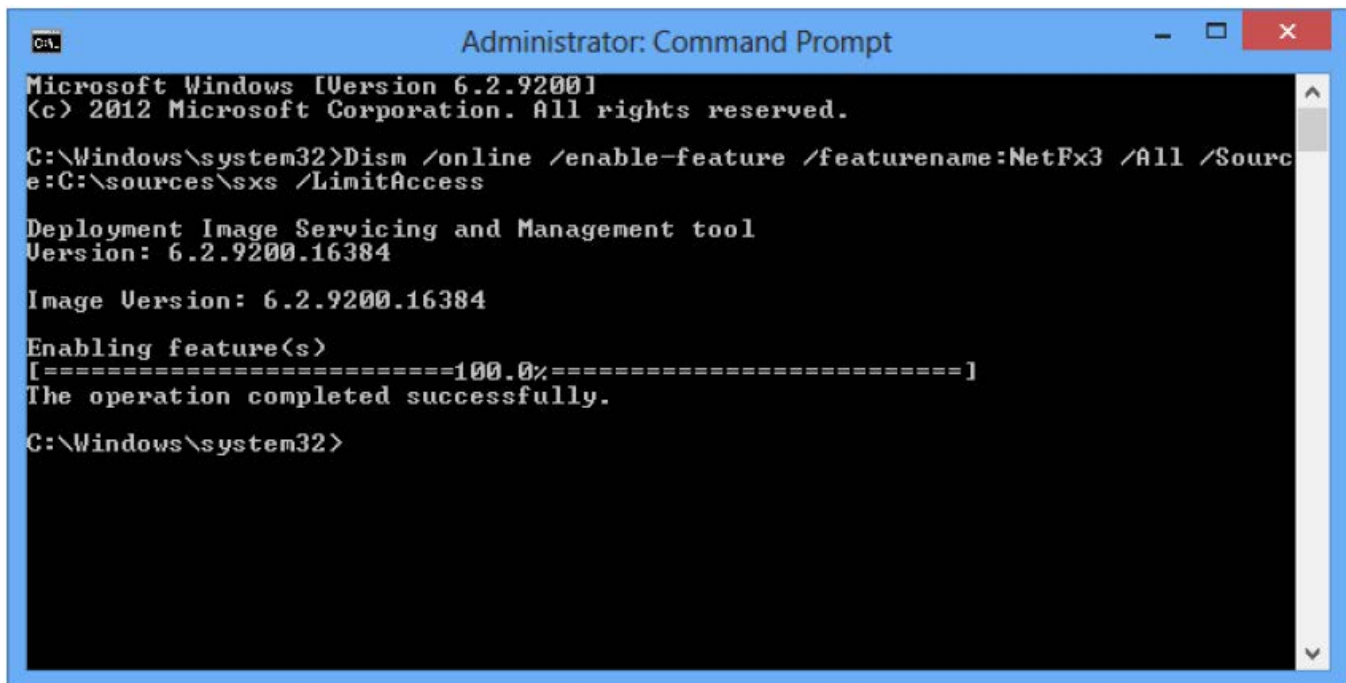


Figure 73. Method 2 (Screen 4)

Method 2 source: <http://support.microsoft.com/kb/2785188>

Table 7. Troubleshoot and Links

Description	Link
Installing the .NET Framework 3.5 on Windows 8 or 8.1	http://msdn.microsoft.com/library/hh506443(v=VS.110).aspx
Enable .NET Framework 3.5 on Windows 8 in Offline Mode	http://support.microsoft.com/kb/2785188
.NET Framework 3.5 installation error: 0x800F0906, 0x800F081F, 0x800F0907	http://support.microsoft.com/kb/2734782
Helpful link	http://comps-tech-solution.blogspot.in/2013/09/how-to-install-net-framework-35-in.html

9.6 COM Port

It has been observed that on certain machines, the GUI will not work for lower COM ports. When the GUI and the USB drivers are installed correctly and the *Device Manager* shows the AFE44x0SPO2EVM recognized as a virtual COM port, but the GUI cannot establish communication to the AFE44x0SPO2EVM and shows the Device Communication Error, change the COM port to a higher number (greater than 25).

10 Bill of Materials

Table 8 lists the bill of materials.

Table 8. AFE4900 EVM Bill of Materials⁽¹⁾

Designator	Description	Manufacturer	Part Number	Quantity
!PCB	Printed Circuit Board	Any	MHR042	1
C1, C4, C46, C48	CAP, CERM, 22 μ F, 10 V, $\pm$ 20%, X5R, 0805	Murata	GRM21BR61A226ME44L	4
C2, C9, C14	CAP, CERM, 1 μ F, 50 V, $\pm$ 10%, X5R, 0603	Murata	GRM188R61H105KAALD	3
C3, C21, C23	CAP, CERM, 1000 pF, 10 V, $\pm$ 10%, X5R, 0402	Murata	GRM155R61A102KA01D	3
C5, C12, C13	CAP, CERM, 10 μ F, 6.3 V, $\pm$ 20%, X5R, 0603	Murata	GRM188R60J106ME47D	3
C6, C7, C11	CAP, CERM, 10 μ F, 25 V, $\pm$ 20%, X5R, 0603	TDK	C1608X5R1E106M080AC	3
C10	CAP, TA, 22 μ F, 6.3 V, $\pm$ 20%, 5.4 ohm, SMD	AVX	TLJN226M006R5400	1
C15, C16	CAP, CERM, 10 μ F, 6.3 V, $\pm$ 20%, X5R, 0402	Murata	GRM155R60J106ME44D	2
C17	CAP, CERM, 100 pF, 50 V, $\pm$ 1%, C0G/NP0, 0603	AVX	06035A101FAT2A	1
C20, C22, C26, C27, C28, C29, C32, C33, C35, C36, C37, C38, C41, C42, C49, C51, C62, C71, C74, C79	CAP, CERM, 0.1 μ F, 16 V, $\pm$ 10%, X7R, 0402	Murata	GRM155R71C104KA88D	20
C30	CAP, CERM, 2200 pF, 50 V, $\pm$ 10%, X7R, 0402	Murata	GRM155R71H222KA01D	1
C39, C40	CAP, CERM, 4700 pF, 50 V, $\pm$ 10%, X7R, 0603	Murata	GRM188R71H472KA01D	2
C44, C52, C53, C54, C55, C56, C65	CAP, CERM, 10 μ F, 16 V, $\pm$ 10%, X5R, 0805	Murata	GRM21BR61C106KE15L	7
C57, C61	CAP, CERM, 0.01 μ F, 50 V, $\pm$ 10%, C0G/NP0, 0402	Murata	GCM155R71H103KA55D	2
C63	CAP, CERM, 4.7 μ F, 10 V, $\pm$ 10%, X5R, 0402	TDK	C1005X5R1A475K050BC	1
C69, C70	CAP, CERM, 12 pF, 50 V, $\pm$ 5%, C0G/NP0, 0402	Murata	GRM1555C1H120JA01D	2
C72	CAP, CERM, 2.2 μ F, 10 V, $\pm$ 10%, X5R, 0603	Kemet	C0603C225K8PACTU	1
C73	CAP, CERM, 0.47 μ F, 10 V, $\pm$ 10%, X5R, 0402	Murata	GRM155R61A474KE15D	1
C75, C76, C77, C78	CAP, CERM, 10 pF, 50 V, $\pm$ 5%, C0G/NP0, 0402	Murata	GRM1555C1H100JA01D	4
D1	Diode, Schottky, 40 V, 0.35 A, SOD-123	Diodes Inc.	SD103AW-7-F	1
H1, H2	Will add component to BOM. Useful for cables, nuts, and so forth, not in libraries	Used in BOM report	Used in BOM report	2
J1, J2, J4, J5, J6, J7, J10, J11, J13, J14, J15, J16, J17, J18, J19	Header, 2.54 mm, 2x1, Tin, TH	Samtec	TSW-102-07-T-S	15
J3, J9	Header, 2.54 mm, 3x1, Tin, TH	Samtec	TSW-103-07-T-S	2
J8	Header, 100 mil, 2x2, Gold, TH	Samtec	TSW-102-07-G-D	1
J12	Connector, Receptacle, USB Micro B, R/A, SMT	FCI	10118193-0001LF	1
J20	Receptacle, D-Sub, 9 Position, R/A, TH	TE Connectivity	1734354-1	1
J22	Receptacle, 50mil, 10x1, Gold, R/A, TH	Mill-Max	851-43-010-20-001000	1
L1	Inductor, Wirewound, Ferrite, 2.2 μ H, 1.8 A, 0.08 Ω , SMD	Würth Elektronik	74404024022	1
L2	Inductor, Shielded, Powdered Iron, 1.5 μ H, 2.1 A, 0.095 Ω , SMD	Murata Toko	1286AS-H-1R5M=P2	1
L3, L7	Coupled inductor, 0.37 A, 0.3 Ω , SMD	Murata	DLW31SN900SQ2L	2
L4, L5, L6, L10	Ferrite Bead, 120 Ω @ 100 MHz, 0.5 A, 0603	Murata	BLM18AG121SN1D	4
L11	Inductor, Shielded, Ferrite, 10 μ H, 0.48 A, 0.54 Ω , SMD	Coilcraft	LPS3010-103MRB	1
LED1, LED3	LED, Green, SMD	Lumex	SML-LX0603GW-TR	2
LED2	LED, Super Red, SMD	Lumex	SML-LX0603SRW-TR	1

⁽¹⁾ Substitutions made according to Alternate Part Number must meet data sheet parameters, not just the Description specified.

Table 8. AFE4900 EVM Bill of Materials⁽¹⁾ (continued)

Designator	Description	Manufacturer	Part Number	Quantity
Q1	Transistor, NPN, 25 V, 0.05 A, SOT-23	ON Semiconductor	MMBT5089LT1G	1
R2	RES, 1.00 M, 1%, 0.1 W, 0603	Vishay-Dale	CRCW06031M00FKEA	1
R3	RES, 255 k, 1%, 0.1 W, 0603	Vishay-Dale	CRCW0603255KFKEA	1
R4, R11, R13	RES, 0, 5%, 0.1 W, 0603	Panasonic	ERJ-3GEY0R00V	3
R5, R8, R9, R40, R57, R63, R71, R84, R119, R133, R134	RES, 0, 5%, 0.063 W, 0402	Vishay-Dale	CRCW04020000Z0ED	11
R10	RES, 47, 5%, 0.1 W, 0603	Vishay-Dale	CRCW060347R0JNEA	1
R12, R17, R23, R33, R34, R41, R43, R96, R97, R98, R99, R100	RES, 0, 5%, 0.1 W, 0603	Vishay-Dale	CRCW06030000Z0EA	12
R14, R118, R128, R130	RES, 4.7 k, 5%, 0.063 W, 0402	Vishay-Dale	CRCW04024K70JNED	4
R19, R20, R64	RES, 100 k, 1%, 0.1 W, 0603	Vishay-Dale	CRCW0603100KFKEA	3
R21, R22	RES, 5.1 M, 5%, 0.1 W, 0603	Vishay-Dale	CRCW06035M10JNEA	2
R24, R27, R29, R30, R32, R35, R36, R37, R38, R46, R47, R48, R49, R52, R53, R56, R58, R61, R66, R69, R70	RES, 0, 5%, 0.05 W, 0201	Panasonic	ERJ-1GE0R00C	21
R26	RES, 499, 1%, 0.063 W, 0402	Vishay-Dale	CRCW0402499RFKED	1
R39, R85, R104	RES, 47 k, 5%, 0.063 W, 0402	Vishay-Dale	CRCW040247K0JNED	3
R50, R51	RES, 620 k, 1%, 0.1 W, 0603	Yageo America	RC0603FR-07620KL	2
R67, R112	RES, 33 k, 5%, 0.063 W, 0402	Vishay-Dale	CRCW040233K0JNED	2
R74	RES, 10.0 k, 1%, 0.063 W, 0402	Vishay-Dale	CRCW040210K0FKED	1
R75, R129, R131	RES, 10.0, 1%, 0.063 W, 0402	Vishay-Dale	CRCW040210R0FKED	3
R76	RES, 1.40 k, 1%, 0.063 W, 0402	Vishay-Dale	CRCW04021K40FKED	1
R77, R78, R82, R83, R86, R87, R88, R92, R93, R95	RES, 10.0 k, 1%, 0.1 W, 0402	Panasonic	ERJ-2RKF1002X	10
R79, R132	RES, 1.00 M, 1%, 0.063 W, 0402	Vishay-Dale	CRCW04021M00FKED	2
R80	RES, 953 k, 1%, 0.063 W, 0402	Vishay-Dale	CRCW0402953KFKEA	1
R81	RES, 180 k, 5%, 0.063 W, 0402	Vishay-Dale	CRCW0402180JNED	1
R102	RES, 75.0 k, 1%, 0.063 W, 0402	Vishay-Dale	CRCW040275K0FKED	1
R113, R125	RES, 100, 1%, 0.063 W, 0402	Vishay-Dale	CRCW0402100RFKED	2
R116	RES, 130, 5%, 0.063 W, 0402	Vishay-Dale	CRCW0402130RJNED	1
R117	RES, 220, 5%, 0.063 W, 0402	Vishay-Dale	CRCW0402220RJNED	1
R122, R123	RES, 33, 5%, 0.063 W, 0402	Vishay-Dale	CRCW040233R0JNED	2
S1, S2	Switch, Tactile, SPST-NO, 0.05 A, 12 V, SMD	C and K Components	RS-032G05A3-SM RT	2
SH-J1, SH-J3, SH-J5, SH-J6, SH-J7	Shunt, 100 mil, Gold plated, Black	3M	969102-0000-DA	5
TP1, TP2, TP10, TP14	Test Point, Miniature, Black, TH	Keystone	5001	4
TP3, TP4, TP5, TP6, TP7, TP8, TP9, TP11, TP12, TP13	Test Point, Miniature, White, TH	Keystone	5002	10
U1, U2, U3, U4, U5, U6, U10, U11	ESD in 0402 Package with 10 pF Capacitance and 6 V Breakdown, 1 Channel, -40 to +125 °C, 2-pin X2SON (DPY), Green (RoHS & no Sb/Br)	Texas Instruments	TPD1E10B06DPYR	8
U7	Tiny Single Inductor Buck Boost Converter, YFF0012AFAP (DSBGA-12)	Texas Instruments	TPS63050YFFR	1
U8	Synchronous Boost Converter with 800 nA Ultra-Low Quiescent Current, YFF0006AFAD (DSBGA-6)	Texas Instruments	TPS61099YFFR	1

Table 8. AFE4900 EVM Bill of Materials⁽¹⁾ (continued)

Designator	Description	Manufacturer	Part Number	Quantity
U9	250-mA, Ultralow IQ, Fast Transient Response, RF Low-Dropout Linear Regulator, YFF0004AUAB_A1 (DSBGA-4)	Texas Instruments	TPS72718YFFR	1
U12	Ultra-Small, Integrated AFE with FIFO for Wearable, Optical Heart-Rate Monitoring and Bio-Sensing, YZ0030-C01 (DSBGA-30)	Texas Instruments	AFE4900YZR	1
U13	2-Bit Bidirectional Voltage-Level Shifter with Auto Direction Sensing and ± 15 -kV ESD Protect, YZP0008ADAD (DSBGA-8)	Texas Instruments	TXB0102YZPR	1
U14, U15	4-Bit Bidirectional Voltage-Level Shifter with Auto Direction Sensing and ± 15 kV ESD Protect, YZT0012ACAC (DSBGA-12)	Texas Instruments	TXB0104YZTR	2
U16	2-Bit Bidirectional Voltage-Level Shifter for Open-Drain and Push-Pull Application, YZP0008ADCD (DSBGA-8)	Texas Instruments	TXS0102YZPR	1
U17	Dual Input Li-Ion Charger with Dynamic Power Path, Output regulated to 4.4 V, Vbat 4.2 V, 4.3 V, -40 to 85 °C, 20-pin QFN (RHL), Green (RoHS & no Sb/Br)	Texas Instruments	BQ24032ARHLR	1
U18	Single Output High PSRR LDO, 150 mA, Adjustable 1.2 to 33 V Output, 3 to 36 V Input, with Ultra-Low Noise, 8-pin MSOP (DGN), -40 to 125 °C, Green (RoHS & no Sb/Br)	Texas Instruments	TPS7A4901DGNT	1
U20	ESD-Protection Array for High-Speed Data Interfaces, 4 Channels, -40 to +85 °C, 6-pin SON (DRY), Green (RoHS & no Sb/Br)	Texas Instruments	TPD4E004DRYR	1
U21	Voltage Monitor with Manual Reset, DBV0005A (SOT-5)	Texas Instruments	TPS3825-33DBVR	1
U22	25 MHz Mixed Signal Microcontroller with 128 KB Flash, 8192 B SRAM and 63 GPIOs, -40 to 85 °C, 80-pin QFP (PN), Green (RoHS & no Sb/Br)	Texas Instruments	MSP430F5529IPN	1
Y1	CRYSTAL, 32.768 kHz, 12.5PF, SMD	Abracon Corporation	ABS07-32.768KHZ-T	1
Y2	Crystal, 24 MHz, 10 pF, SMD	Abracon Corporation	ABM3B-24.000MHZ-10-1-U-T	1

[Table 9](#) lists the sensor board bill of materials.

Table 9. Sensor Board Bill of Materials

Item	Designator	Description	Manufacturer	Part Number	Quantity
1	C1	CAP, CERM, 33 μ F, 10 V, $\pm$ 20%, X5R, 0805	TDK	C2012X5R1A336M125AC	1
2	FID1, FID2, FID3	Fiducial mark. There is nothing to buy or mount.	N/A	N/A	3
3	H1	UC-Rubber Cover custom fitted to sensor: Post Build	3M	SJ5832	1
4	J1	Header, 50mil, 10x1, R/A, TH	Mill-Max	850-10-010-20-001000	1
5	R1, R2	RES, 0, 5%, 0.05 W, 0201	Panasonic	ERJ-1GE0R00C	2
6	U1	BioMon Sensor, SMD	OSRAM	SFH7072	1

11 PCB Layouts and Schematics

11.1 AFE4900 EVM PCB Layouts

Figure 74 through Figure 79 show the EVM PCB layouts.

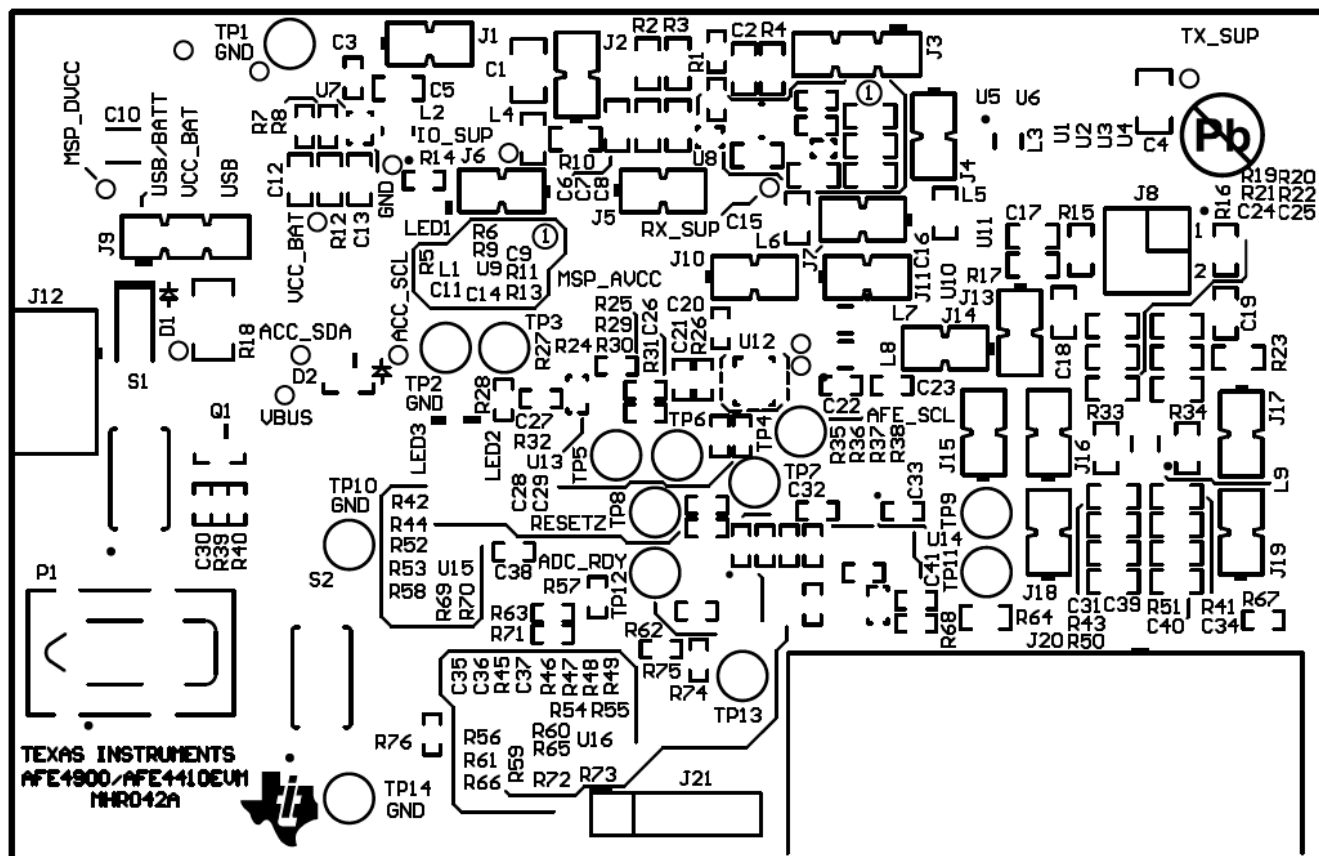


Figure 74. AFE4900 EVM Top Overlay

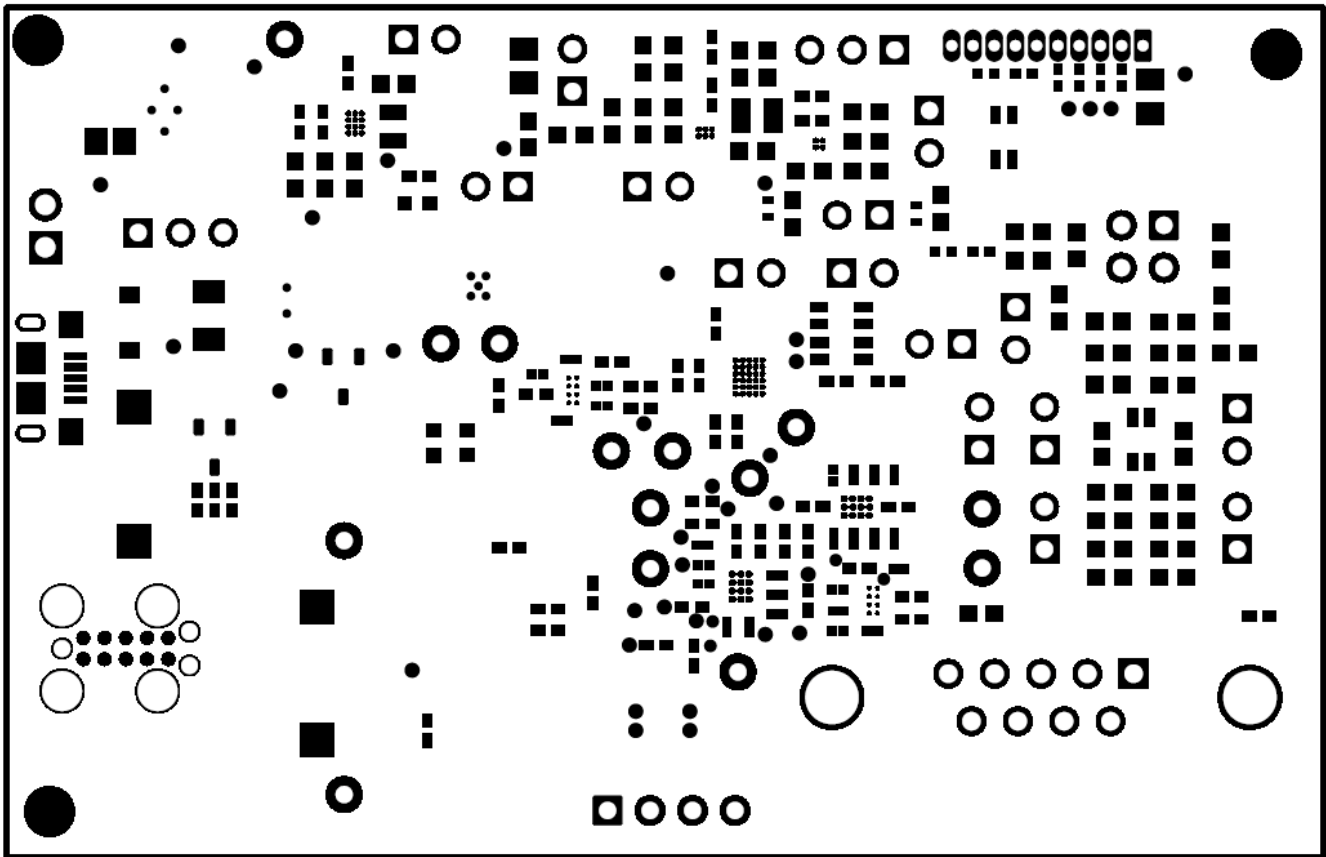


Figure 75. Top Solder

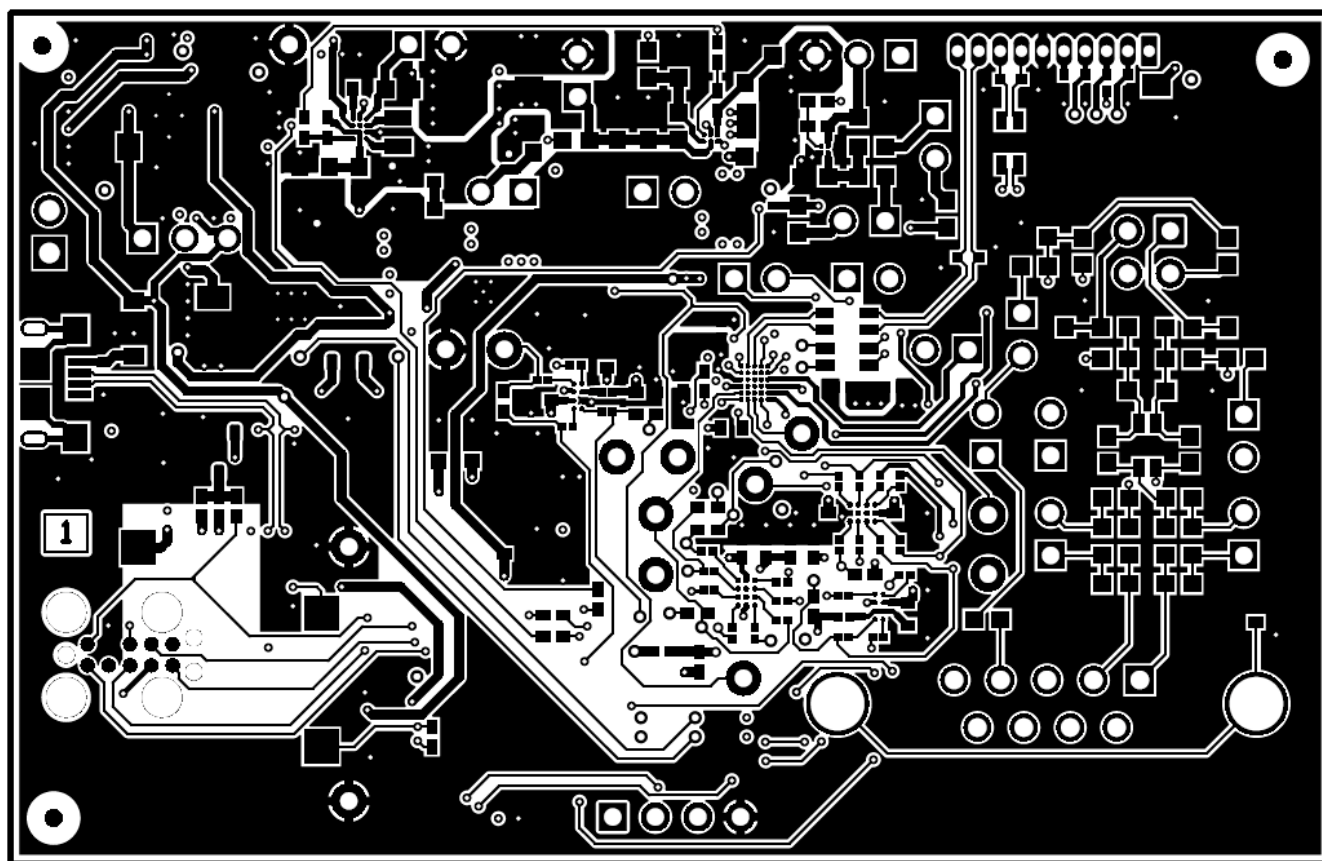


Figure 76. Top Copper (Layer 1)

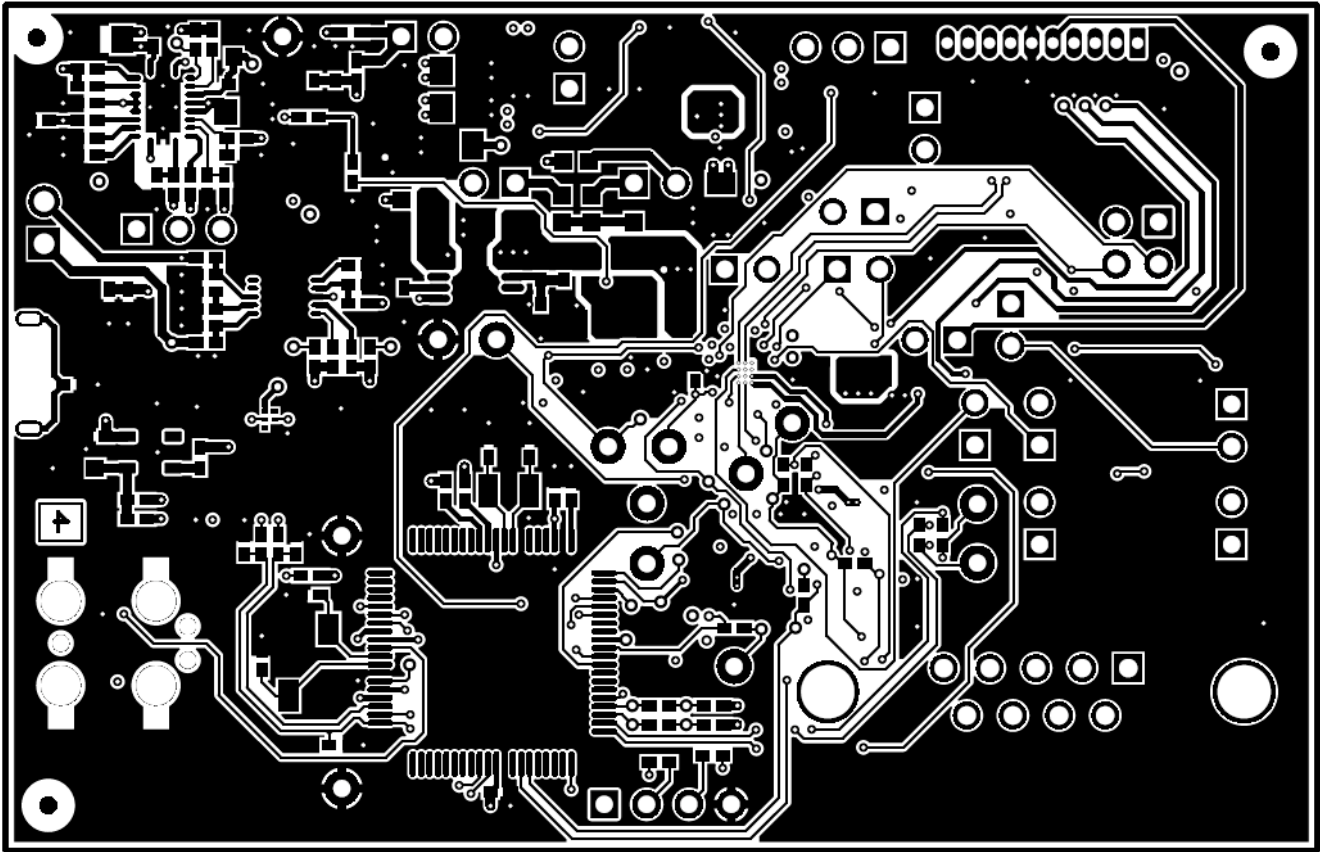


Figure 77. Bottom Copper (Layer 4)

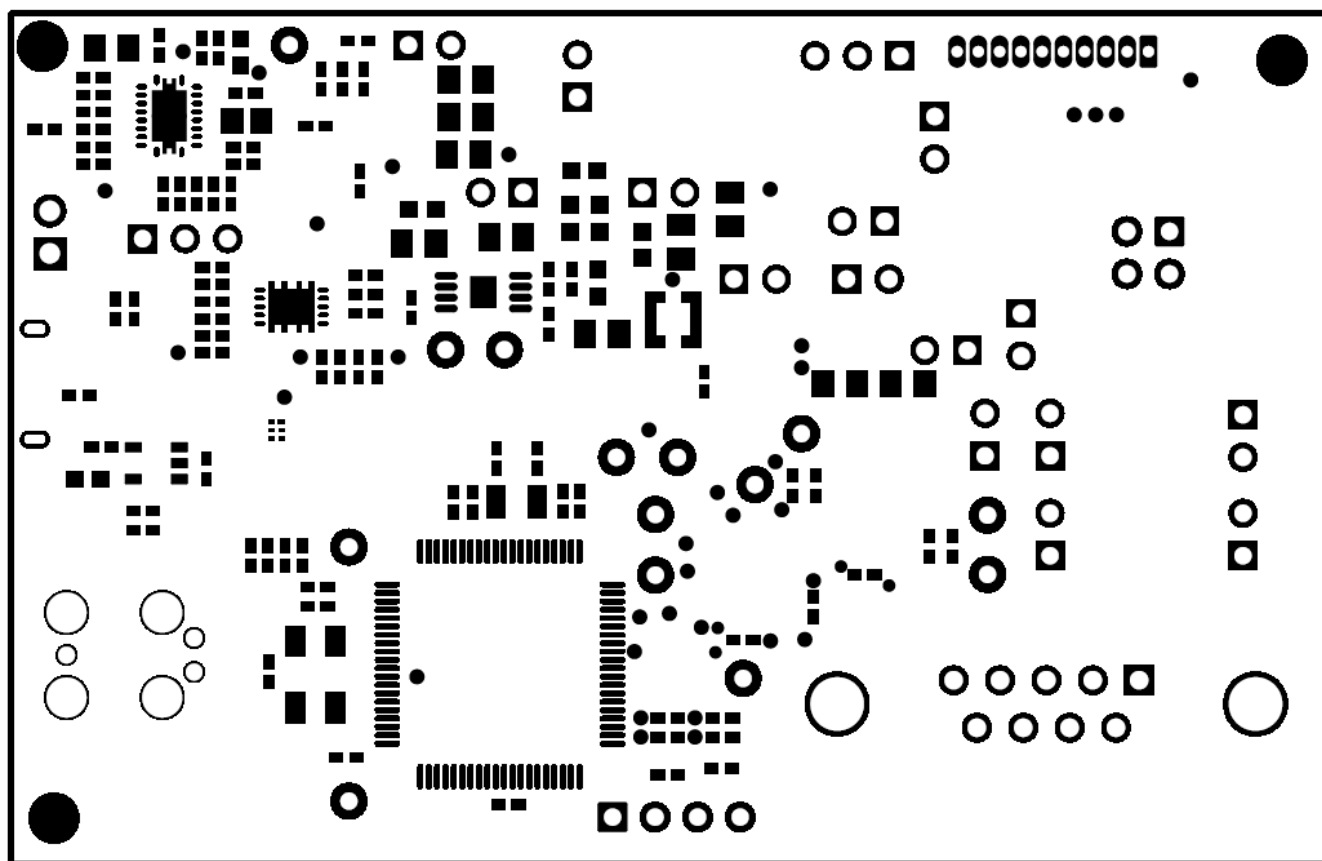


Figure 78. Bottom Solder

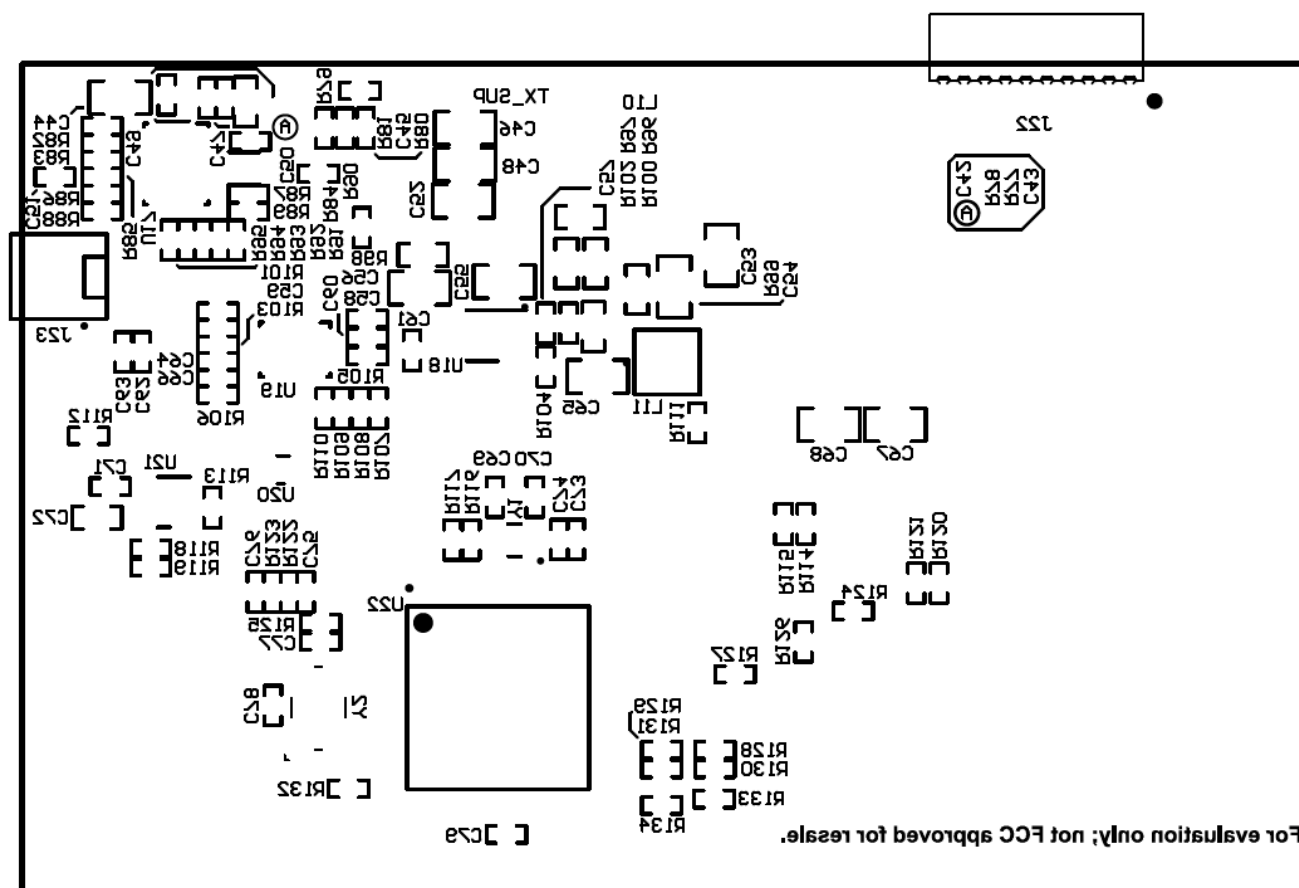


Figure 79. Bottom Overlay

11.2 Sensor Board Layouts

Figure 80 through Figure 85 show the sensor board layouts.

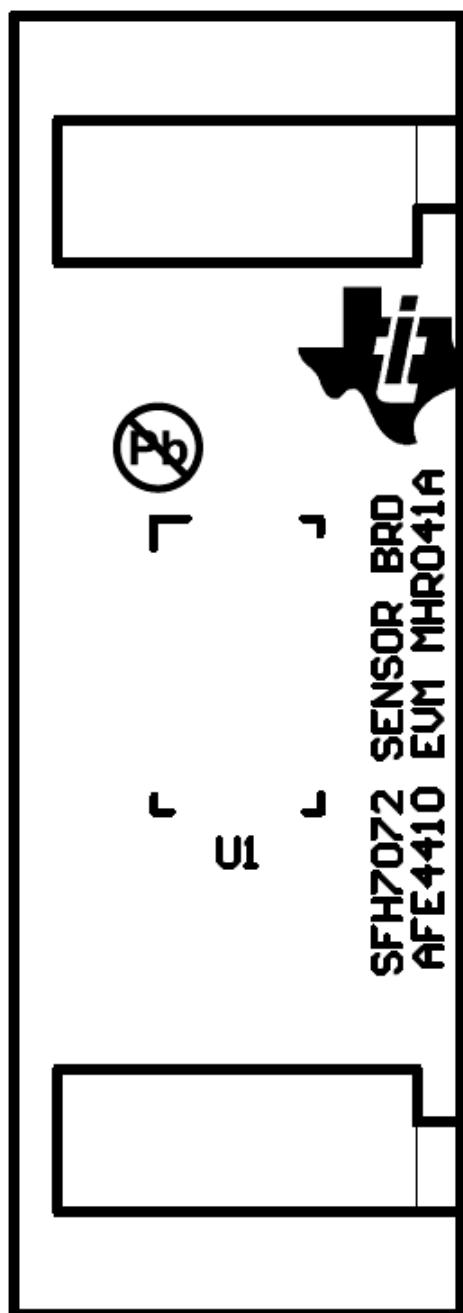


Figure 80. Sensor Board Top Silk Screen

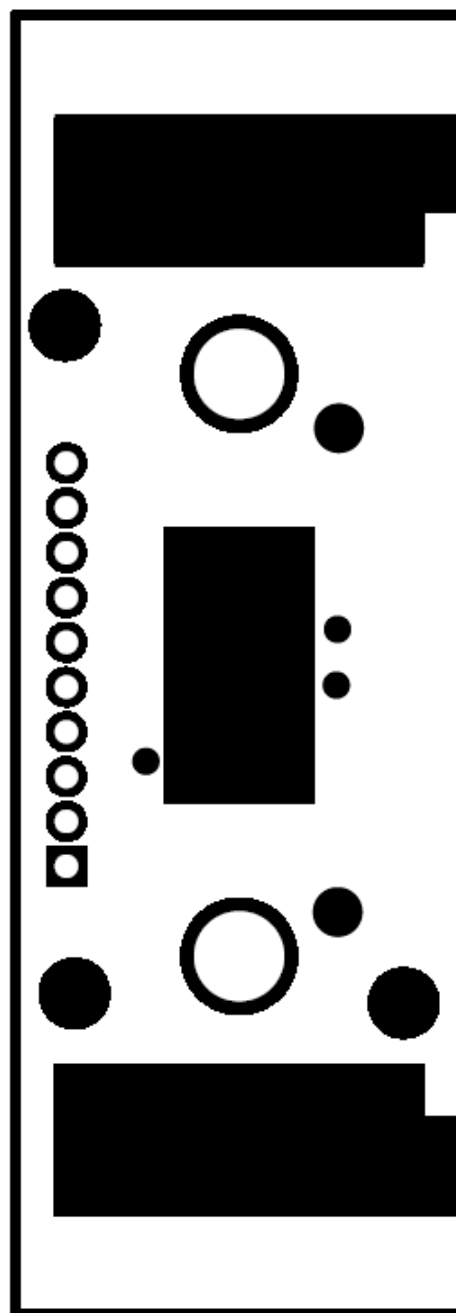


Figure 81. Sensor Board Top Solder Mask

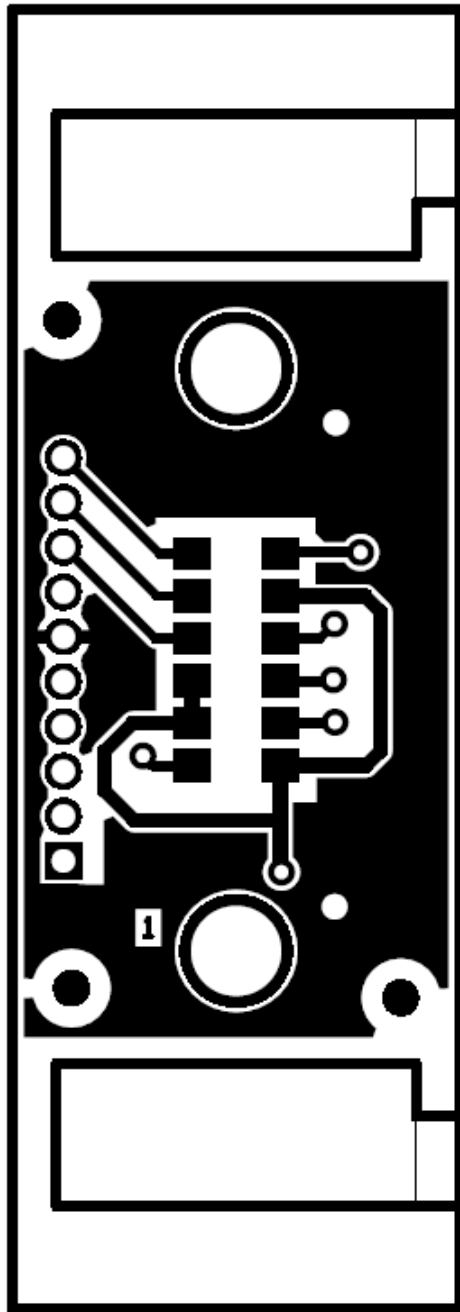


Figure 82. Sensor Board Top Copper

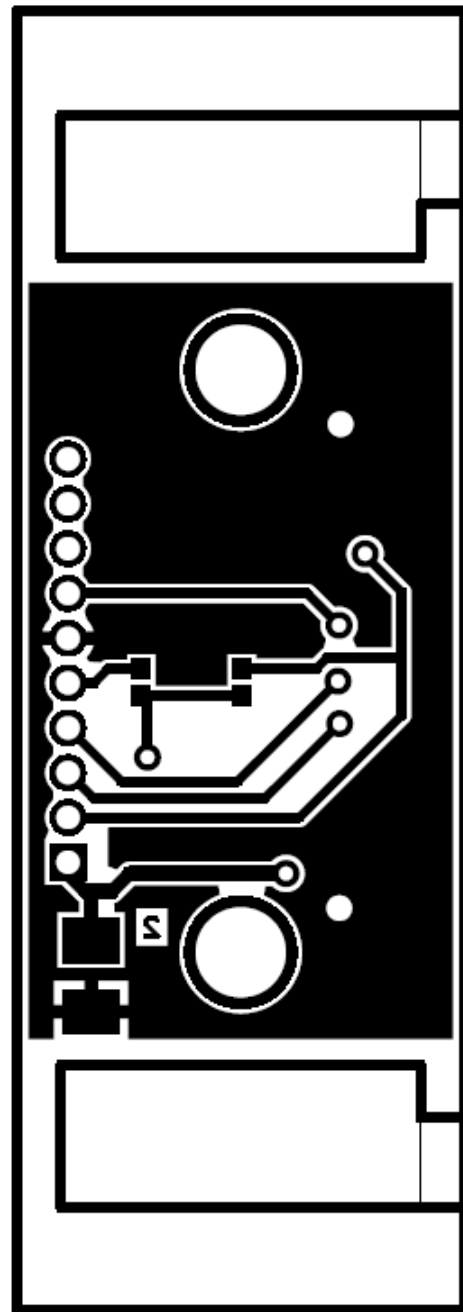


Figure 83. Sensor Board Bottom Copper

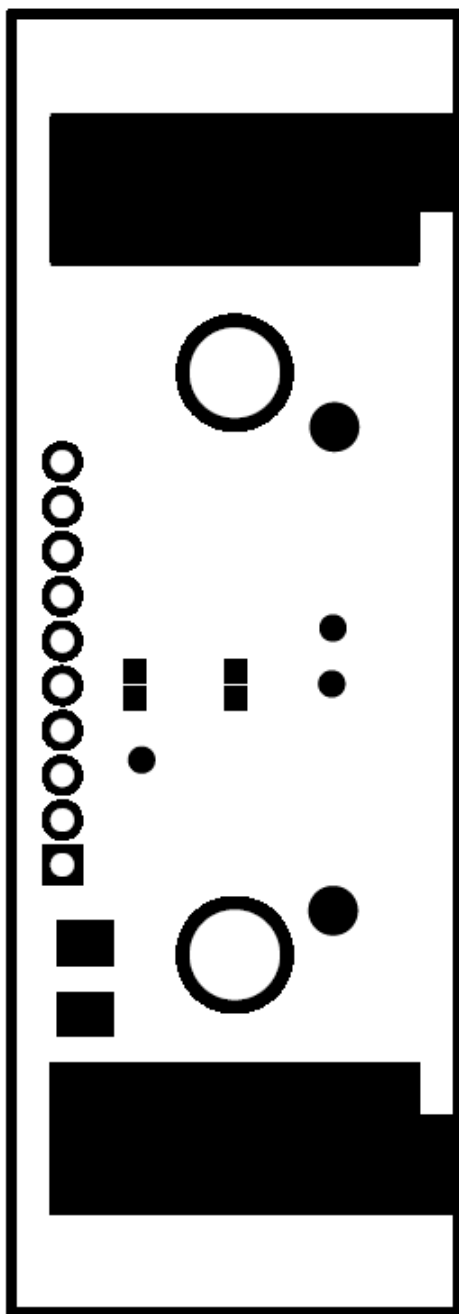


Figure 84. Sensor Board Bottom Solder Mask

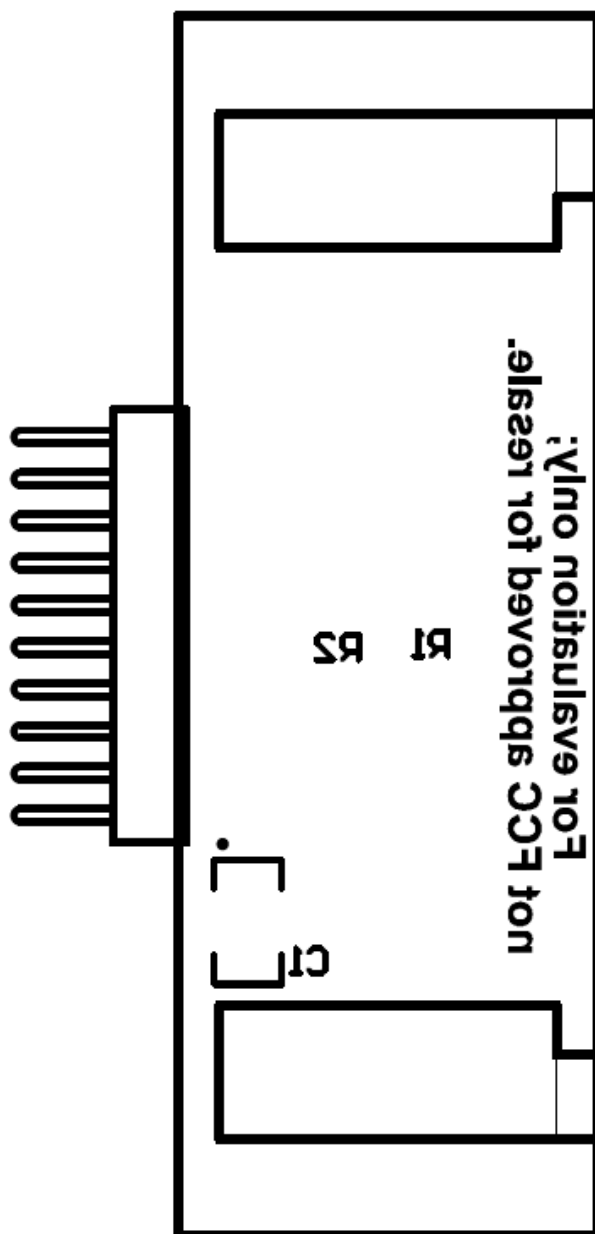


Figure 85. Sensor Board Bottom Silk Screen

11.3 Schematics

This section includes the AFE4900 EVM and sensor board schematics.

11.3.1 AFE4900 EVM Schematics

Figure 86 through Figure 90 illustrate the AFE4900 EVM schematics.

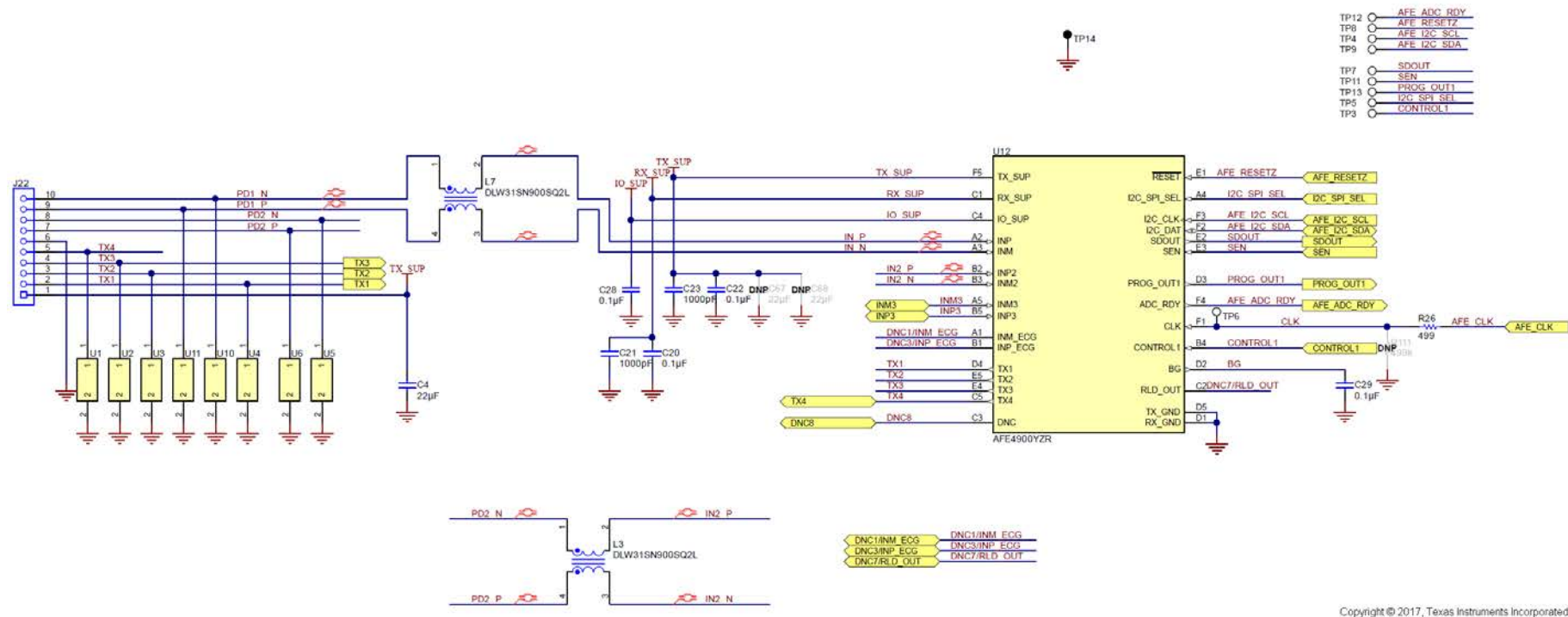
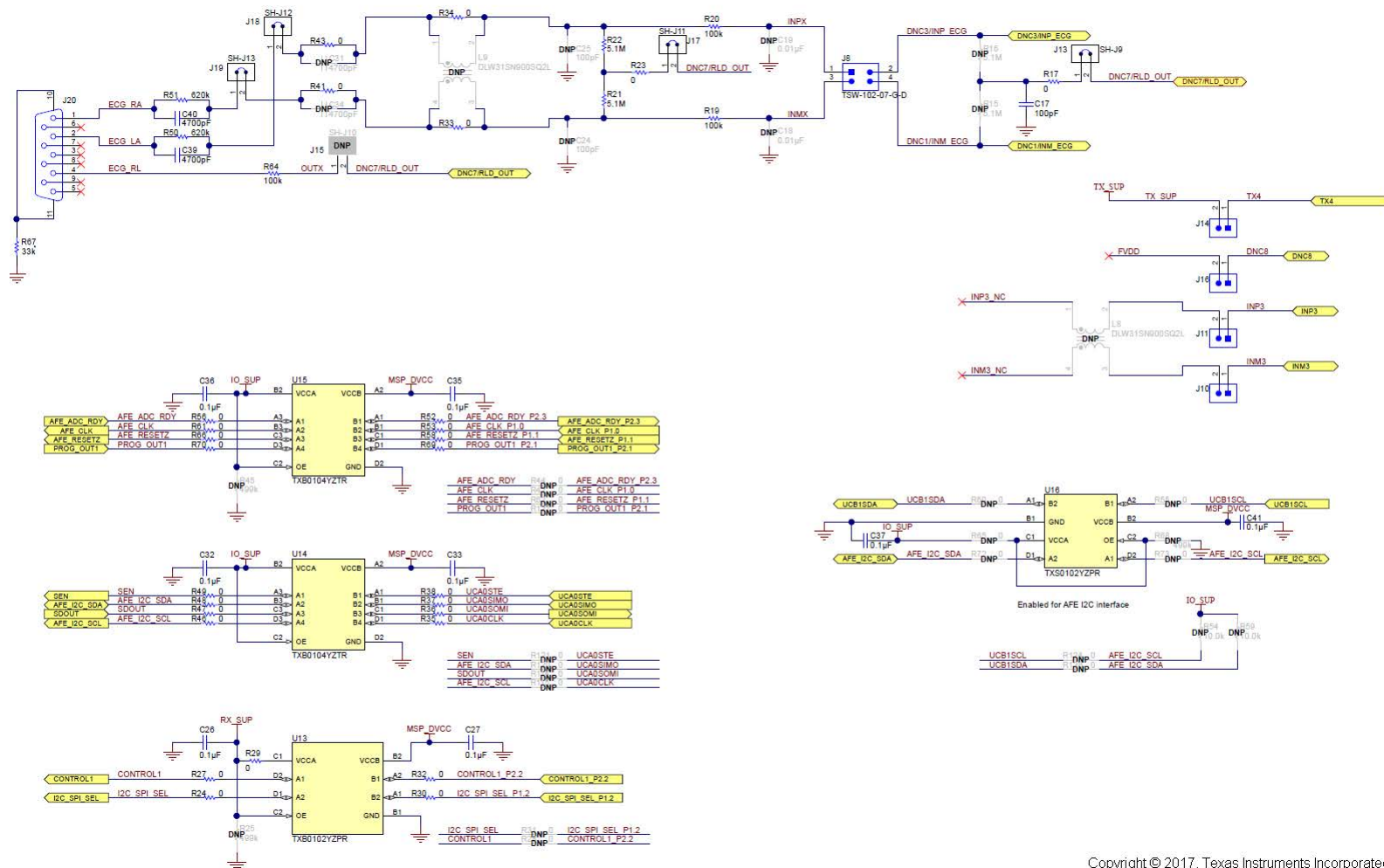


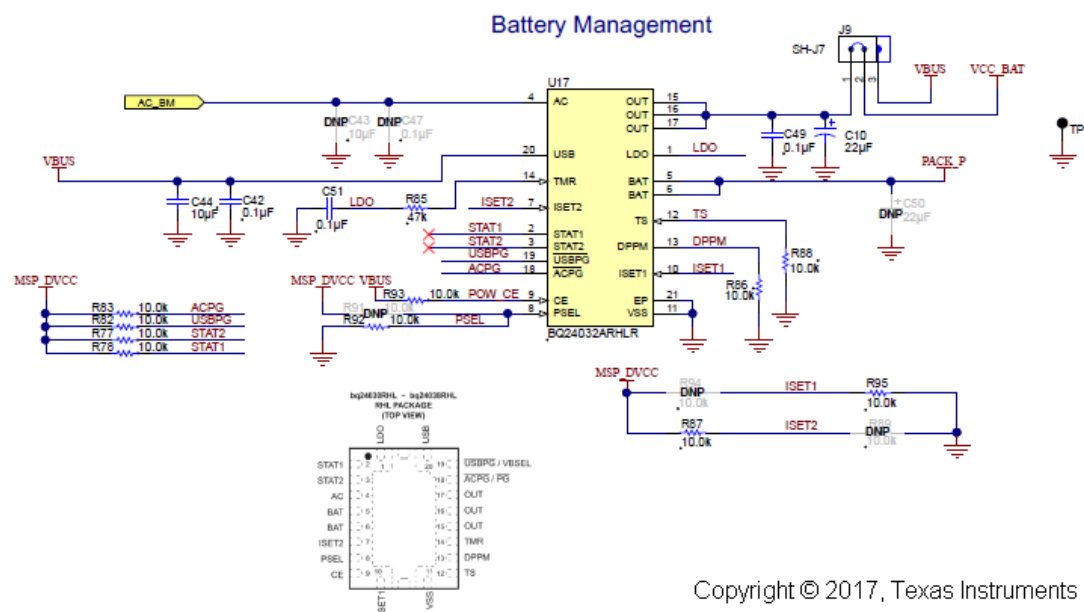
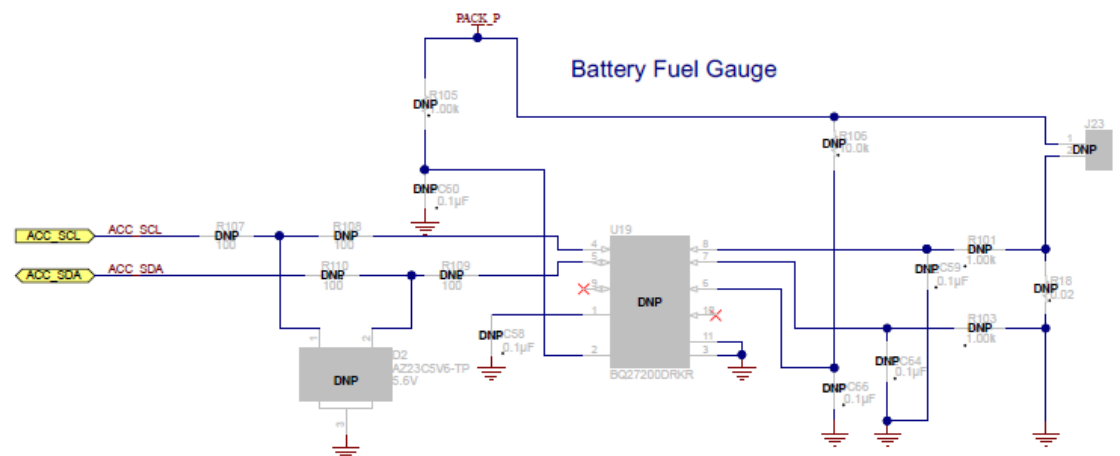
Figure 86. AFE4900 EVM Schematics (1 of 5)



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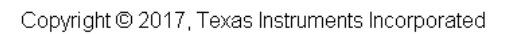
Figure 87. AFE4900 EVM Schematics (2 of 5)

82 *AFE4900 Evaluation Module*



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Figure 89. AFE4900 EVM Schematics (4 of 5)

84 *AFE4900 Evaluation Module*

11.4 Sensor Board Schematic

Figure 91 illustrates the sensor board schematic.

Optical Sensor

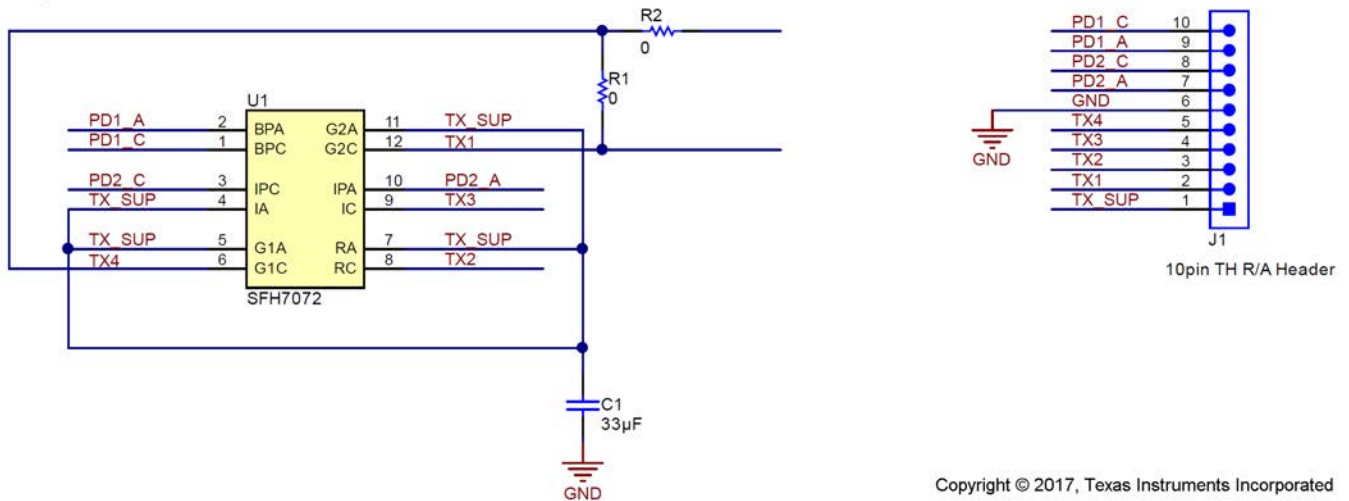


Figure 91. Sensor Board Schematic

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1. **Delivery:** TI delivers TI evaluation boards, kits, or modules, including any accompanying demonstration software, components, and/or documentation which may be provided together or separately (collectively, an "EVM" or "EVMs") to the User ("User") in accordance with the terms set forth herein. User's acceptance of the EVM is expressly subject to the following terms.
 - 1.1 EVMs are intended solely for product or software developers for use in a research and development setting to facilitate feasibility evaluation, experimentation, or scientific analysis of TI semiconductor products. EVMs have no direct function and are not finished products. EVMs shall not be directly or indirectly assembled as a part or subassembly in any finished product. For clarification, any software or software tools provided with the EVM ("Software") shall not be subject to the terms and conditions set forth herein but rather shall be subject to the applicable terms that accompany such Software
 - 1.2 EVMs are not intended for consumer or household use. EVMs may not be sold, sublicensed, leased, rented, loaned, assigned, or otherwise distributed for commercial purposes by Users, in whole or in part, or used in any finished product or production system.
2. **Limited Warranty and Related Remedies/Disclaimers:**
 - 2.1 These terms do not apply to Software. The warranty, if any, for Software is covered in the applicable Software License Agreement.
 - 2.2 TI warrants that the TI EVM will conform to TI's published specifications for ninety (90) days after the date TI delivers such EVM to User. Notwithstanding the foregoing, TI shall not be liable for a nonconforming EVM if (a) the nonconformity was caused by neglect, misuse or mistreatment by an entity other than TI, including improper installation or testing, or for any EVMs that have been altered or modified in any way by an entity other than TI, (b) the nonconformity resulted from User's design, specifications or instructions for such EVMs or improper system design, or (c) User has not paid on time. Testing and other quality control techniques are used to the extent TI deems necessary. TI does not test all parameters of each EVM. User's claims against TI under this Section 2 are void if User fails to notify TI of any apparent defects in the EVMs within ten (10) business days after delivery, or of any hidden defects with ten (10) business days after the defect has been detected.
 - 2.3 TI's sole liability shall be at its option to repair or replace EVMs that fail to conform to the warranty set forth above, or credit User's account for such EVM. TI's liability under this warranty shall be limited to EVMs that are returned during the warranty period to the address designated by TI and that are determined by TI not to conform to such warranty. If TI elects to repair or replace such EVM, TI shall have a reasonable time to repair such EVM or provide replacements. Repaired EVMs shall be warranted for the remainder of the original warranty period. Replaced EVMs shall be warranted for a new full ninety (90) day warranty period.
3. **Regulatory Notices:**
 - 3.1 **United States**
 - 3.1.1 **Notice applicable to EVMs not FCC-Approved:**

FCC NOTICE: This kit is designed to allow product developers to evaluate electronic components, circuitry, or software associated with the kit to determine whether to incorporate such items in a finished product and software developers to write software applications for use with the end product. This kit is not a finished product and when assembled may not be resold or otherwise marketed unless all required FCC equipment authorizations are first obtained. Operation is subject to the condition that this product not cause harmful interference to licensed radio stations and that this product accept harmful interference. Unless the assembled kit is designed to operate under part 15, part 18 or part 95 of this chapter, the operator of the kit must operate under the authority of an FCC license holder or must secure an experimental authorization under part 5 of this chapter.
 - 3.1.2 **For EVMs annotated as FCC – FEDERAL COMMUNICATIONS COMMISSION Part 15 Compliant:**

CAUTION

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

FCC Interference Statement for Class A EVM devices

NOTE: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

FCC Interference Statement for Class B EVM devices

NOTE: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- *Reorient or relocate the receiving antenna.*
- *Increase the separation between the equipment and receiver.*
- *Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.*
- *Consult the dealer or an experienced radio/TV technician for help.*

3.2 Canada

3.2.1 For EVMs issued with an Industry Canada Certificate of Conformance to RSS-210 or RSS-247

Concerning EVMs Including Radio Transmitters:

This device complies with Industry Canada license-exempt RSSs. Operation is subject to the following two conditions:

(1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device.

Concernant les EVMs avec appareils radio:

Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes: (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

Concerning EVMs Including Detachable Antennas:

Under Industry Canada regulations, this radio transmitter may only operate using an antenna of a type and maximum (or lesser) gain approved for the transmitter by Industry Canada. To reduce potential radio interference to other users, the antenna type and its gain should be so chosen that the equivalent isotropically radiated power (e.i.r.p.) is not more than that necessary for successful communication. This radio transmitter has been approved by Industry Canada to operate with the antenna types listed in the user guide with the maximum permissible gain and required antenna impedance for each antenna type indicated. Antenna types not included in this list, having a gain greater than the maximum gain indicated for that type, are strictly prohibited for use with this device.

Concernant les EVMs avec antennes détachables

Conformément à la réglementation d'Industrie Canada, le présent émetteur radio peut fonctionner avec une antenne d'un type et d'un gain maximal (ou inférieur) approuvé pour l'émetteur par Industrie Canada. Dans le but de réduire les risques de brouillage radioélectrique à l'intention des autres utilisateurs, il faut choisir le type d'antenne et son gain de sorte que la puissance isotrope rayonnée équivalente (p.i.r.e.) ne dépasse pas l'intensité nécessaire à l'établissement d'une communication satisfaisante. Le présent émetteur radio a été approuvé par Industrie Canada pour fonctionner avec les types d'antenne énumérés dans le manuel d'usage et ayant un gain admissible maximal et l'impédance requise pour chaque type d'antenne. Les types d'antenne non inclus dans cette liste, ou dont le gain est supérieur au gain maximal indiqué, sont strictement interdits pour l'exploitation de l'émetteur.

3.3 Japan

3.3.1 *Notice for EVMs delivered in Japan:* Please see http://www.tij.co.jp/lstds/ti_ja/general/eStore/notice_01.page 日本国内に輸入される評価用キット、ボードについては、次のところをご覧ください。
http://www.tij.co.jp/lstds/ti_ja/general/eStore/notice_01.page

3.3.2 *Notice for Users of EVMs Considered "Radio Frequency Products" in Japan:* EVMs entering Japan may not be certified by TI as conforming to Technical Regulations of Radio Law of Japan.

If User uses EVMs in Japan, not certified to Technical Regulations of Radio Law of Japan, User is required to follow the instructions set forth by Radio Law of Japan, which includes, but is not limited to, the instructions below with respect to EVMs (which for the avoidance of doubt are stated strictly for convenience and should be verified by User):

1. Use EVMs in a shielded room or any other test facility as defined in the notification #173 issued by Ministry of Internal Affairs and Communications on March 28, 2006, based on Sub-section 1.1 of Article 6 of the Ministry's Rule for Enforcement of Radio Law of Japan,
2. Use EVMs only after User obtains the license of Test Radio Station as provided in Radio Law of Japan with respect to EVMs, or
3. Use of EVMs only after User obtains the Technical Regulations Conformity Certification as provided in Radio Law of Japan with respect to EVMs. Also, do not transfer EVMs, unless User gives the same notice above to the transferee. Please note that if User does not follow the instructions above, User will be subject to penalties of Radio Law of Japan.

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2. 実験局の免許を取得後ご使用いただく。
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3.4 *European Union*

3.4.1 *For EVMs subject to EU Directive 2014/30/EU (Electromagnetic Compatibility Directive):*

This is a class A product intended for use in environments other than domestic environments that are connected to a low-voltage power-supply network that supplies buildings used for domestic purposes. In a domestic environment this product may cause radio interference in which case the user may be required to take adequate measures.

4 *EVM Use Restrictions and Warnings:*

4.1 EVMS ARE NOT FOR USE IN FUNCTIONAL SAFETY AND/OR SAFETY CRITICAL EVALUATIONS, INCLUDING BUT NOT LIMITED TO EVALUATIONS OF LIFE SUPPORT APPLICATIONS.

4.2 User must read and apply the user guide and other available documentation provided by TI regarding the EVM prior to handling or using the EVM, including without limitation any warning or restriction notices. The notices contain important safety information related to, for example, temperatures and voltages.

4.3 *Safety-Related Warnings and Restrictions:*

4.3.1 User shall operate the EVM within TI's recommended specifications and environmental considerations stated in the user guide, other available documentation provided by TI, and any other applicable requirements and employ reasonable and customary safeguards. Exceeding the specified performance ratings and specifications (including but not limited to input and output voltage, current, power, and environmental ranges) for the EVM may cause personal injury or death, or property damage. If there are questions concerning performance ratings and specifications, User should contact a TI field representative prior to connecting interface electronics including input power and intended loads. Any loads applied outside of the specified output range may also result in unintended and/or inaccurate operation and/or possible permanent damage to the EVM and/or interface electronics. Please consult the EVM user guide prior to connecting any load to the EVM output. If there is uncertainty as to the load specification, please contact a TI field representative. During normal operation, even with the inputs and outputs kept within the specified allowable ranges, some circuit components may have elevated case temperatures. These components include but are not limited to linear regulators, switching transistors, pass transistors, current sense resistors, and heat sinks, which can be identified using the information in the associated documentation. When working with the EVM, please be aware that the EVM may become very warm.

4.3.2 EVMs are intended solely for use by technically qualified, professional electronics experts who are familiar with the dangers and application risks associated with handling electrical mechanical components, systems, and subsystems. User assumes all responsibility and liability for proper and safe handling and use of the EVM by User or its employees, affiliates, contractors or designees. User assumes all responsibility and liability to ensure that any interfaces (electronic and/or mechanical) between the EVM and any human body are designed with suitable isolation and means to safely limit accessible leakage currents to minimize the risk of electrical shock hazard. User assumes all responsibility and liability for any improper or unsafe handling or use of the EVM by User or its employees, affiliates, contractors or designees.

4.4 User assumes all responsibility and liability to determine whether the EVM is subject to any applicable international, federal, state, or local laws and regulations related to User's handling and use of the EVM and, if applicable, User assumes all responsibility and liability for compliance in all respects with such laws and regulations. User assumes all responsibility and liability for proper disposal and recycling of the EVM consistent with all applicable international, federal, state, and local requirements.

5. *Accuracy of Information:* To the extent TI provides information on the availability and function of EVMs, TI attempts to be as accurate as possible. However, TI does not warrant the accuracy of EVM descriptions, EVM availability or other information on its websites as accurate, complete, reliable, current, or error-free.

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